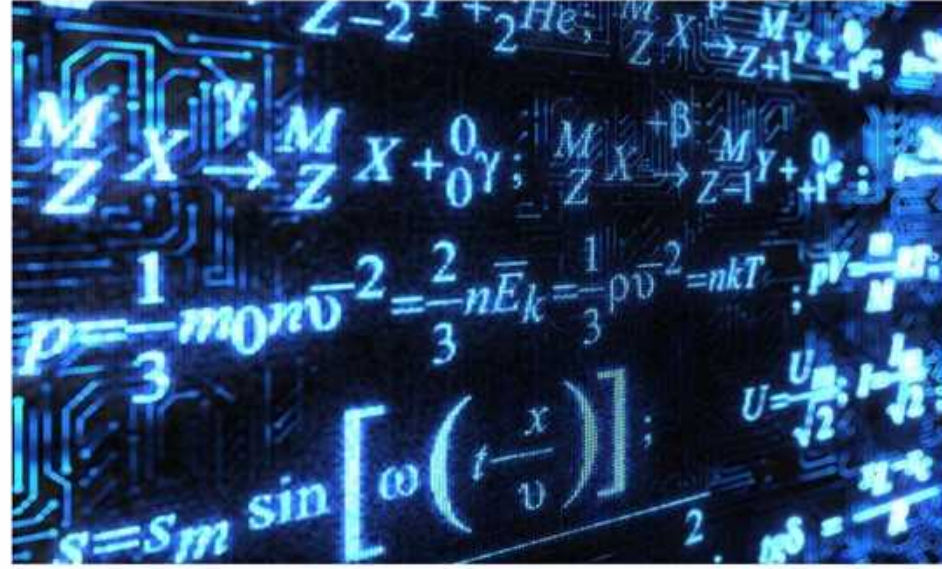




Linear & Polynomial Regression

Session -> 6

Linear Regression



Linear Regression

Machine Learning

```
graph TD; ML[Machine Learning] --> S[Supervised]; ML --> U[Unsupervised]; S --> R[Regression]; S --> C[Classification];
```

Supervised

Unsupervised

Regression

Classification

Linear Regression

Machine Learning

```
graph TD; ML[Machine Learning] --> S[Supervised]; ML --> U[Unsupervised]; S --> R[Regression]; S --> C[Classification];
```

Supervised

Unsupervised

Regression

Classification

Regression in machine learning is a supervised learning method used to predict a continuous numerical value from input data.

Linear Regression

Machine Learning

```
graph TD; ML[Machine Learning] --> S[Supervised]; ML --> U[Unsupervised]; S --> R[Regression]; S --> C[Classification]; R --> RV((Continuous Values)); RV --- RVList["1- House prices<br/>2- Video watch time<br/>3- Estimate medical expenses"]
```

Supervised

Unsupervised

Regression

Classification

Regression in machine learning is a supervised learning method used to predict a continuous numerical value from input data.

Continuous Values

- 1- House prices
- 2- Video watch time
- 3- Estimate medical expenses

Machine Learning

```
graph TD; ML[Machine Learning] --> S[Supervised]; ML --> U[Unsupervised]; S --> R[Regression]; S --> C[Classification]; R --> LR[Linear Regression];
```

Supervised

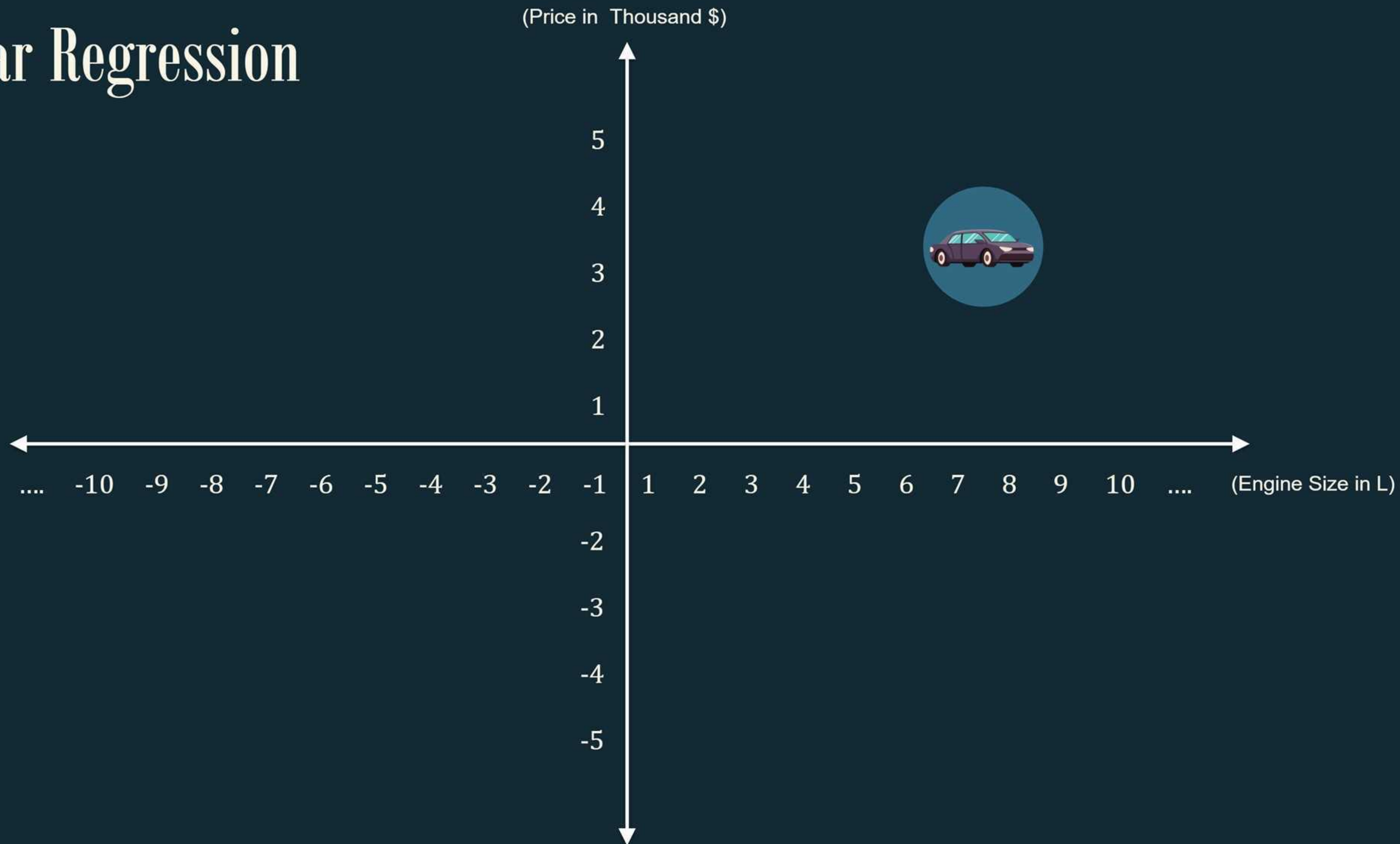
Unsupervised

Regression

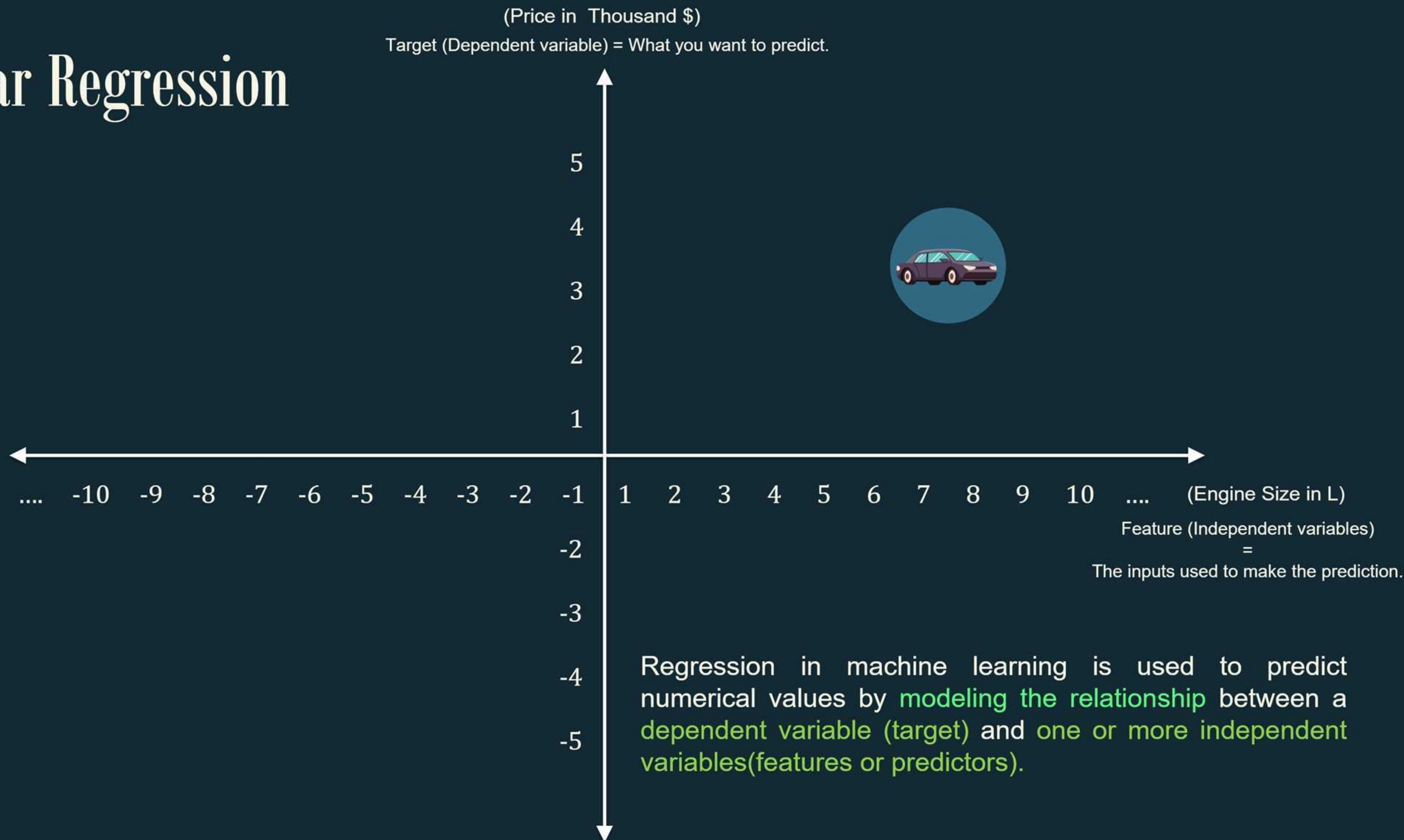
Classification

Linear Regression

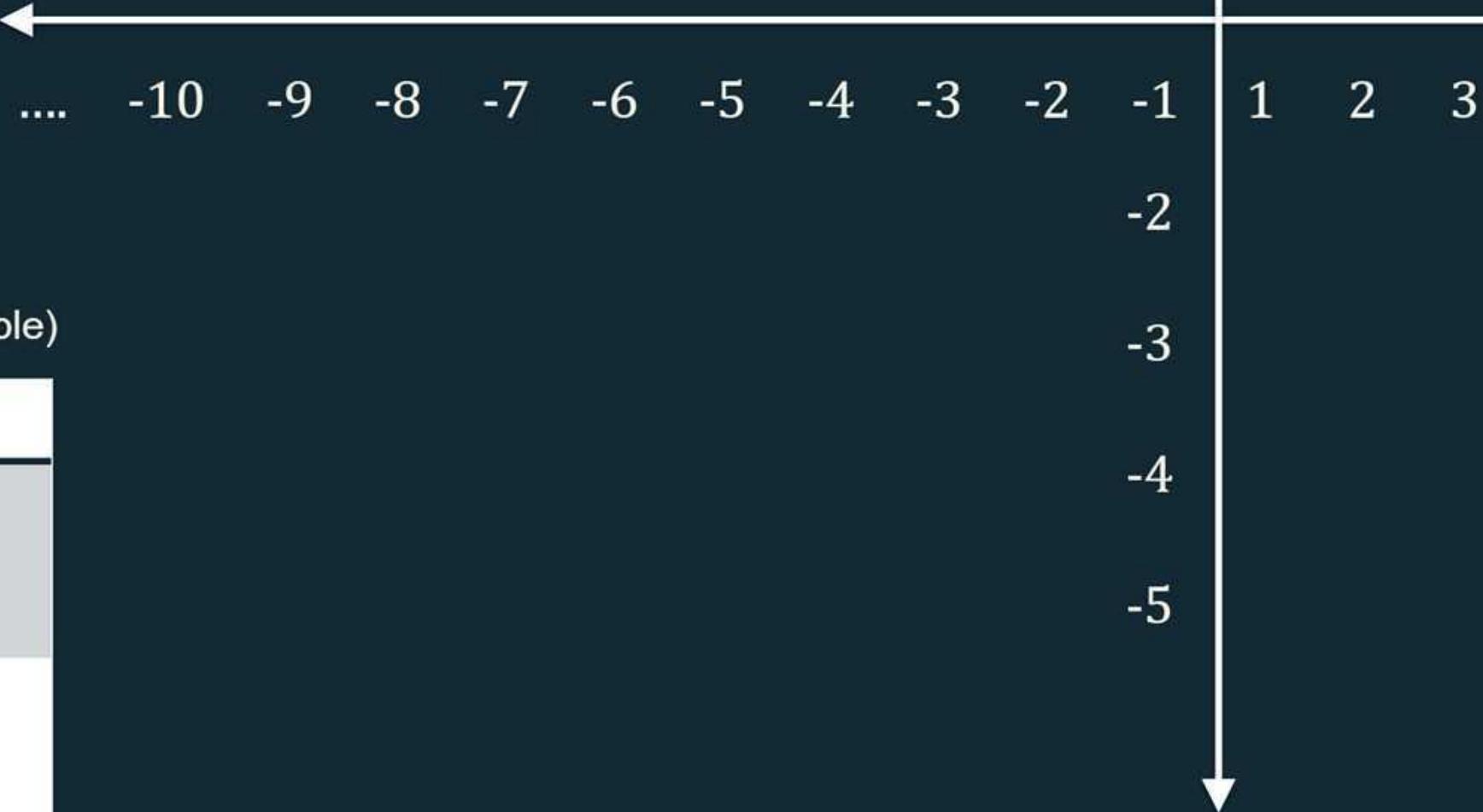
Linear Regression



Linear Regression

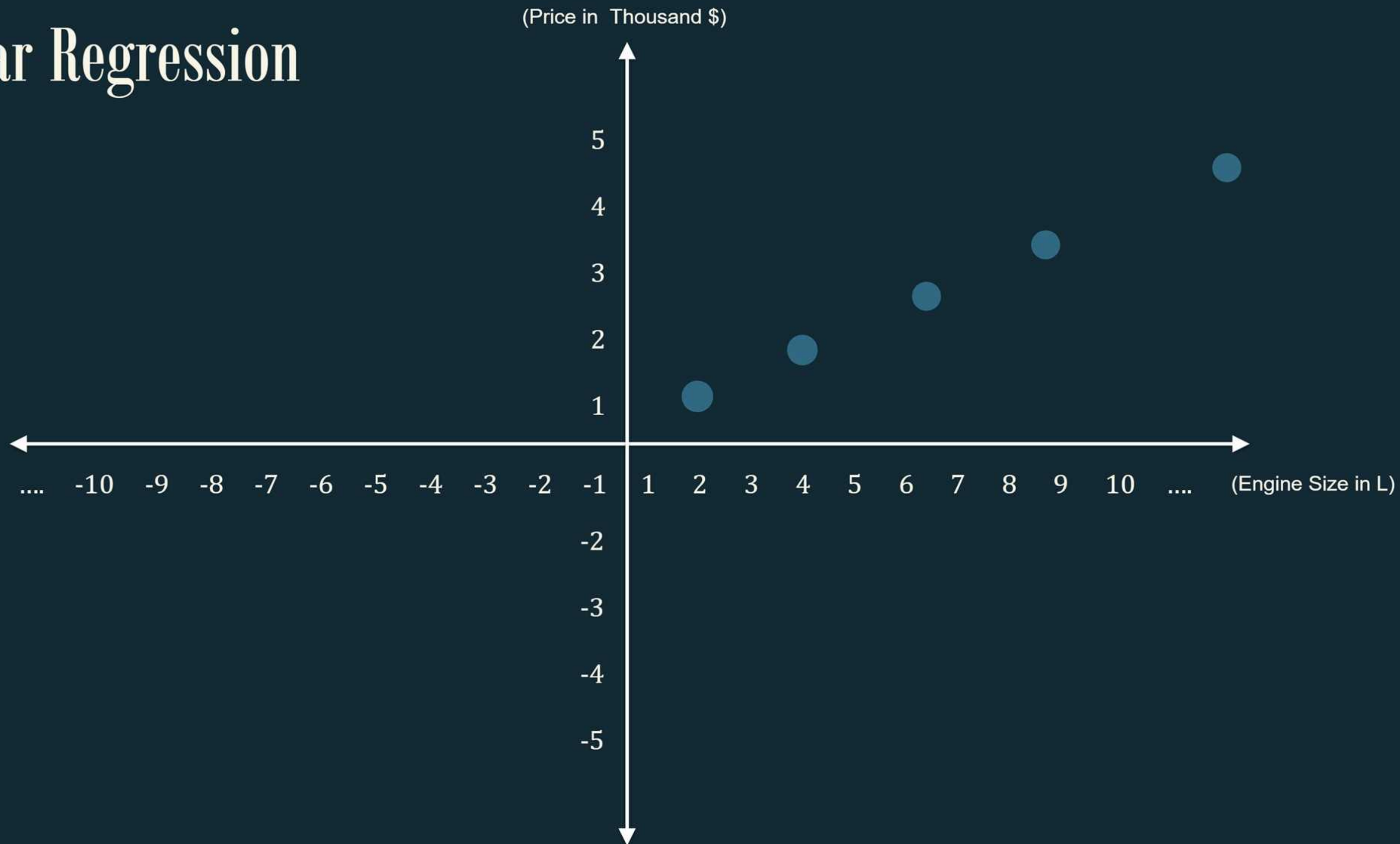


Linear Regression

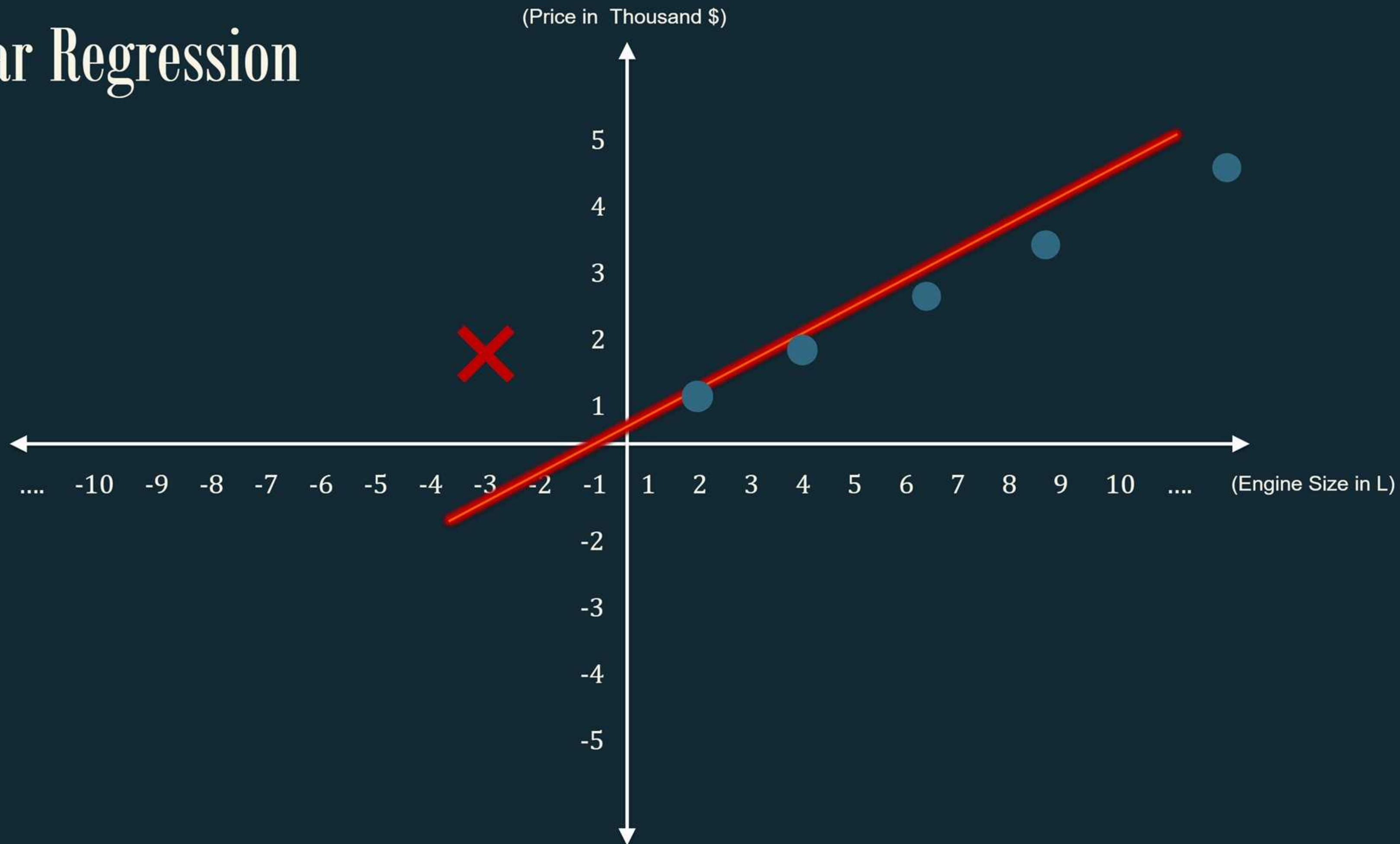


		Feature (Independent variable)	Target (Dependent variable)
		Engine Size	Price
Sample 1		1.2	2.3
Sample 2		2	4.2
Sample 3		1.6	3.45
Sample ...		...	...
Sample n		...	...

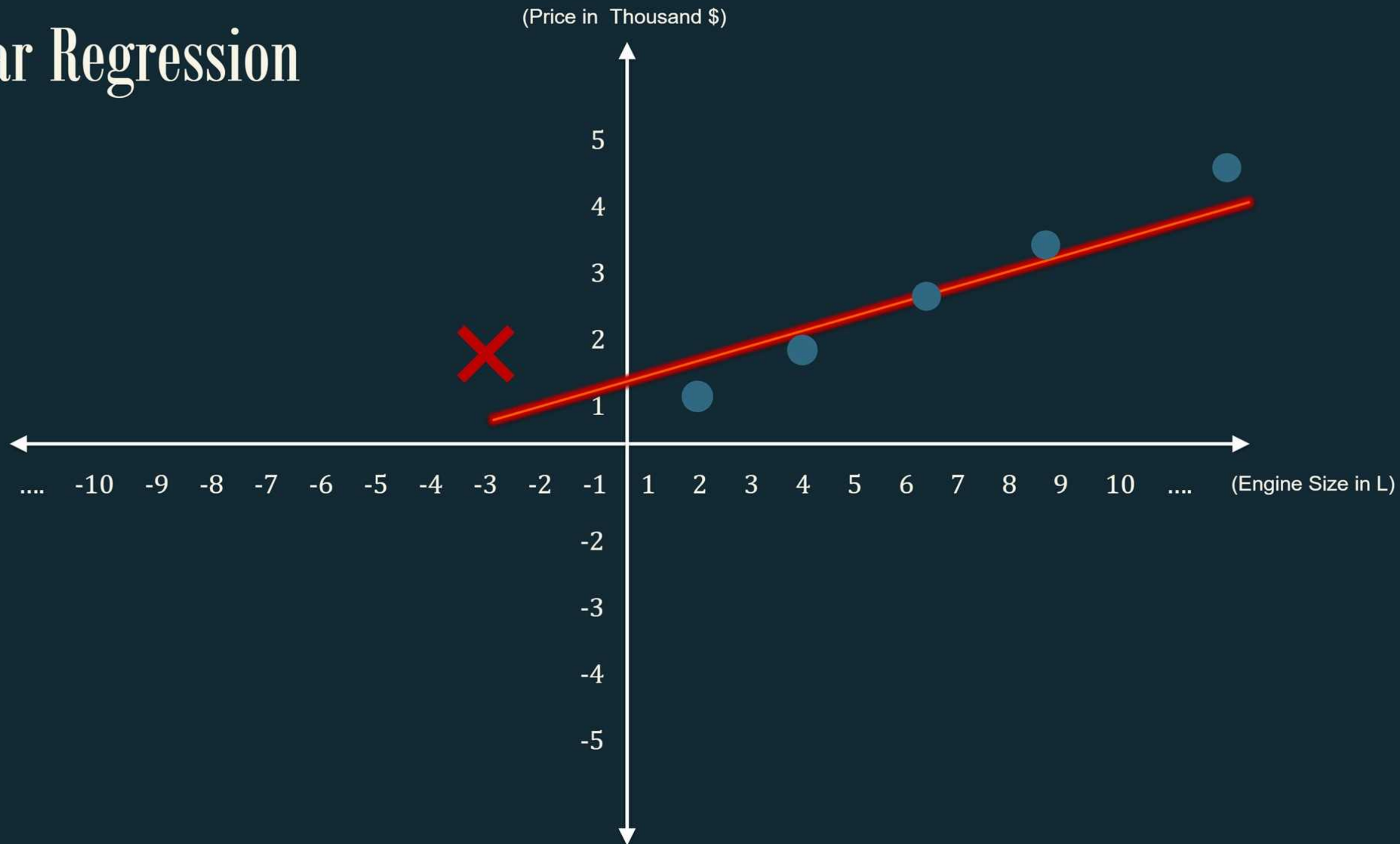
Linear Regression



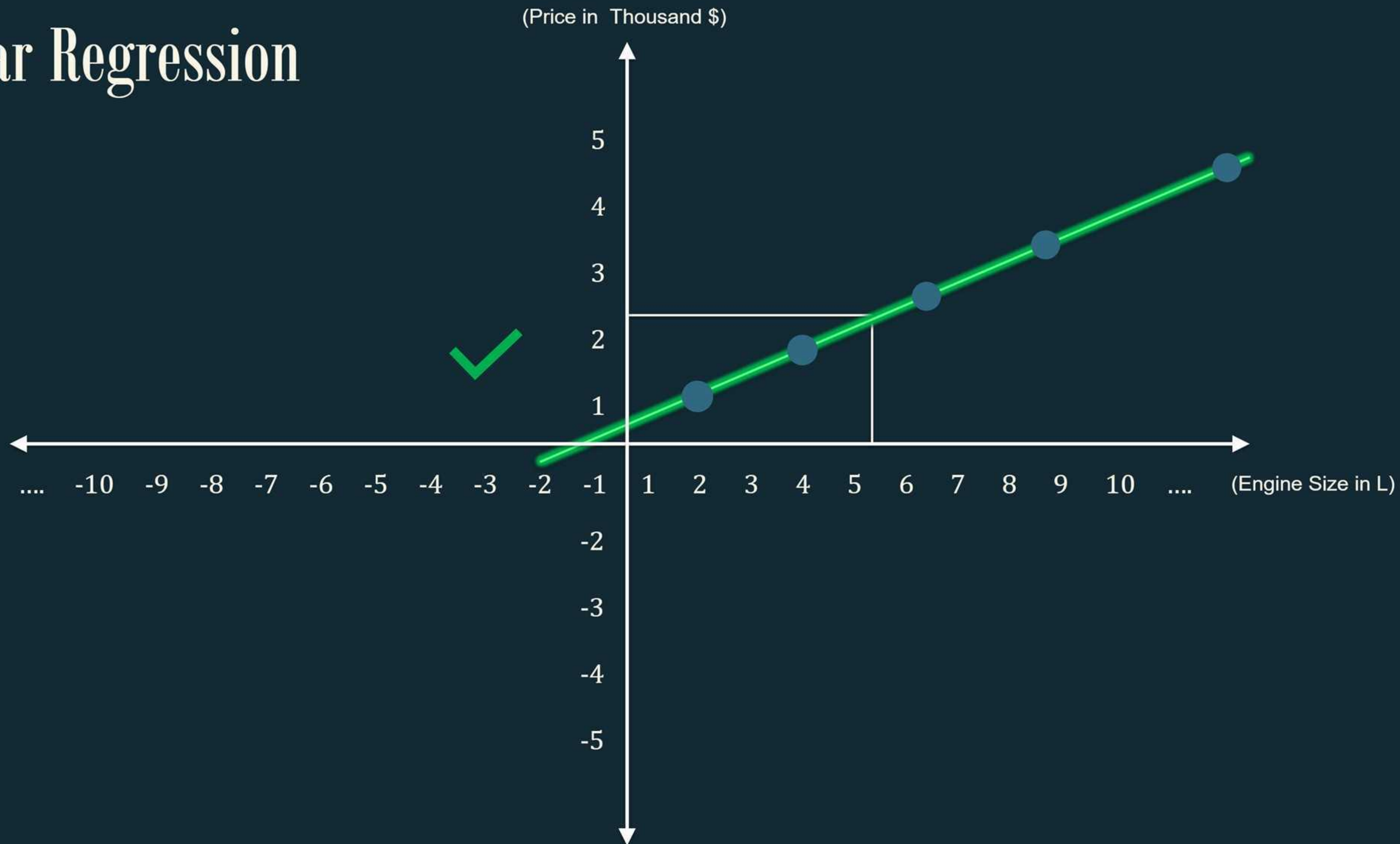
Linear Regression



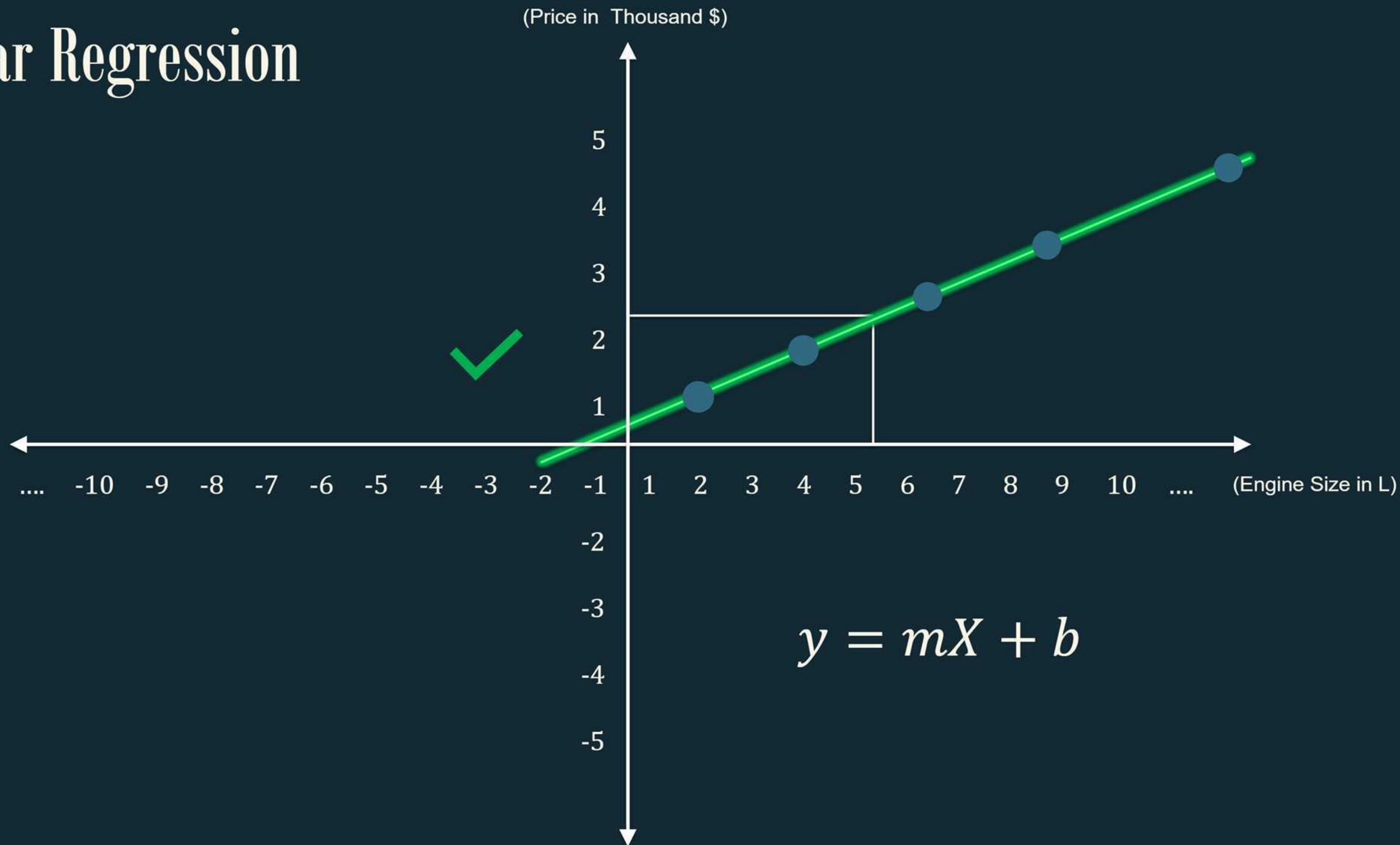
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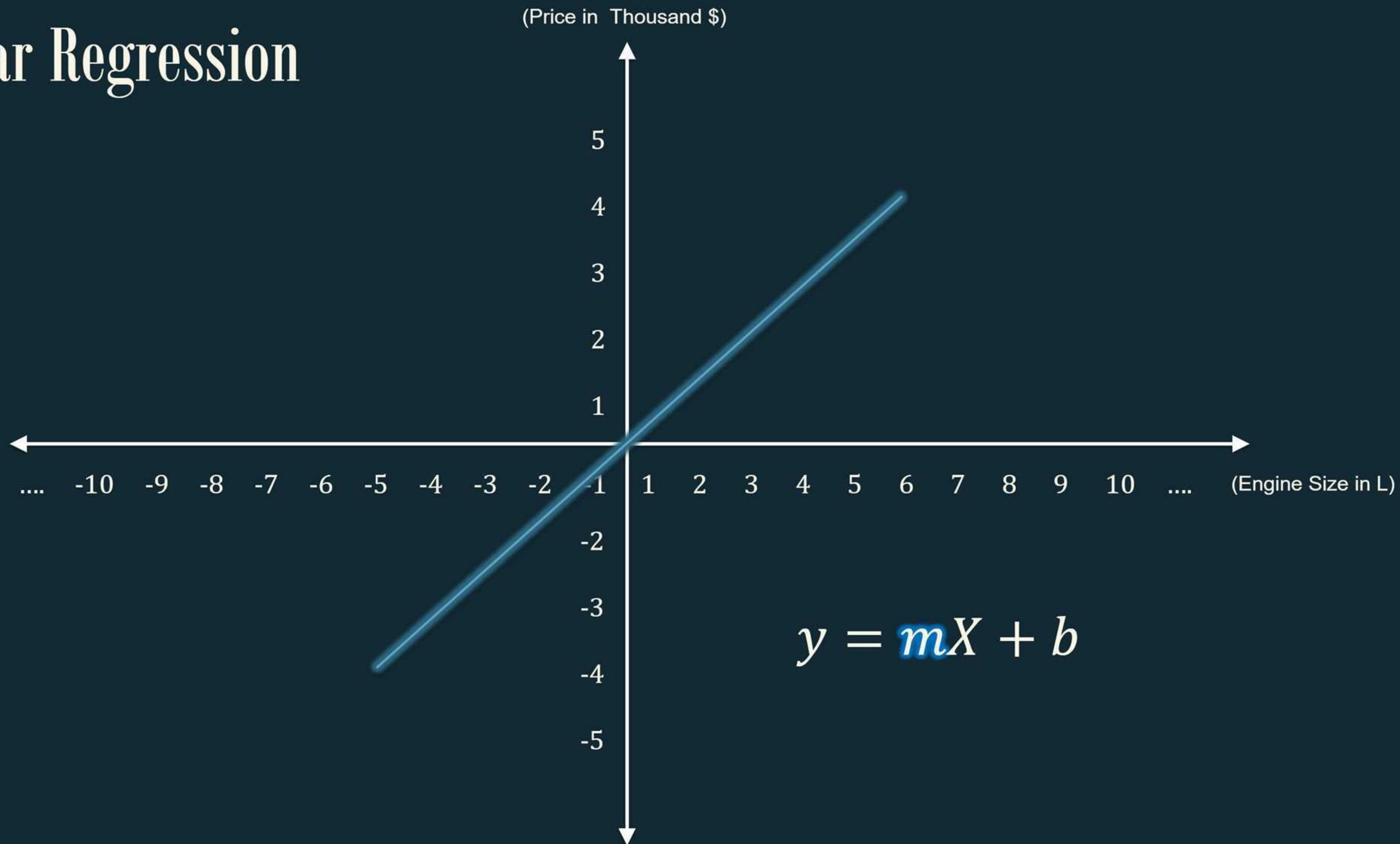
Linear Regression



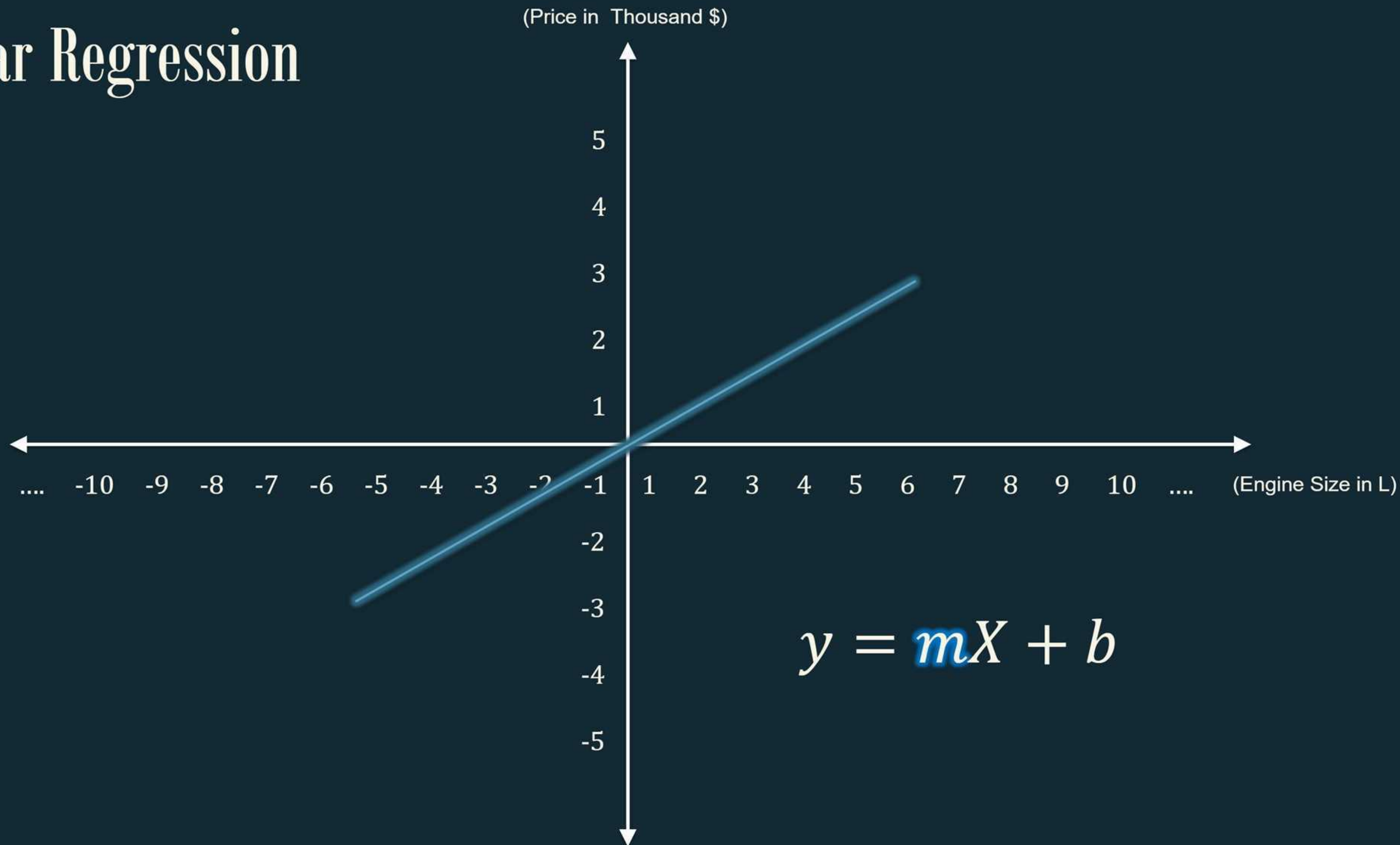
Linear Regression



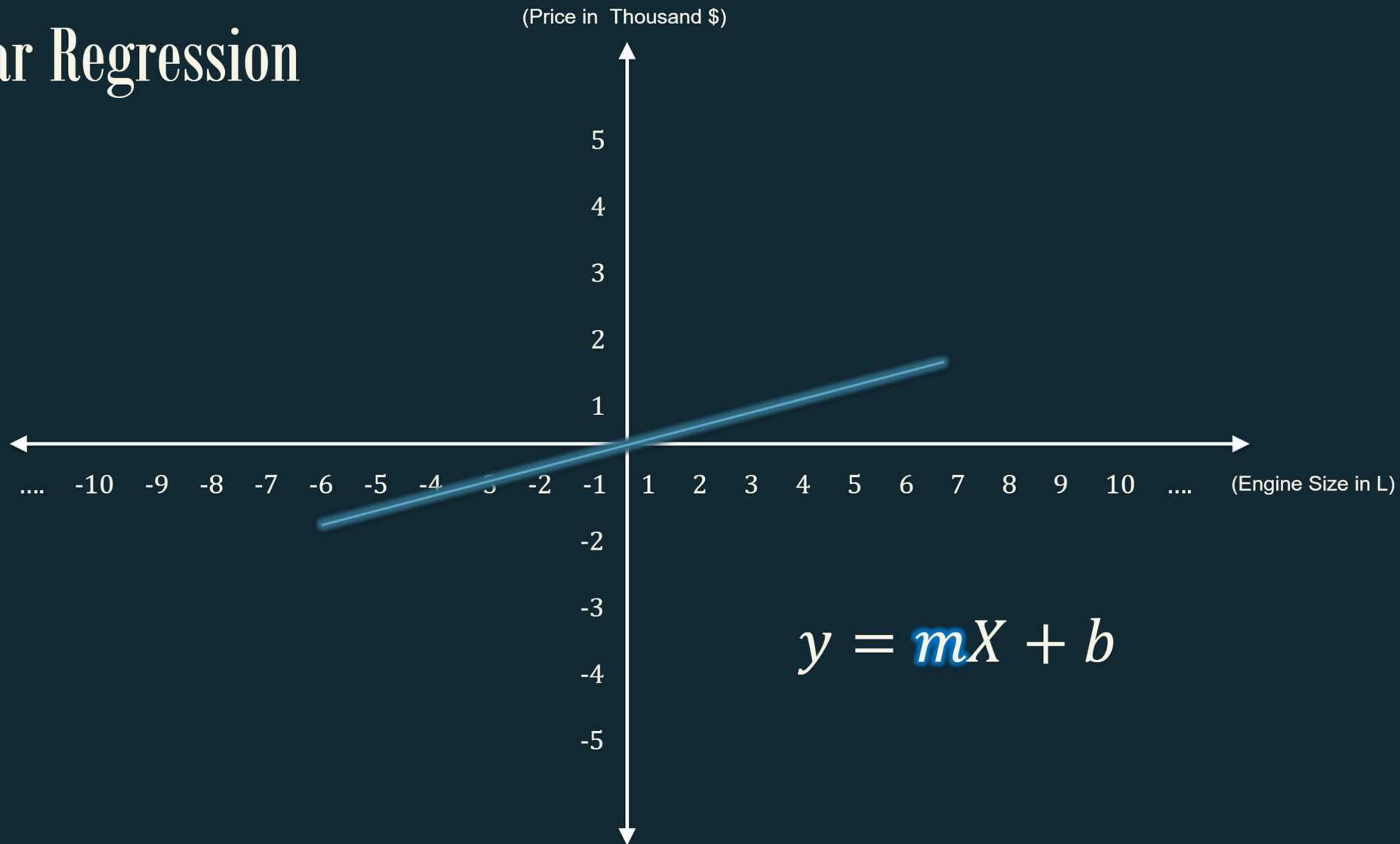
Linear Regression



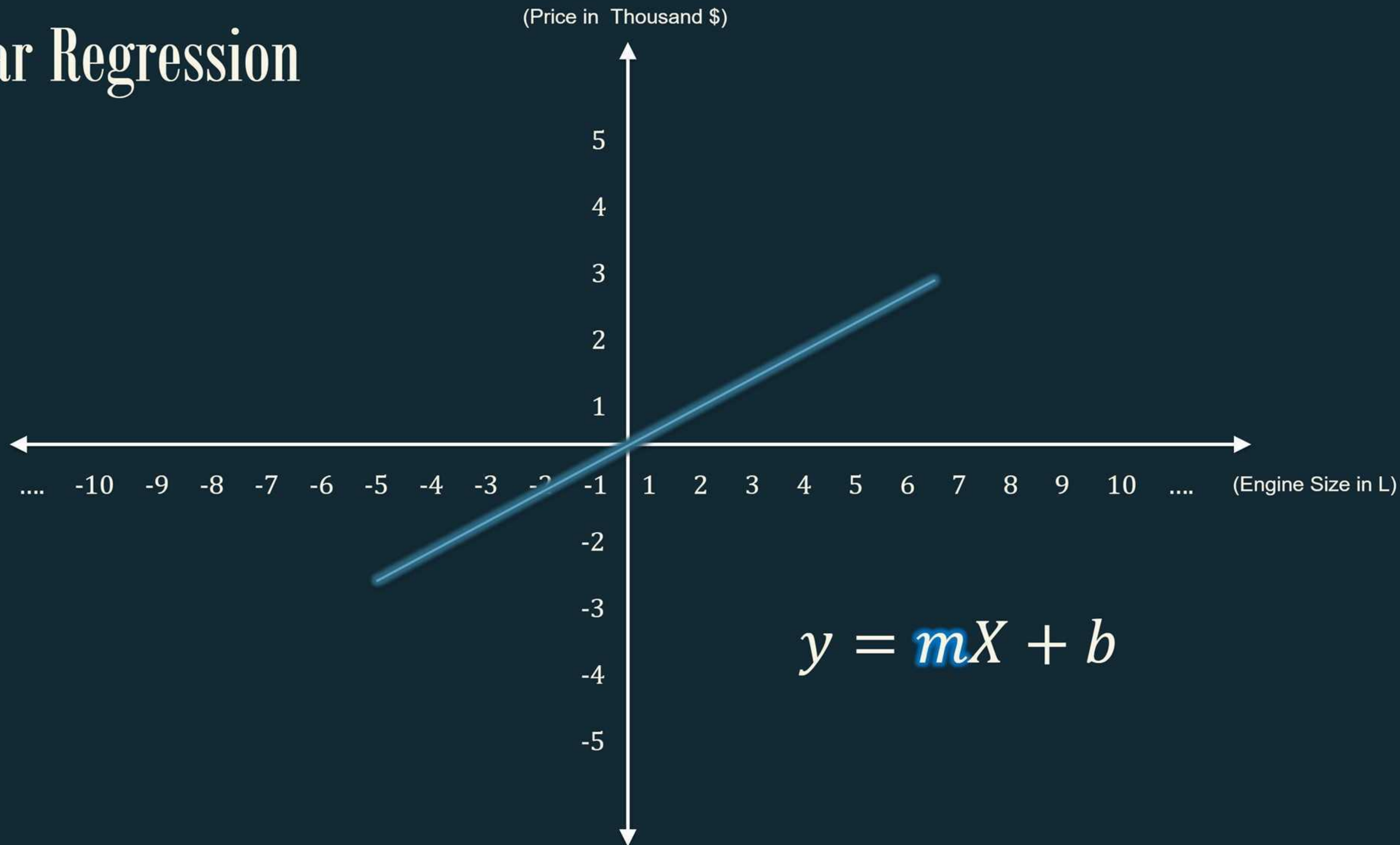
Linear Regression



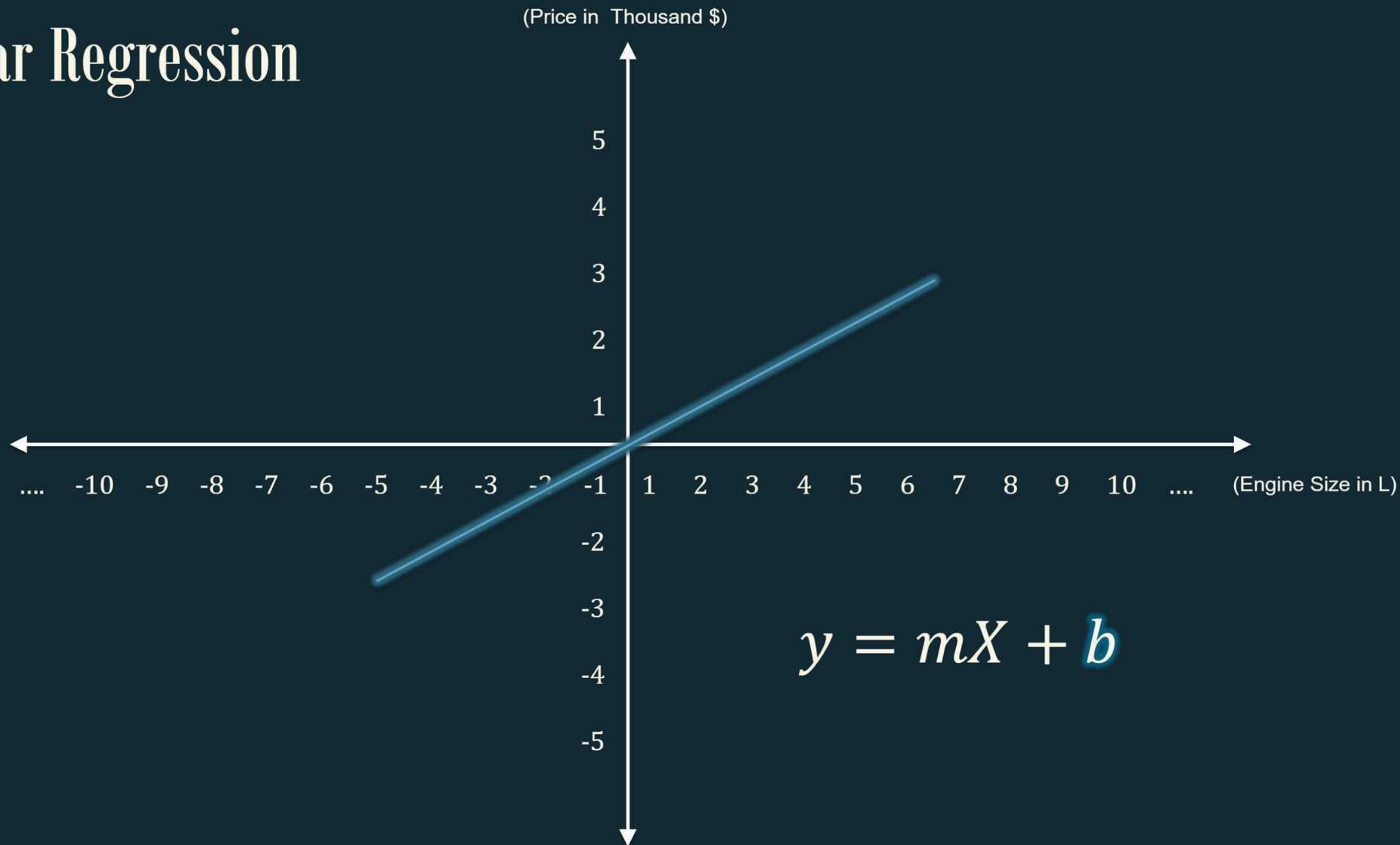
Linear Regression



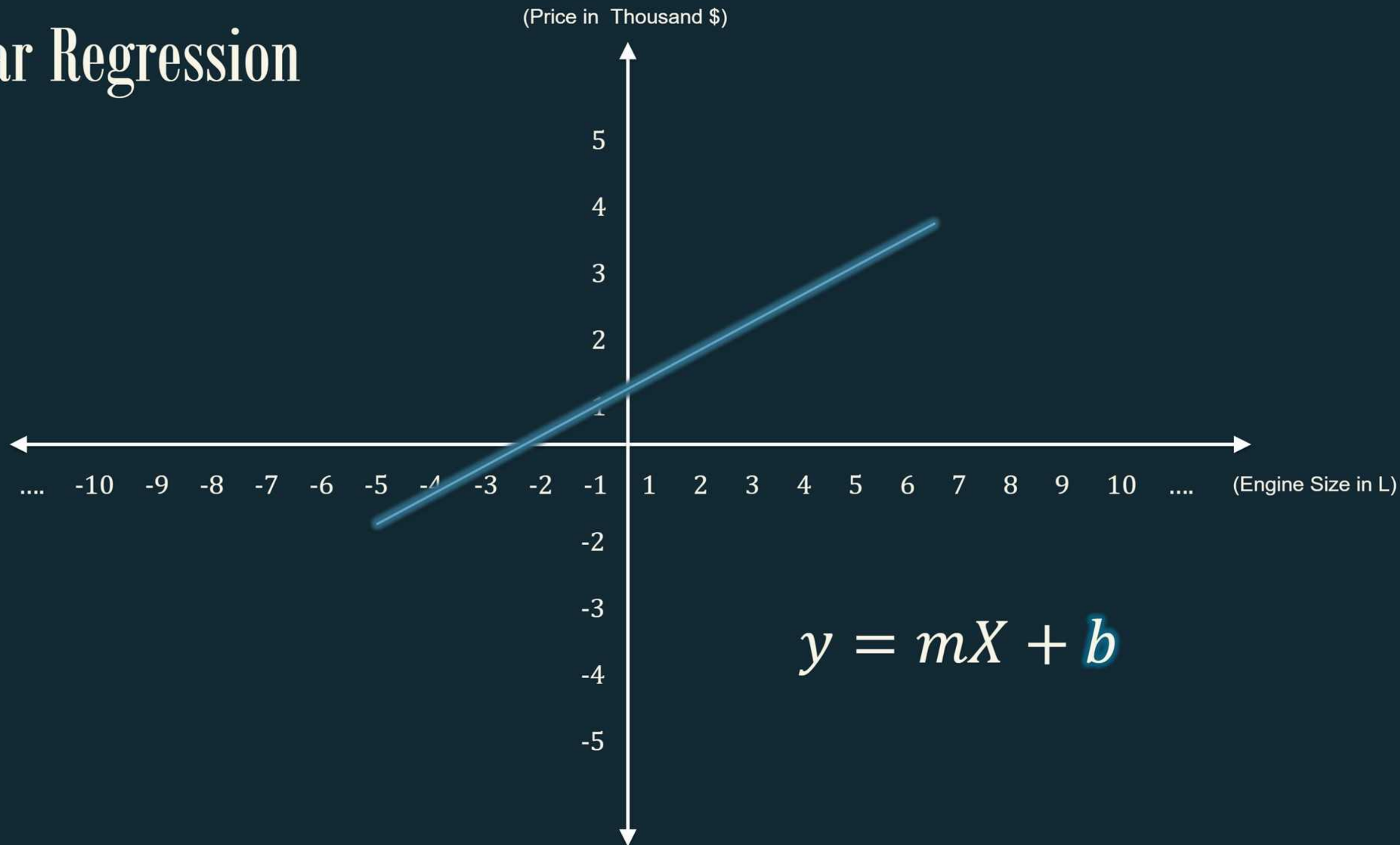
Linear Regression



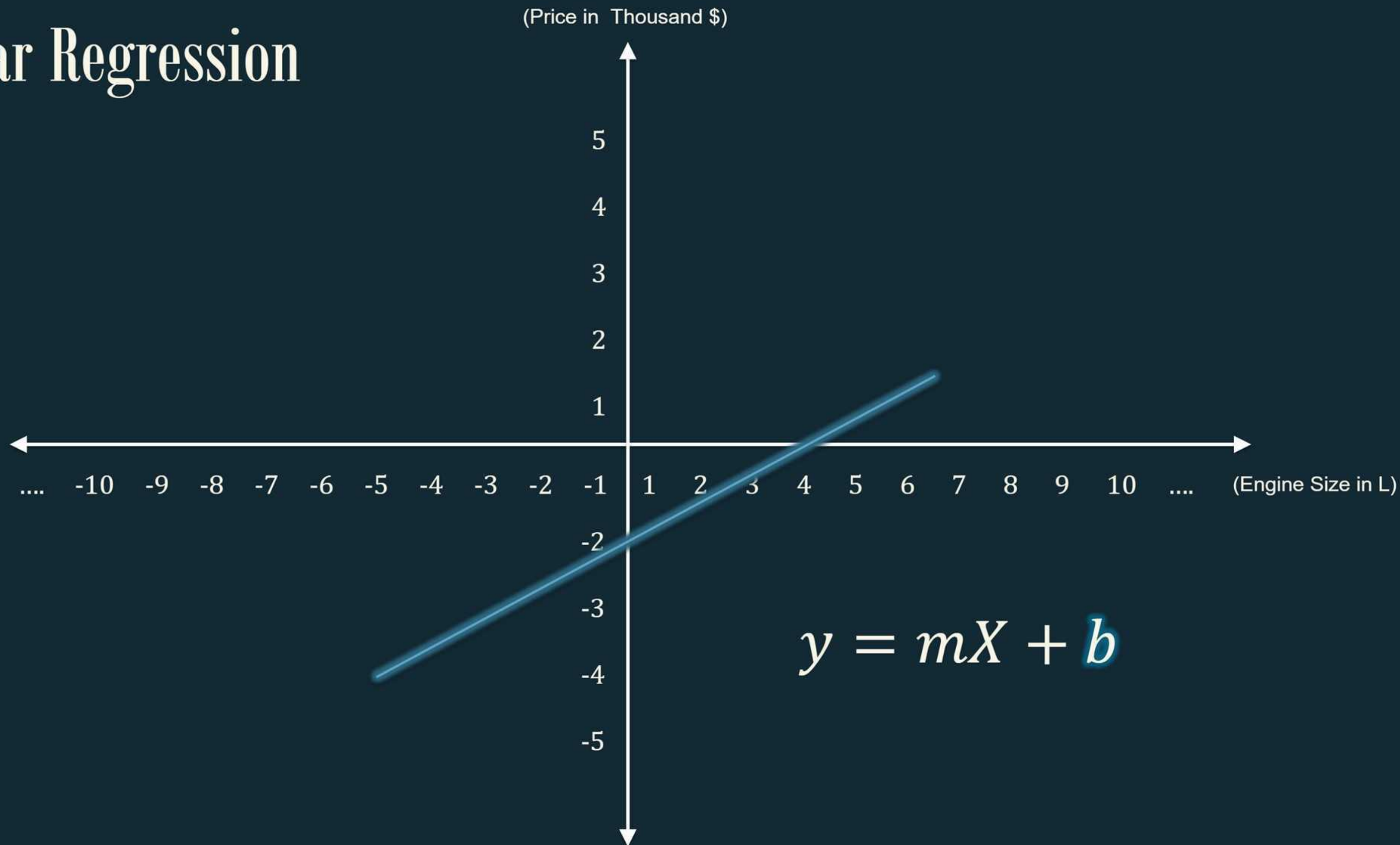
Linear Regression



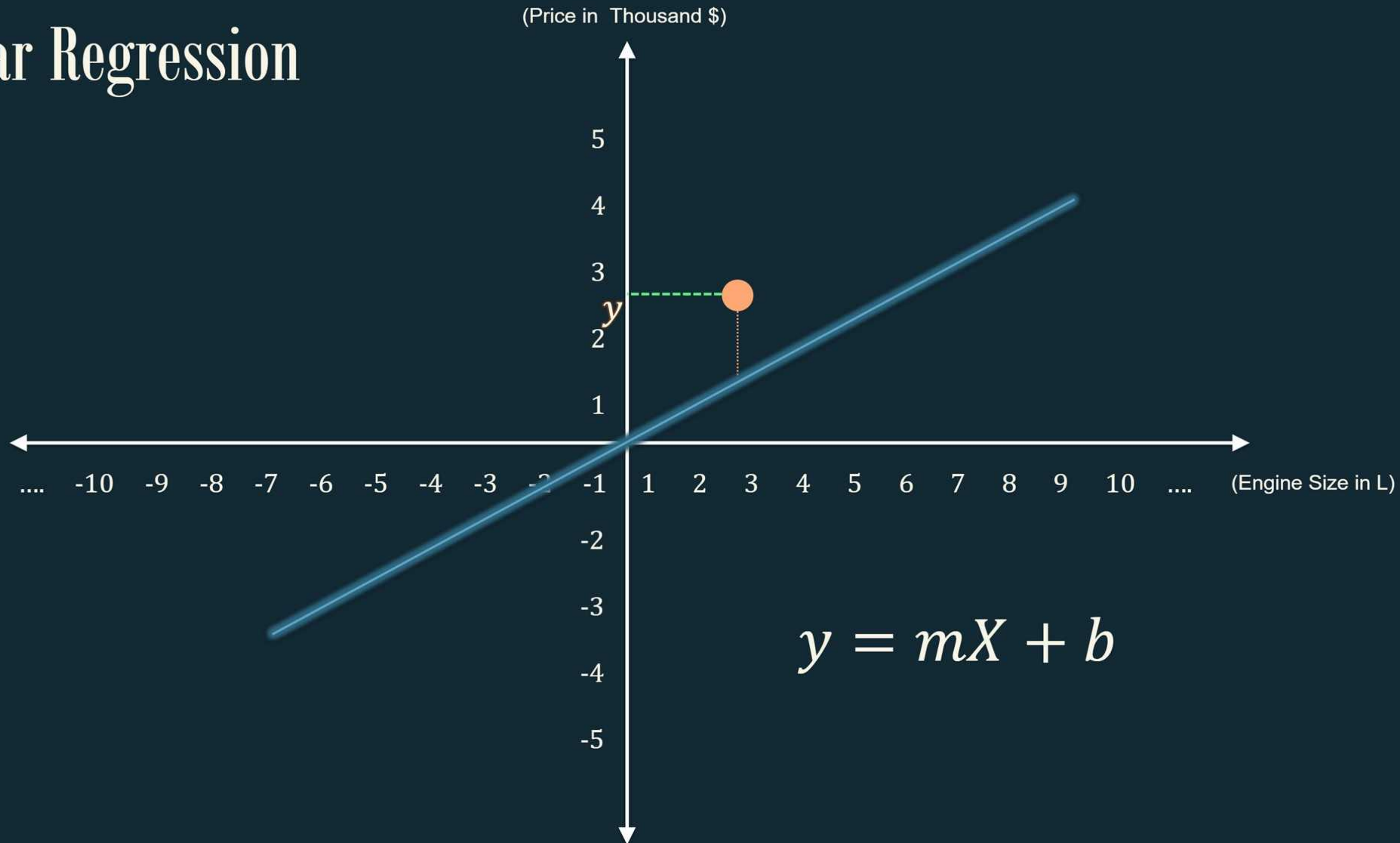
Linear Regression



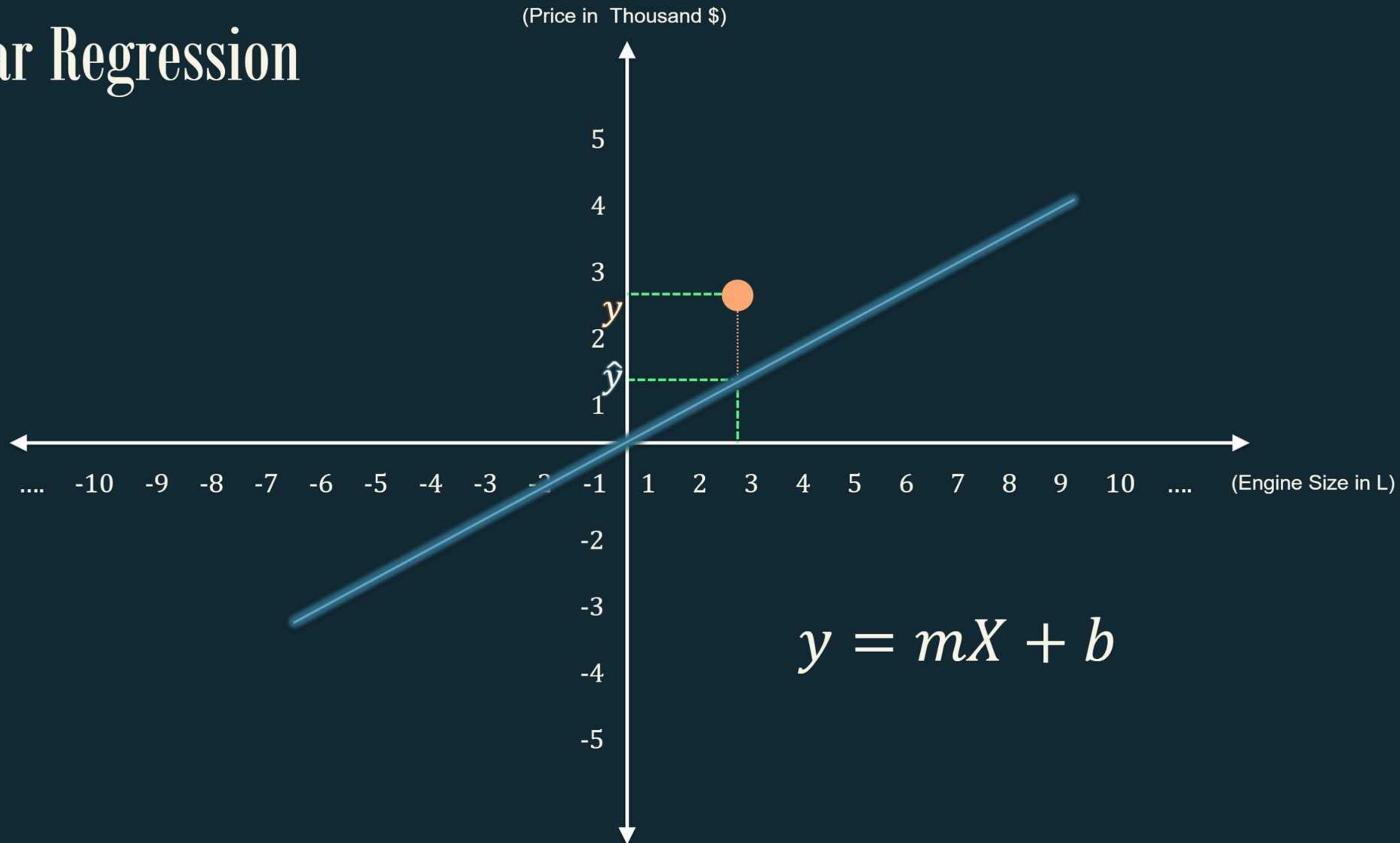
Linear Regression



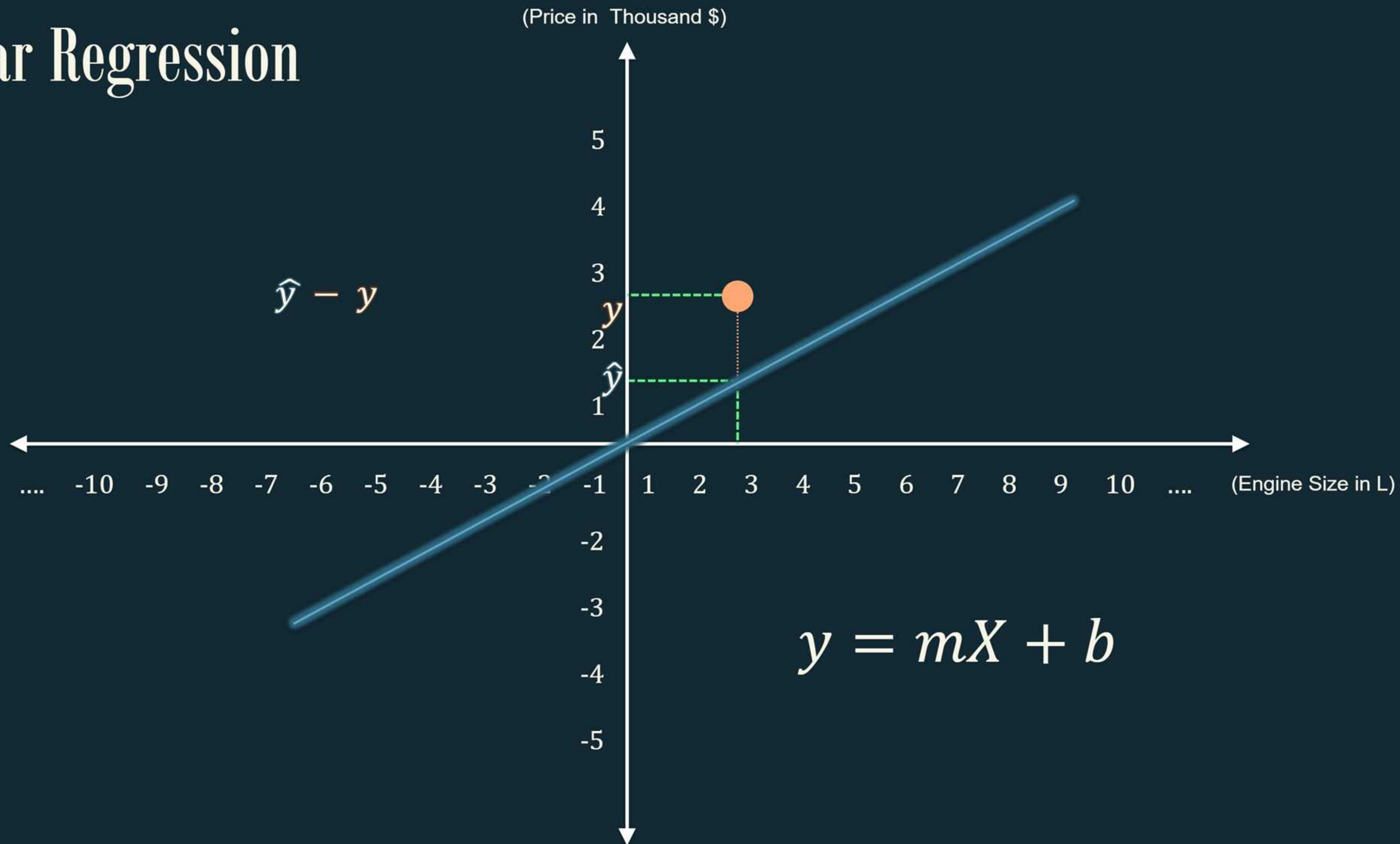
Linear Regression



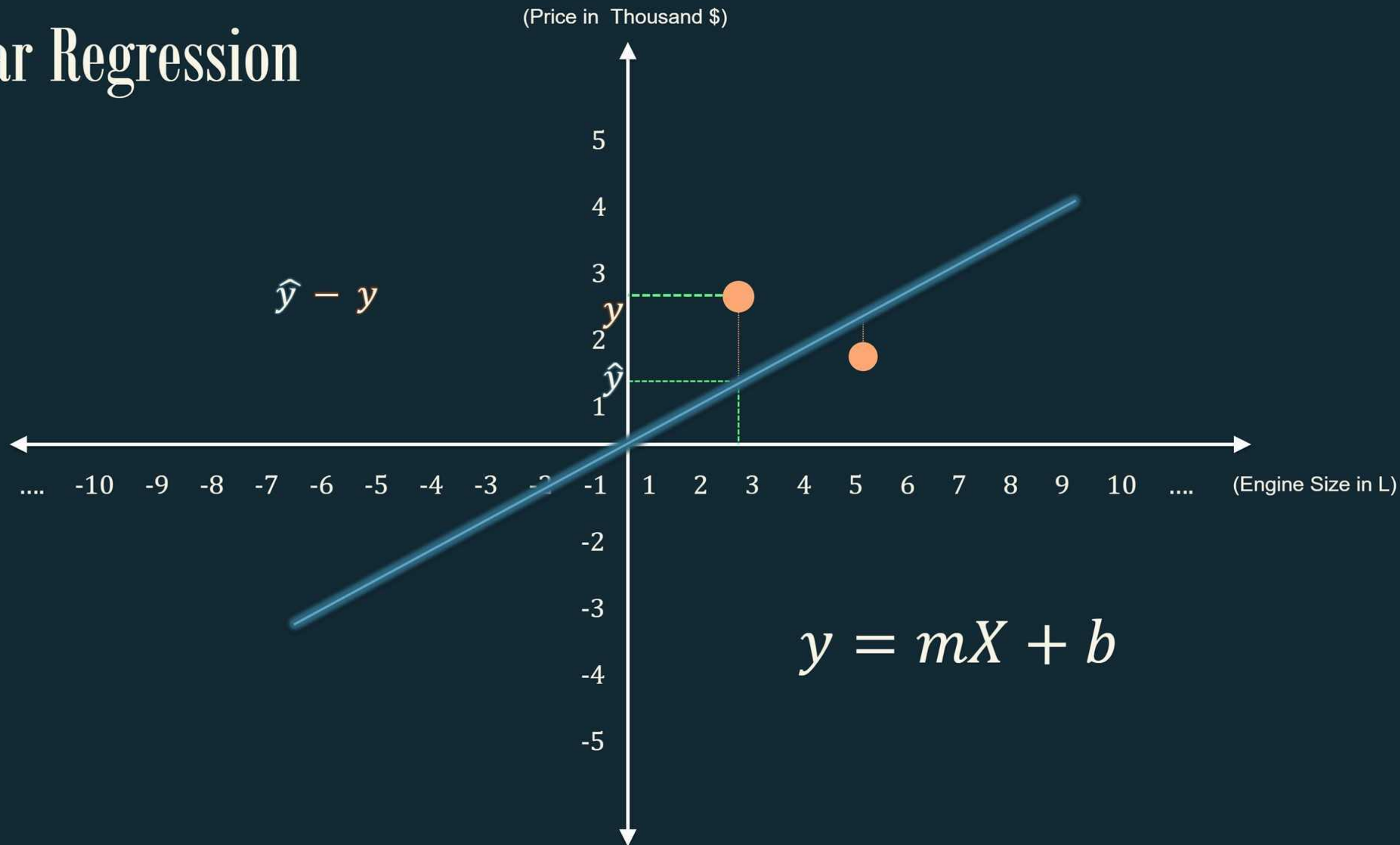
Linear Regression



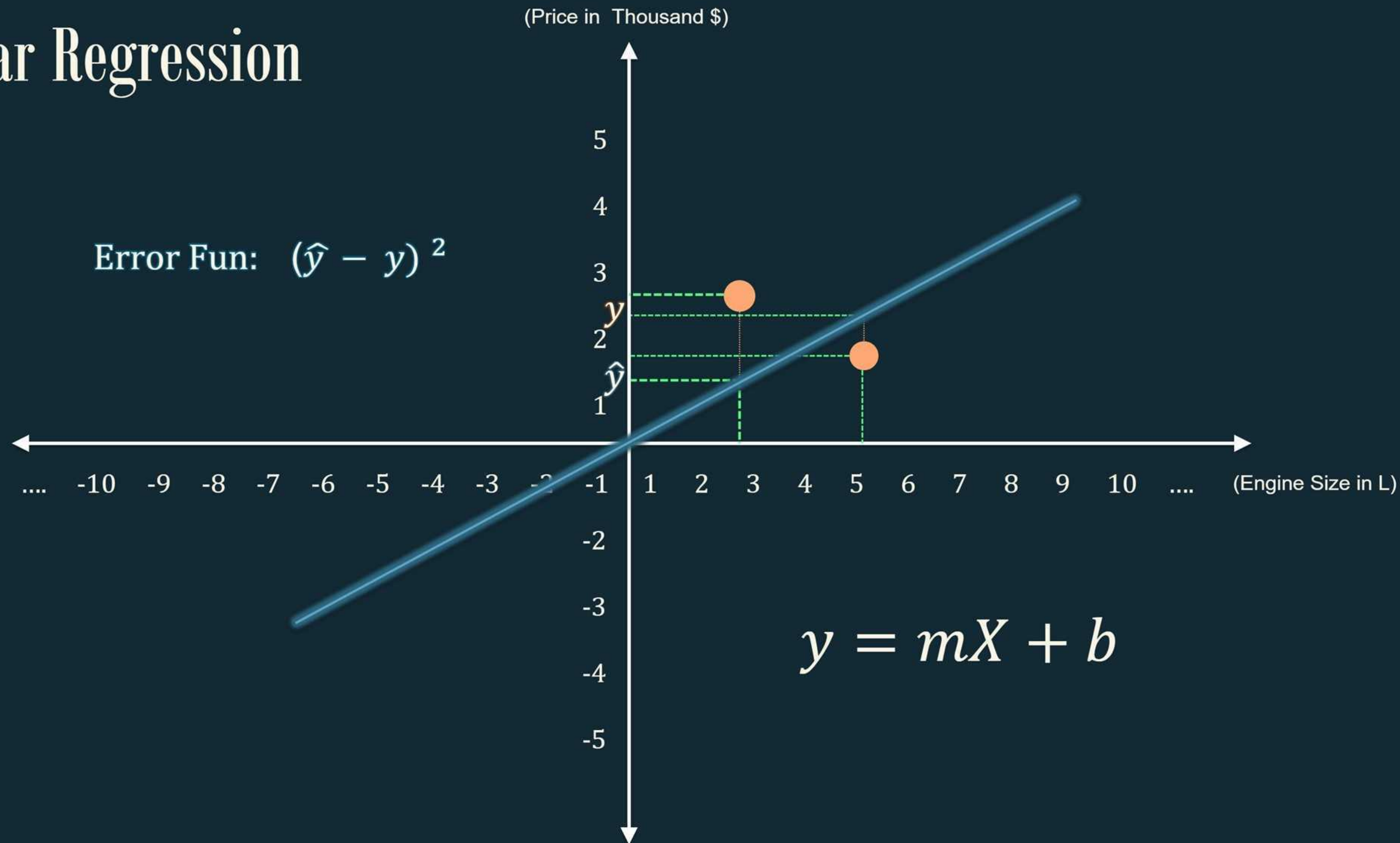
Linear Regression



Linear Regression



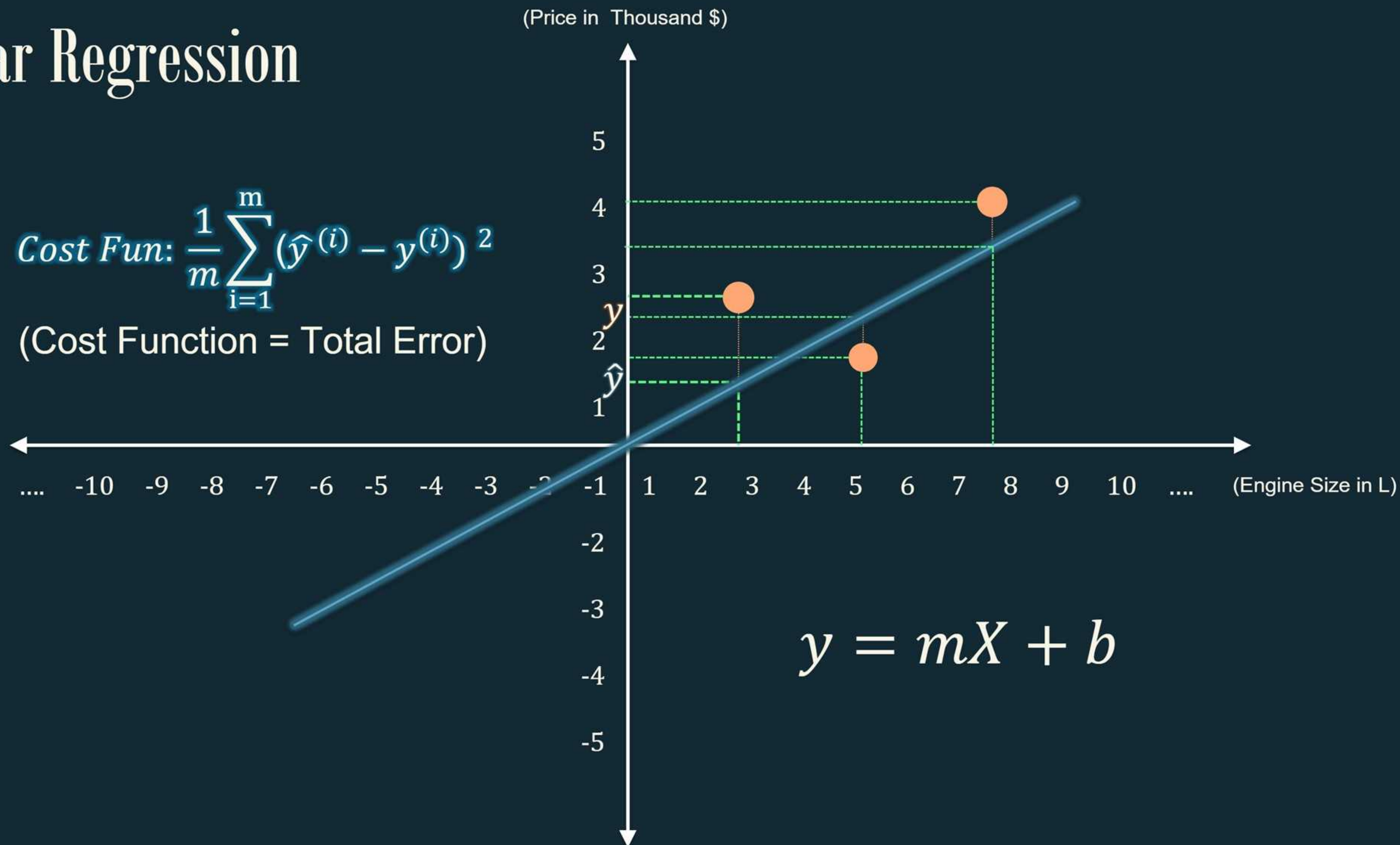
Linear Regression



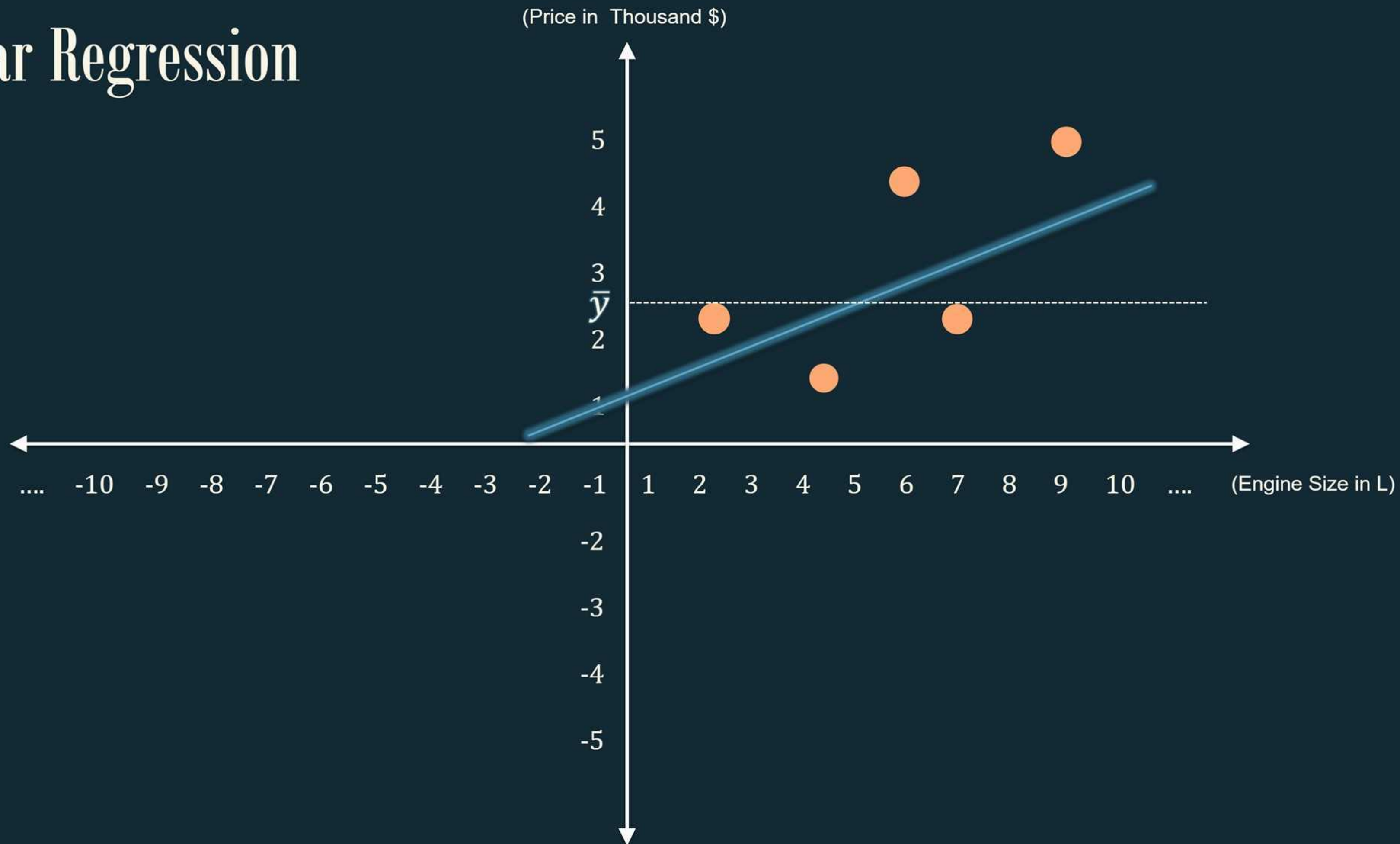
Linear Regression

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2$$

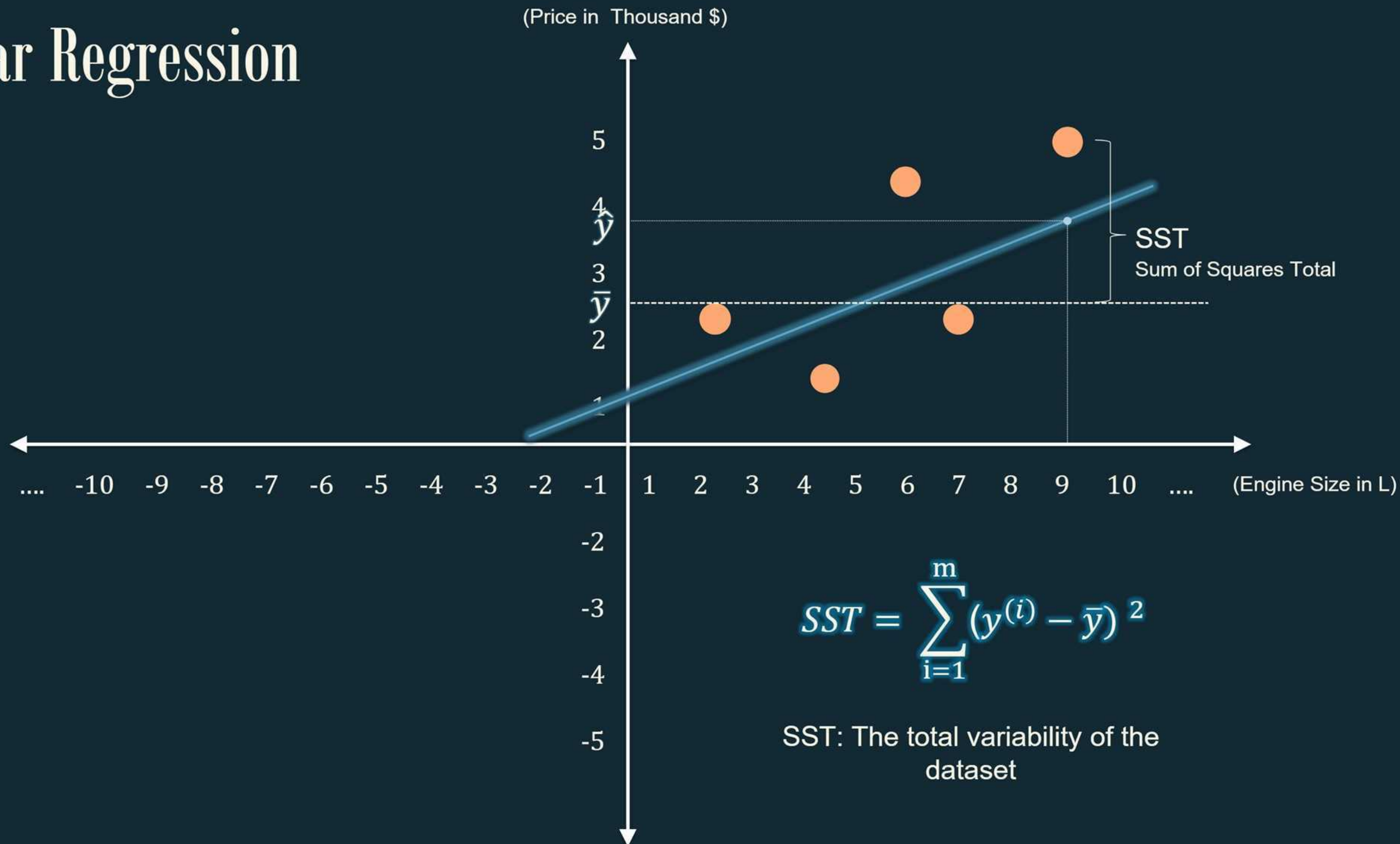
(Cost Function = Total Error)



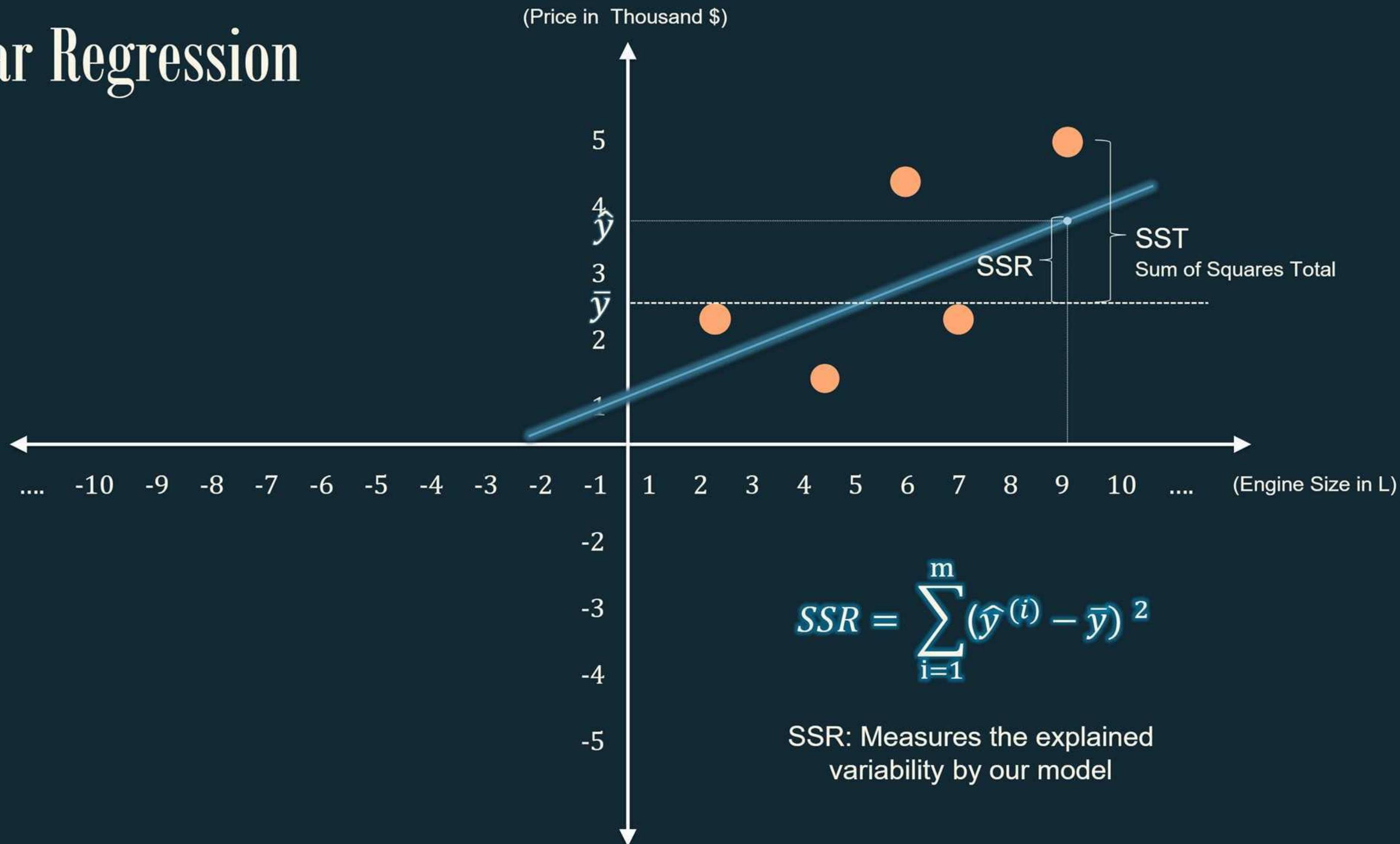
Linear Regression



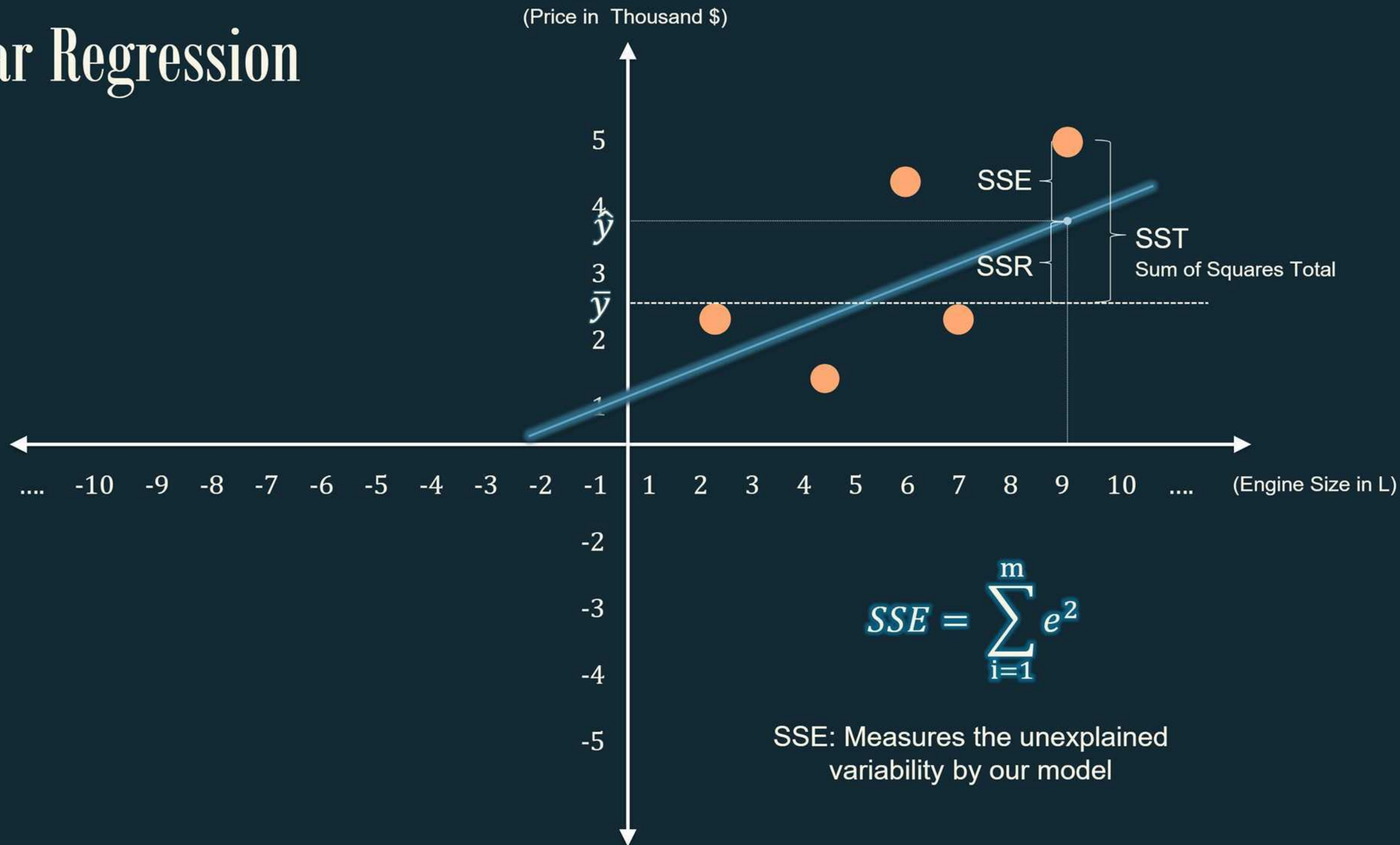
Linear Regression



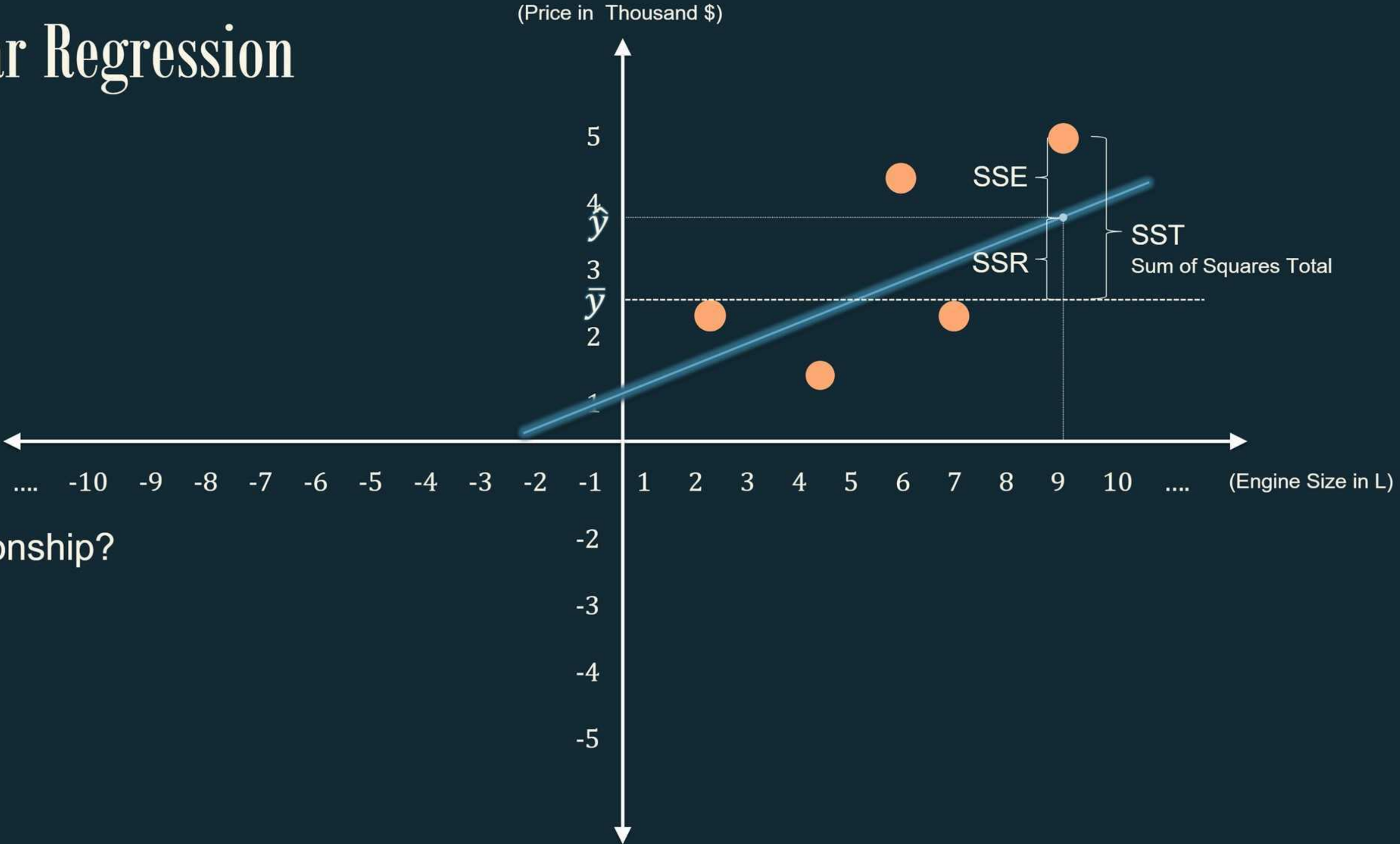
Linear Regression



Linear Regression

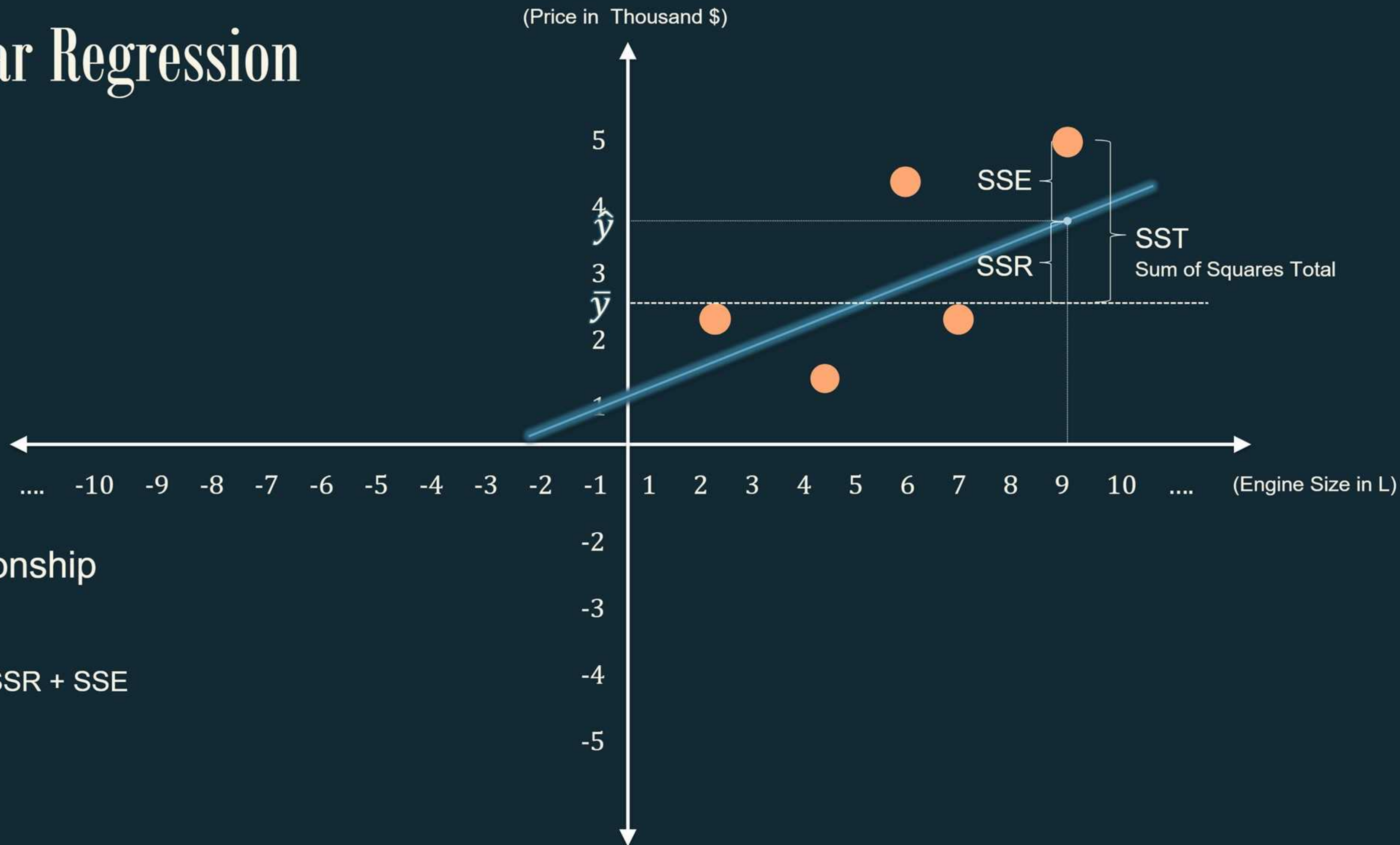


Linear Regression

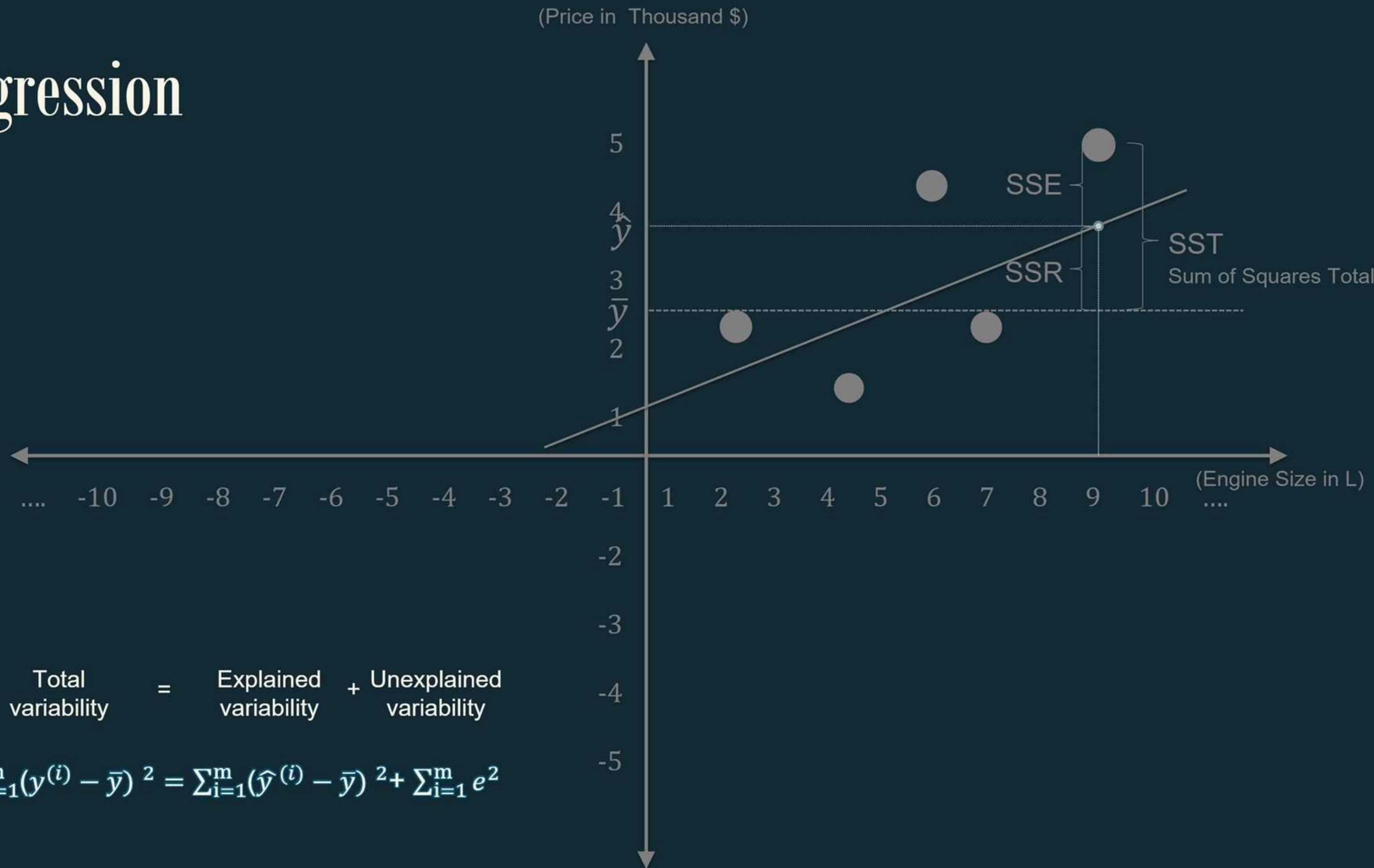


Relationship?

Linear Regression



Linear Regression

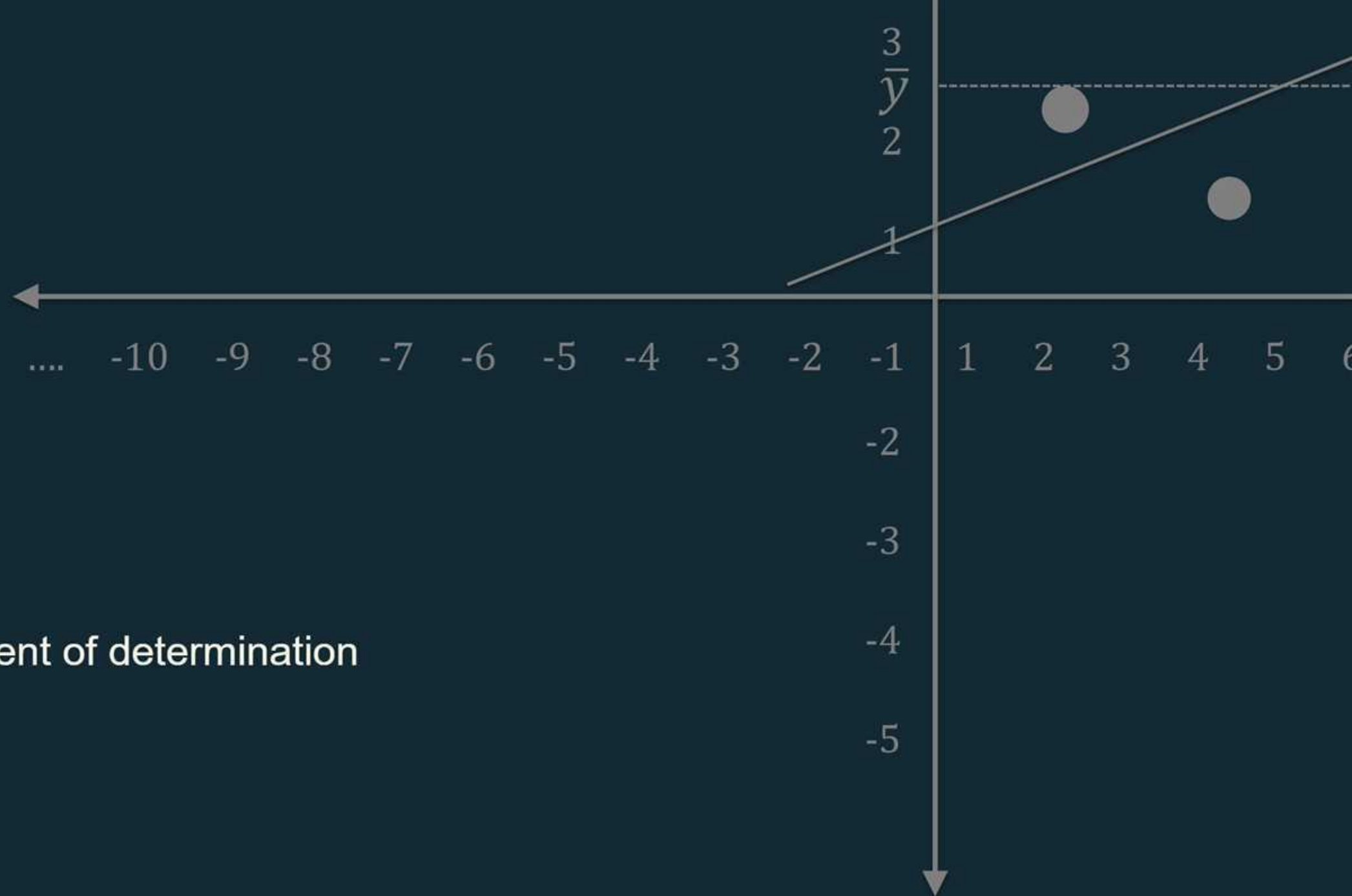


Relationship

Total variability = Explained variability + Unexplained variability

$$SST = SSR + SSE \quad \sum_{i=1}^m (y^{(i)} - \bar{y})^2 = \sum_{i=1}^m (\hat{y}^{(i)} - \bar{y})^2 + \sum_{i=1}^m e^2$$

Linear Regression



Evaluation Metrics

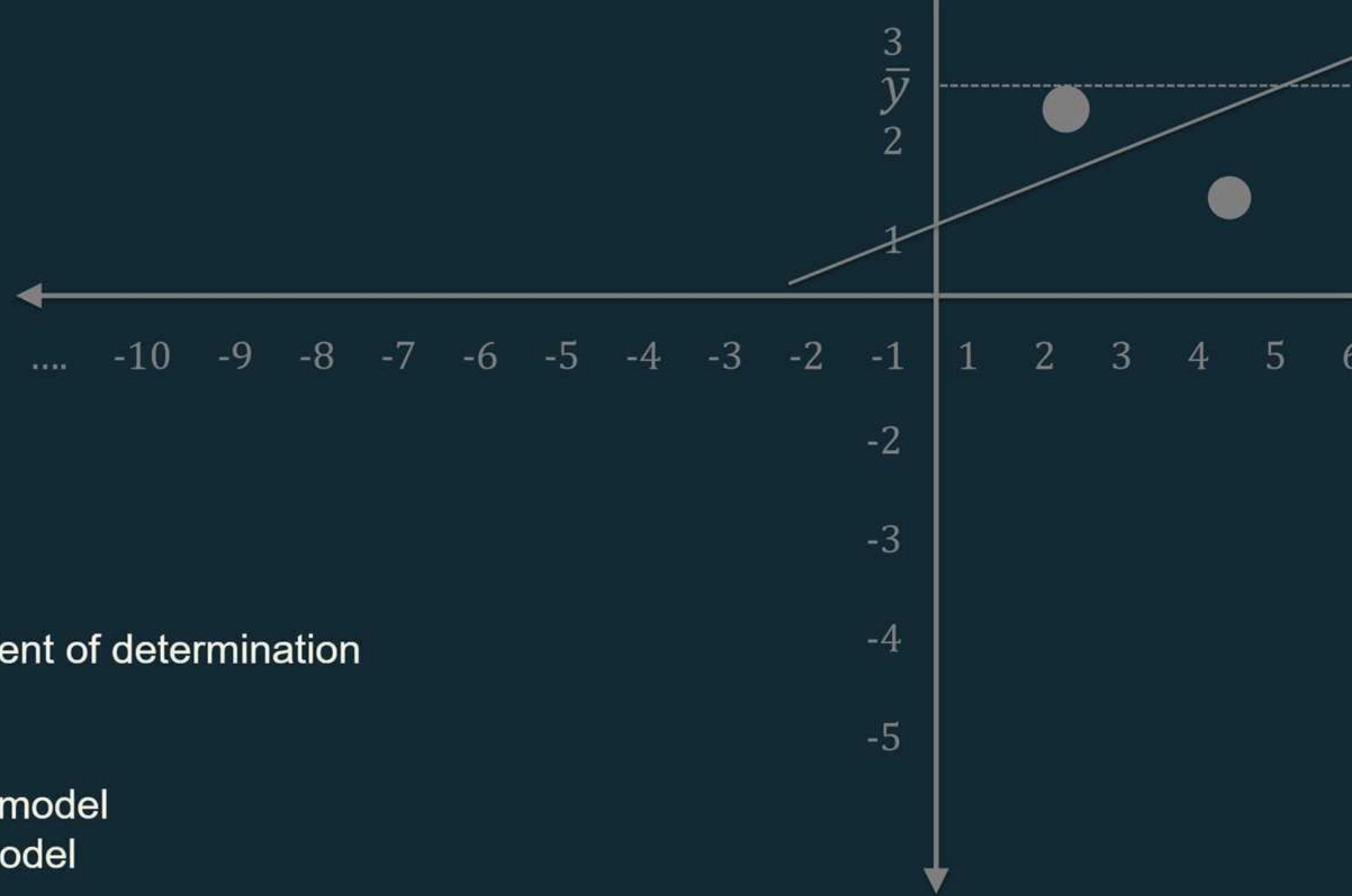
$$R^2 = 1 - \frac{SSE}{SST}$$

Where, R^2 is the coefficient of determination

$$rmse = \sqrt{mse}$$

Root mean square

Linear Regression



Evaluation Metrics

$$R^2 = 1 - \frac{SSE}{SST}$$

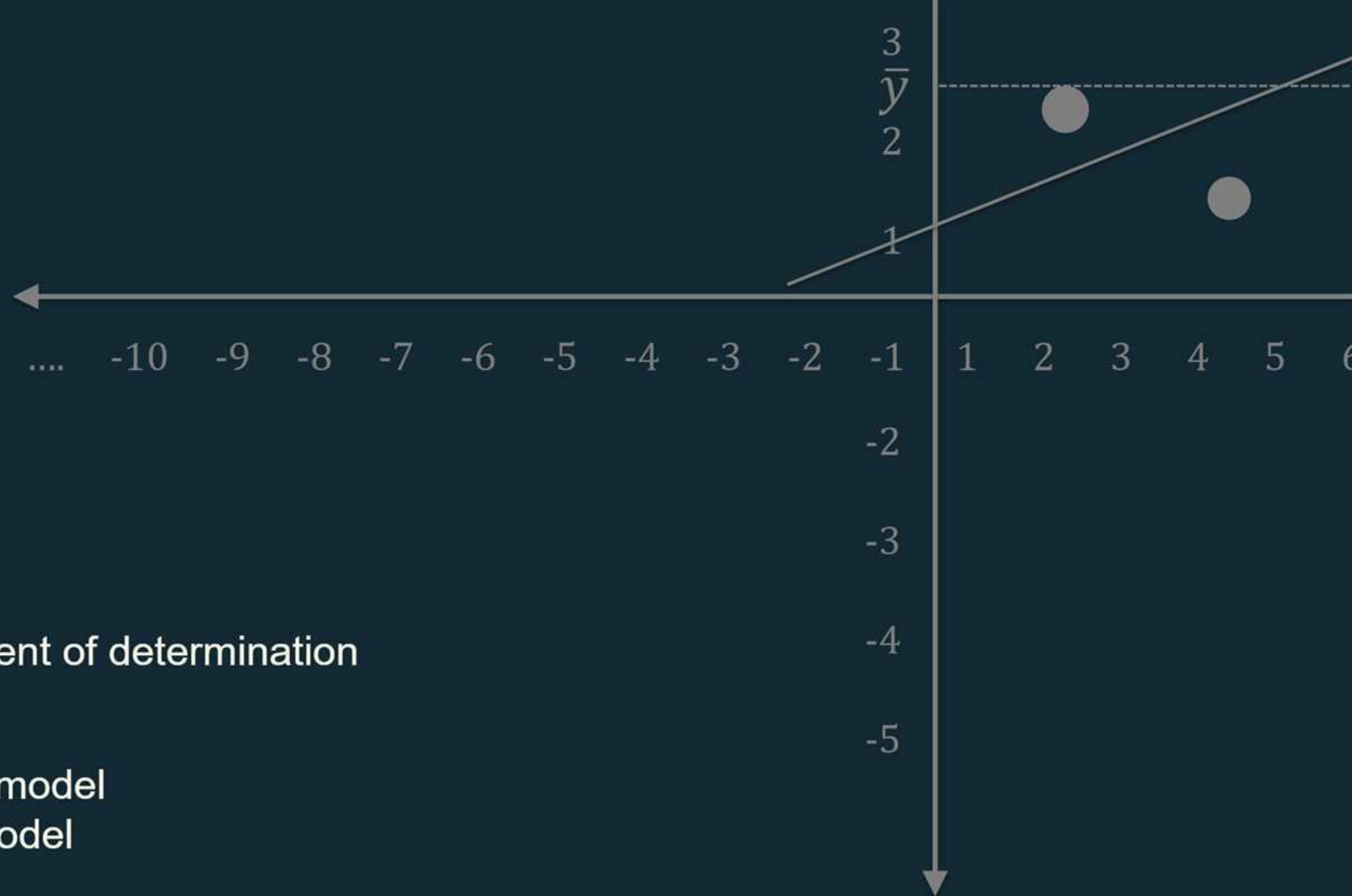
Where, R^2 is the coefficient of determination

A high (R^2) value is not always an indicator of a good model because it only measures explained variance, not model validity, data quality, or predictive accuracy.

$$rmse = \sqrt{mse}$$

Root mean square

Linear Regression



Evaluation Metrics

$$R^2 = 1 - \frac{SSE}{SST}$$

Where, R^2 is the coefficient of determination

A high (R^2) value is not always an indicator of a good model because it only measures explained variance, not model validity, data quality, or predictive accuracy.

Why High (R^2) Misleads

- Overfitting: Adding more variables increases High (R^2), even if those variables are purely random noise.
- No Spurious Protection: Two completely unrelated trends that both move upward over time (like inflation and sea levels) will produce a high High (R^2).

$$rmse = \sqrt{mse}$$

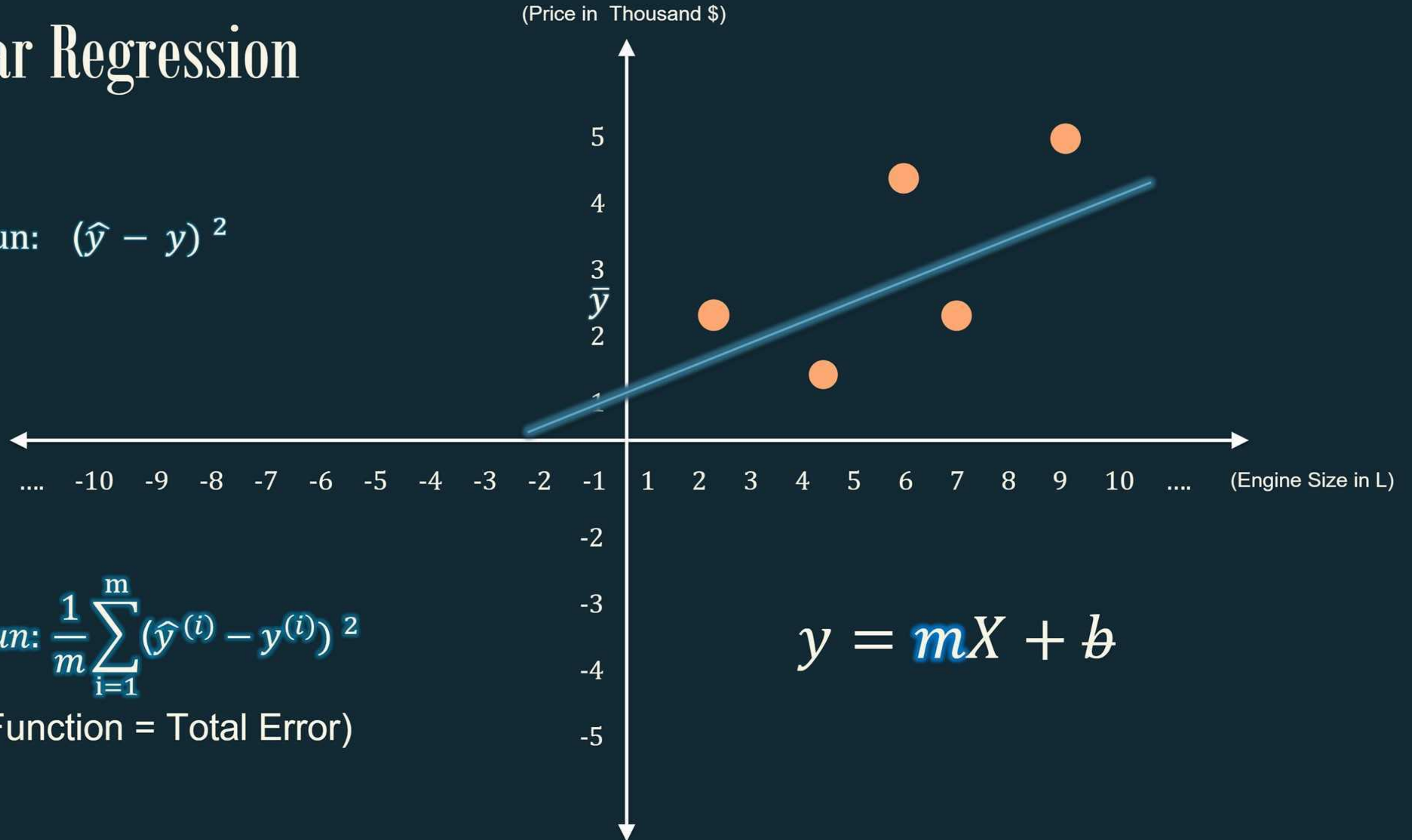
Root mean square

Linear Regression

Error Fun: $(\hat{y} - y)^2$

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2$$

(Cost Function = Total Error)



$$y = mX + b$$

Linear Regression

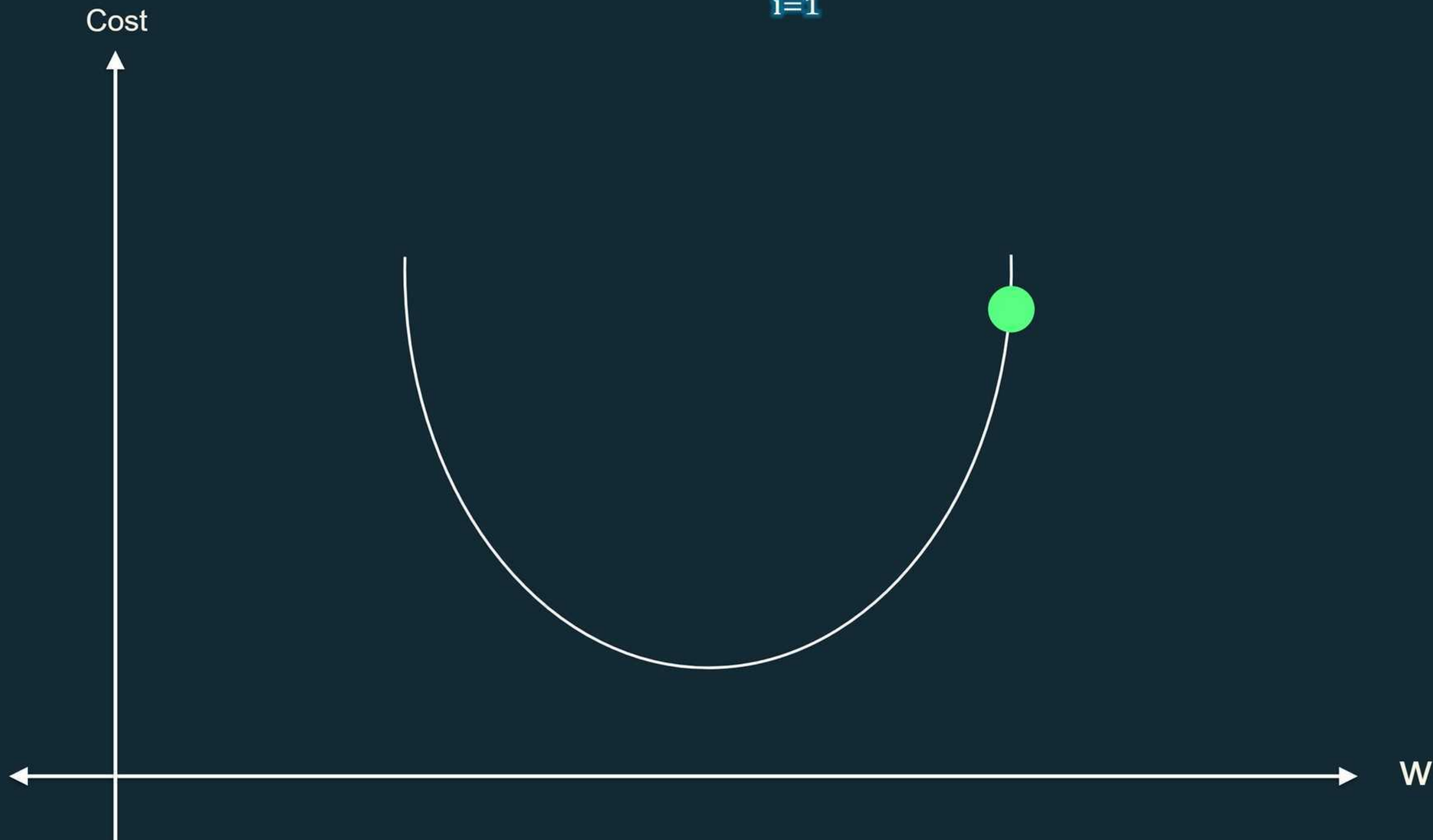
$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2$$

$$\hat{y} = wX + b$$

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$

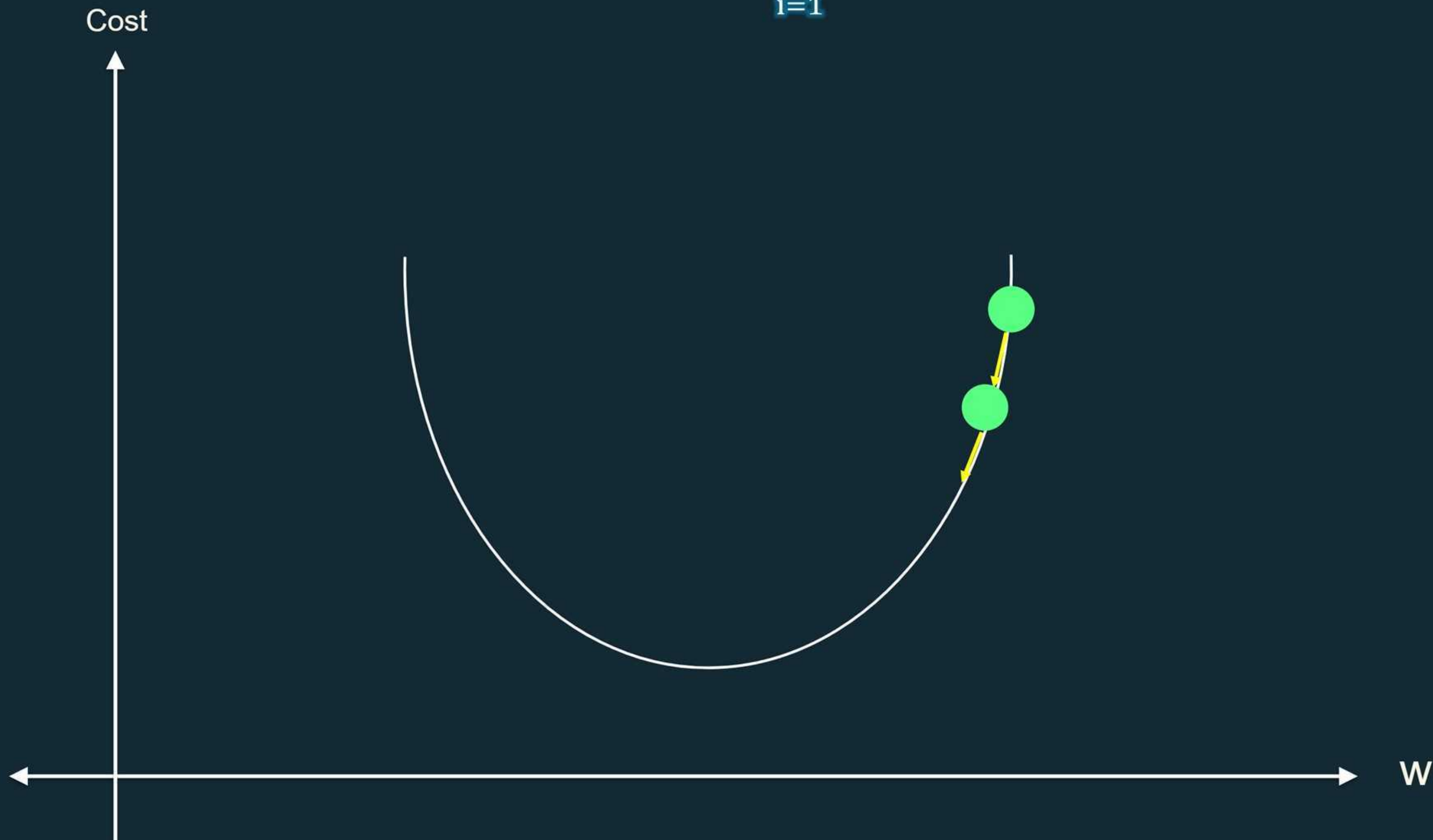
Linear Regression

$$\text{Cost Fun: } \frac{1}{w} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



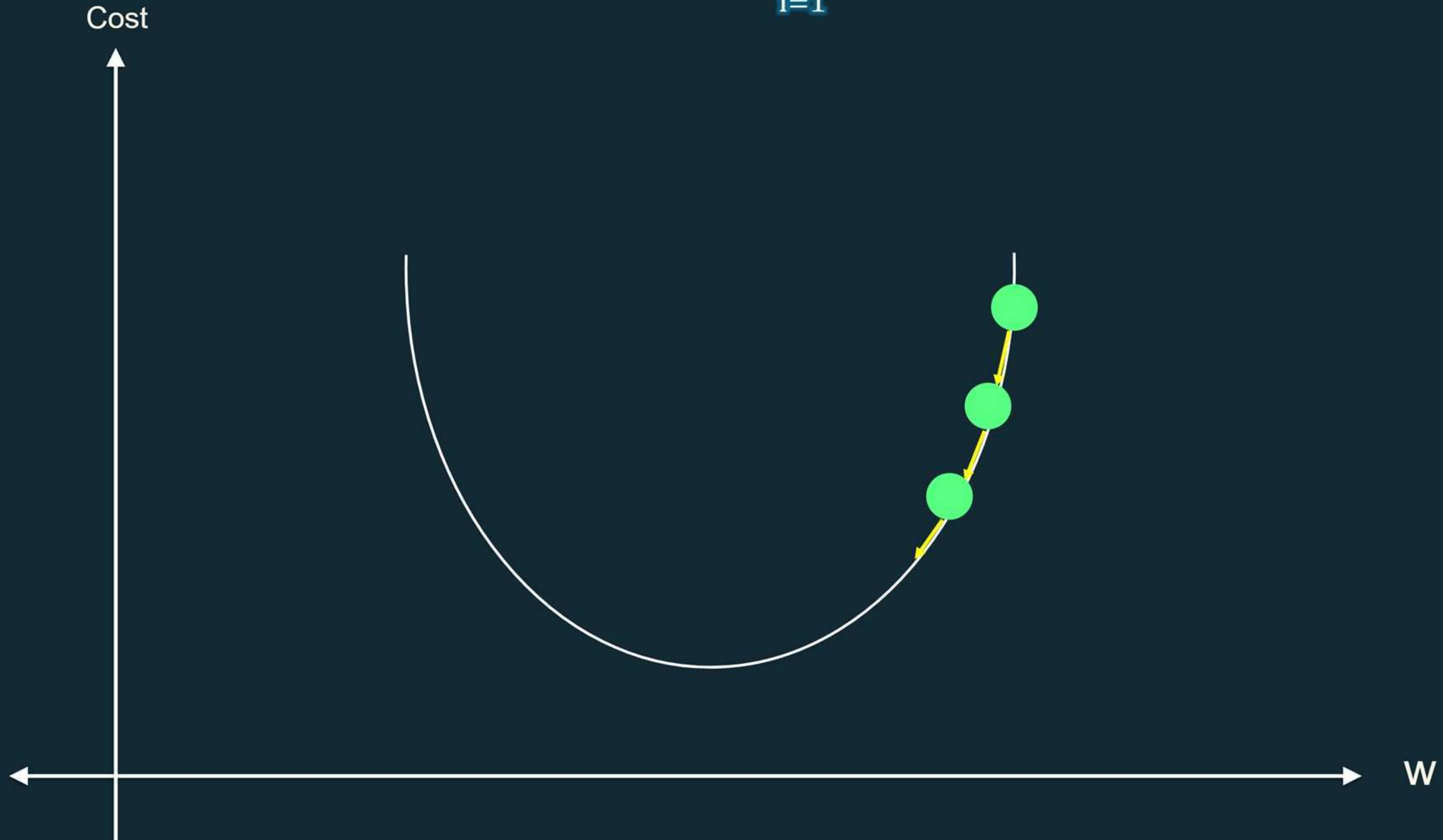
Linear Regression

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



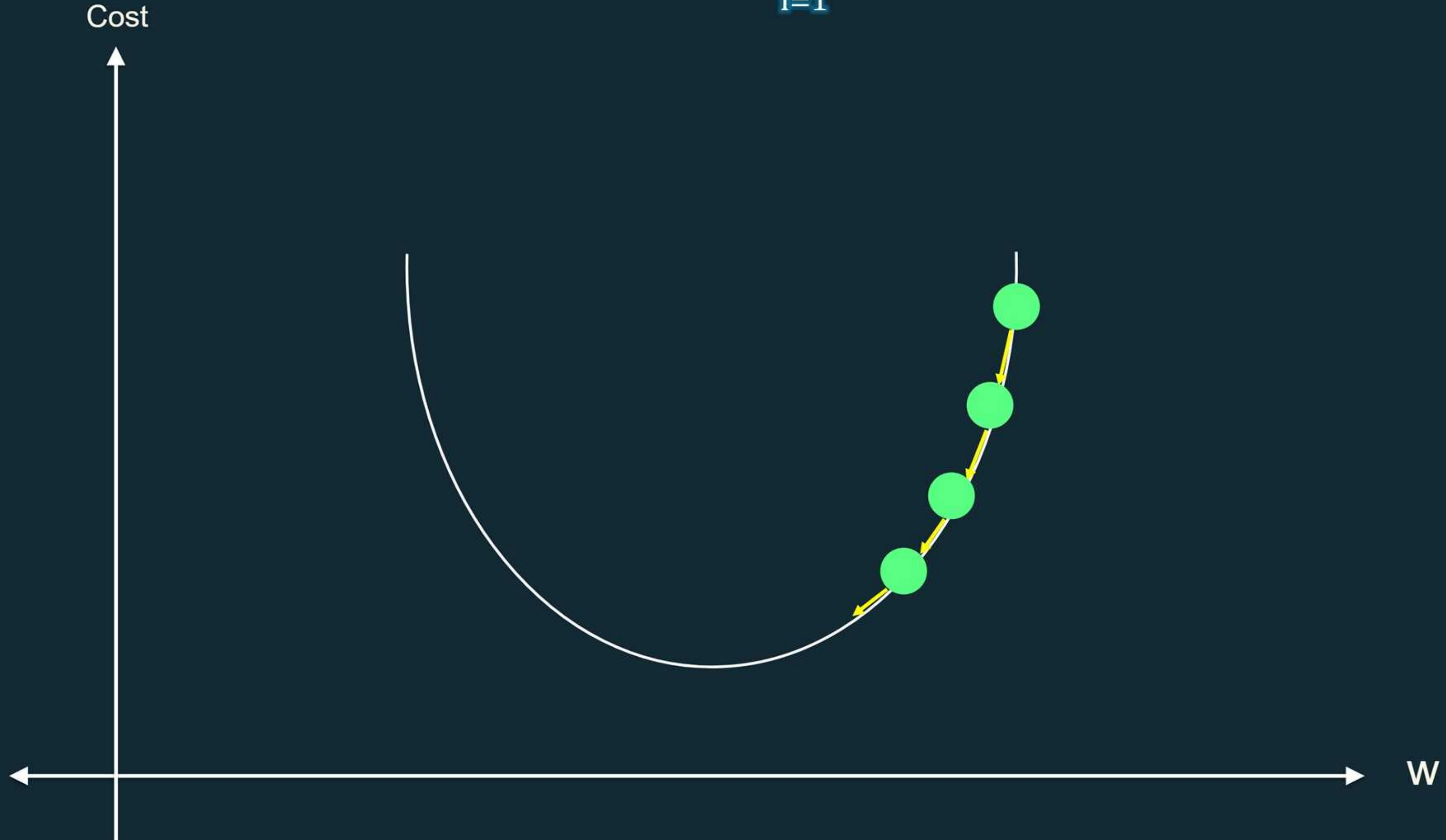
Linear Regression

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



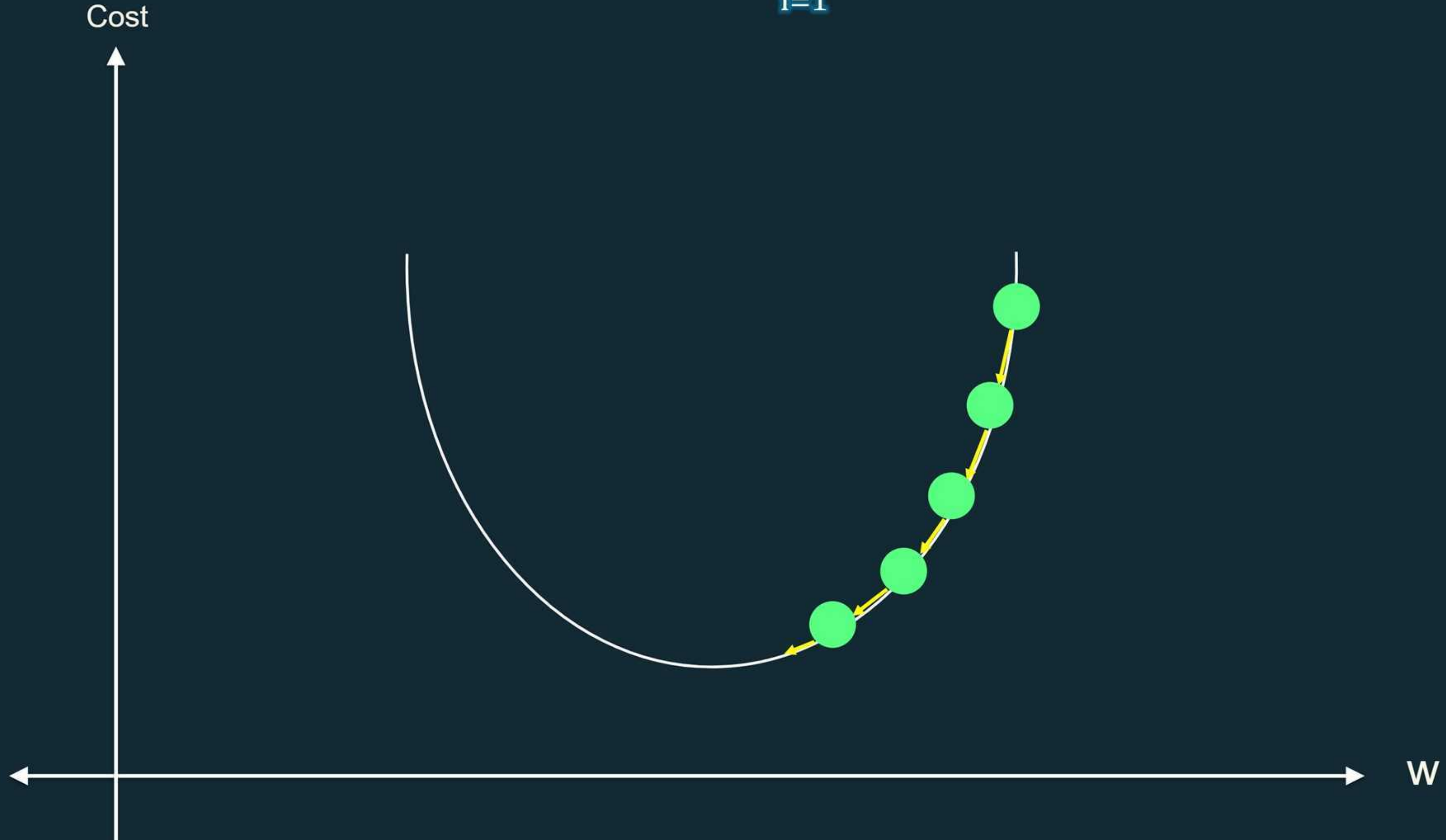
Linear Regression

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



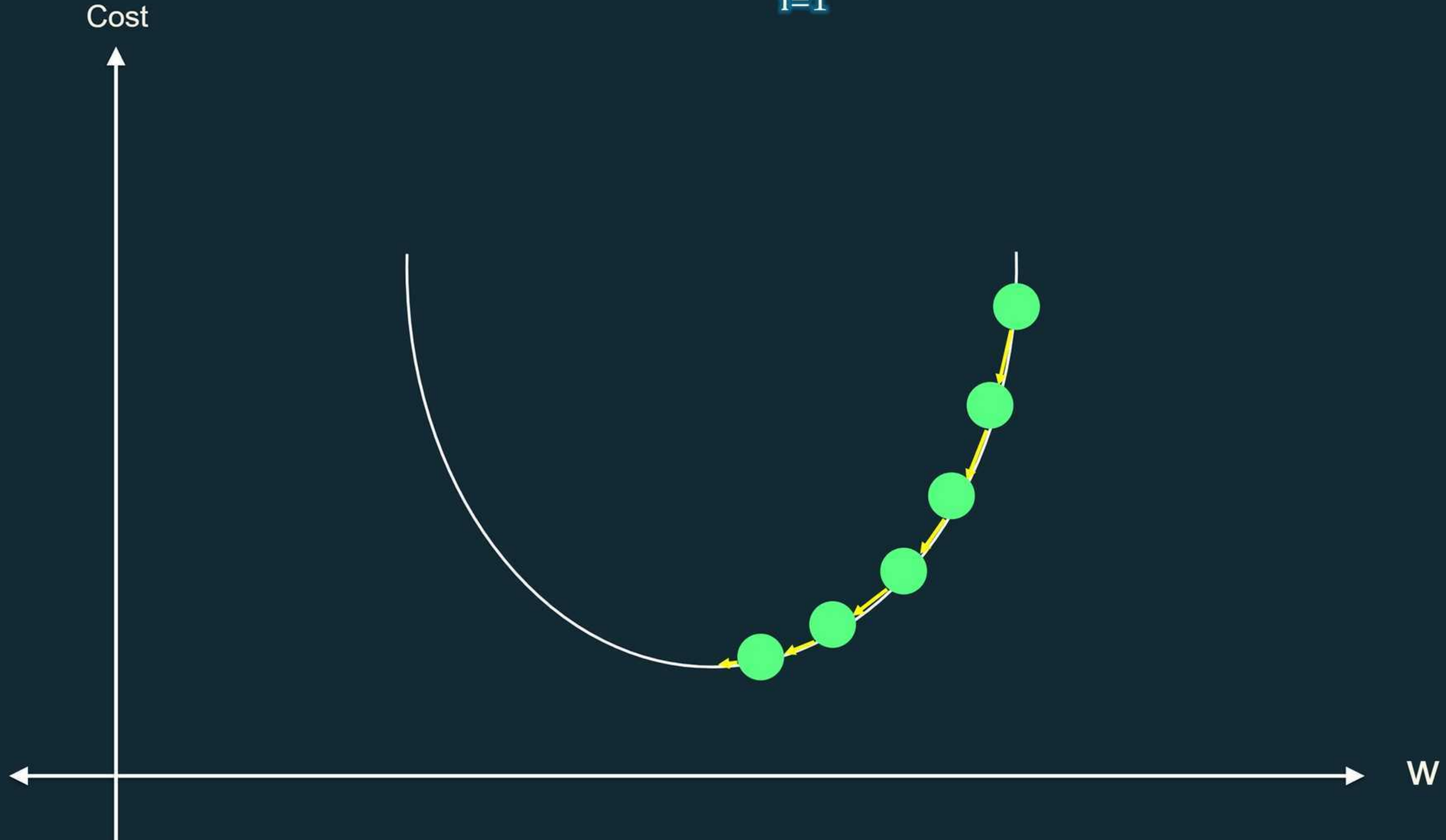
Linear Regression

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



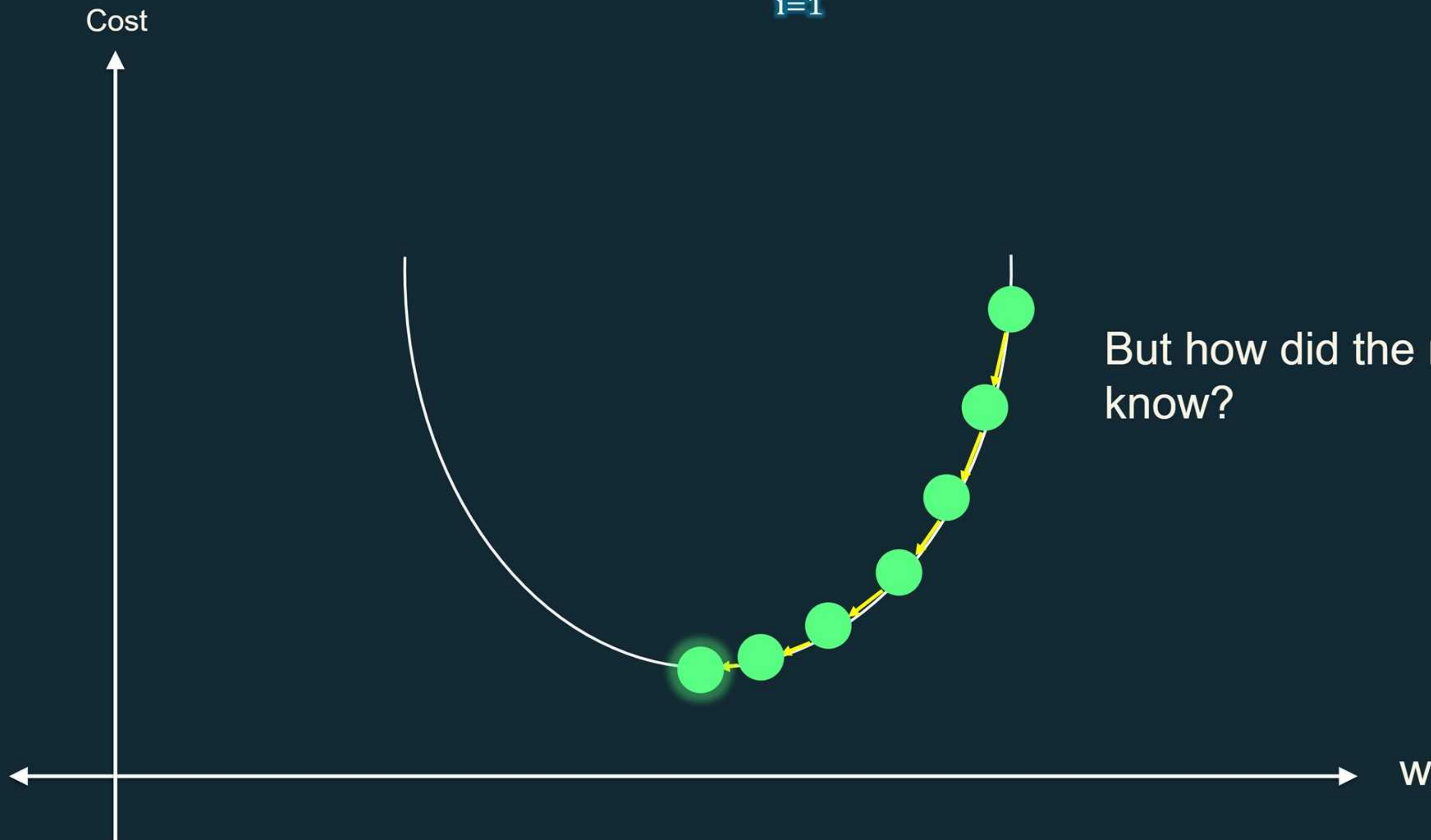
Linear Regression

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



Linear Regression

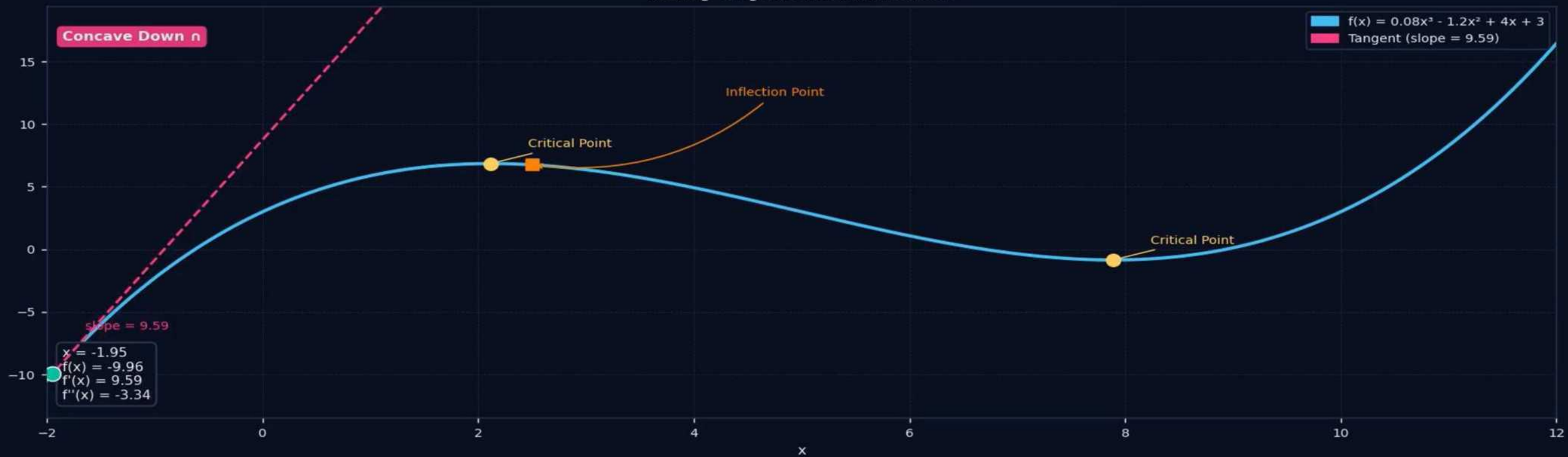
$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



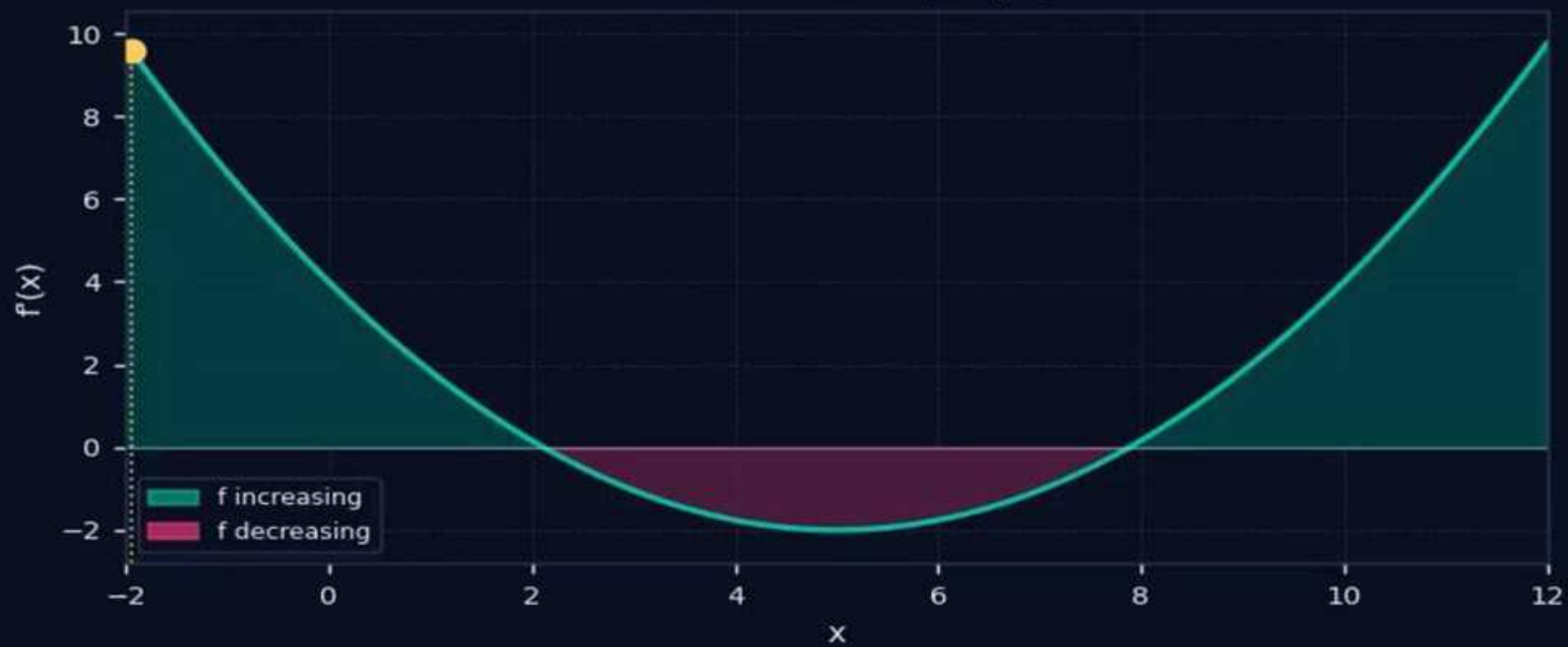
But how did the machine know?

Gradient Descent (GD)

Sliding Tangent Line Visualization



First Derivative (Slope)

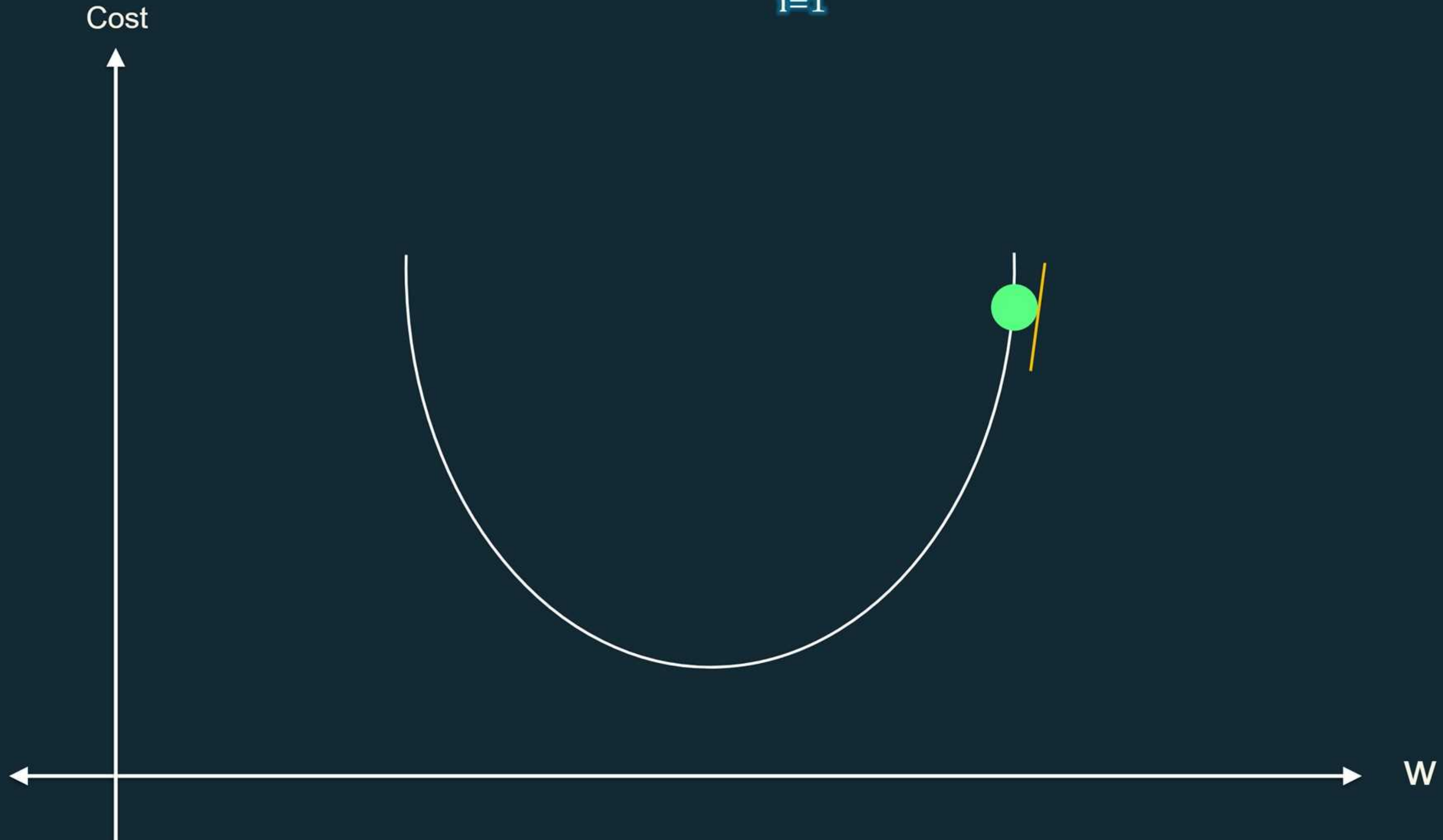


Second Derivative (Curvature)



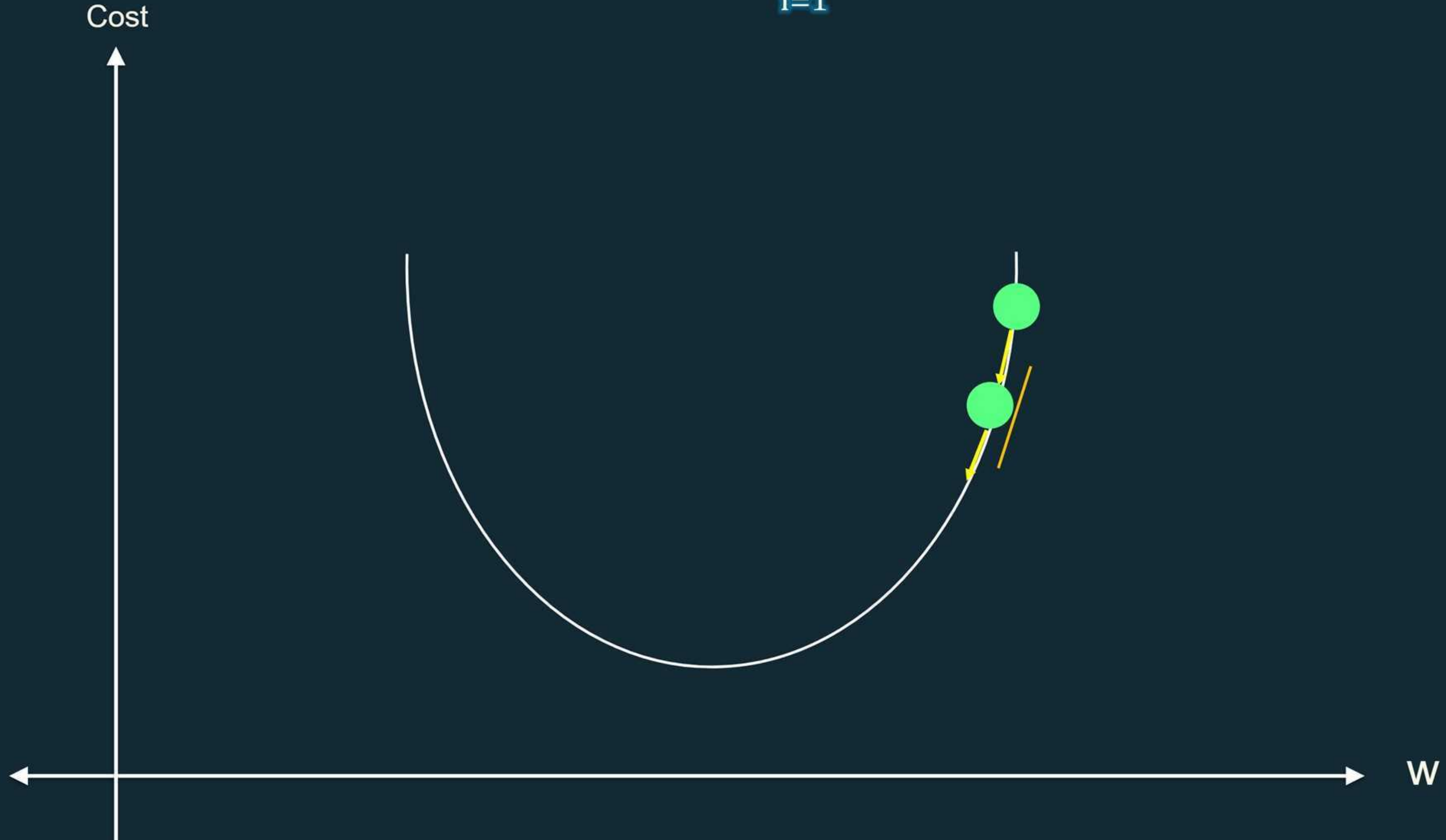
Linear Regression

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



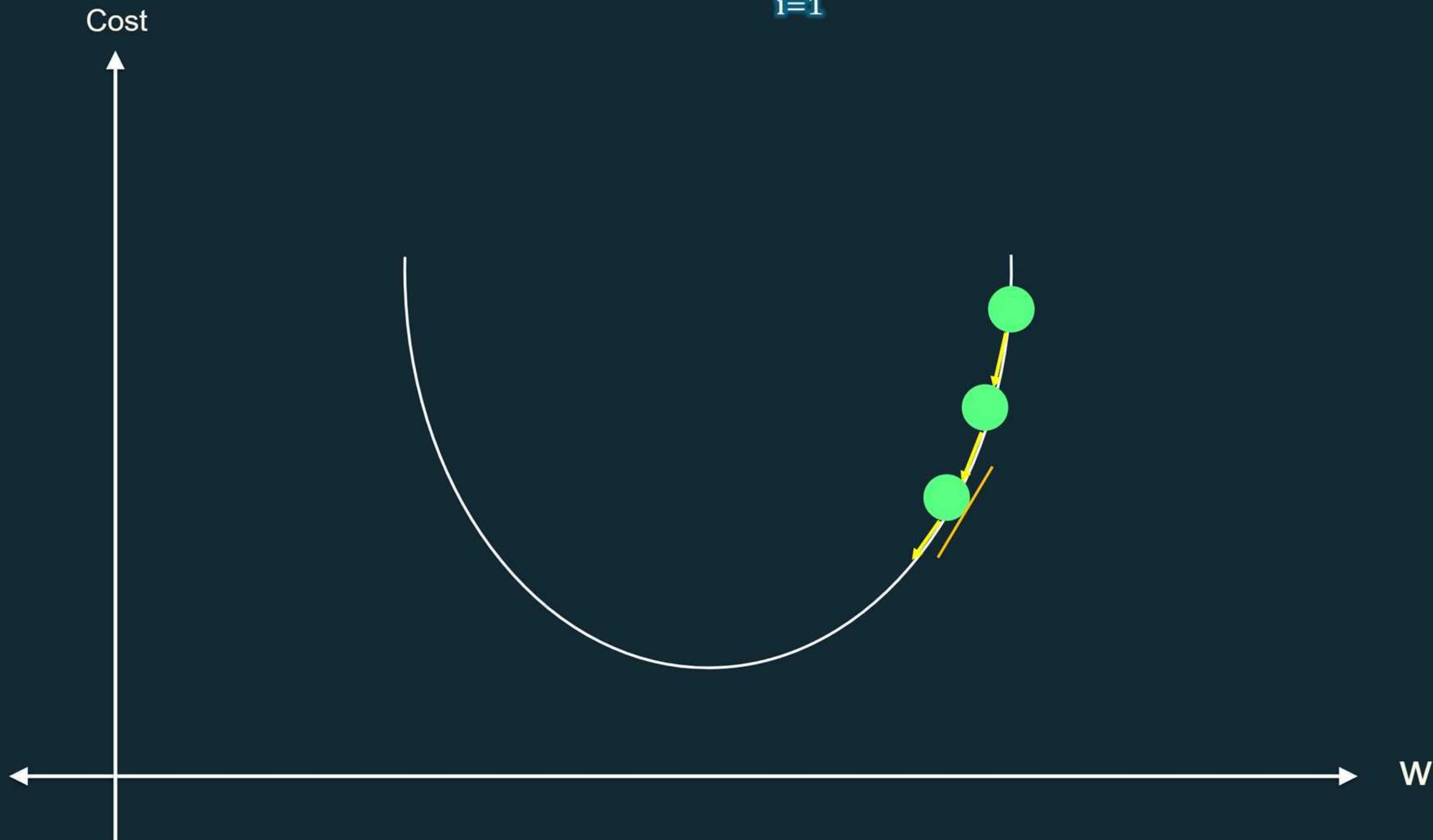
Linear Regression

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



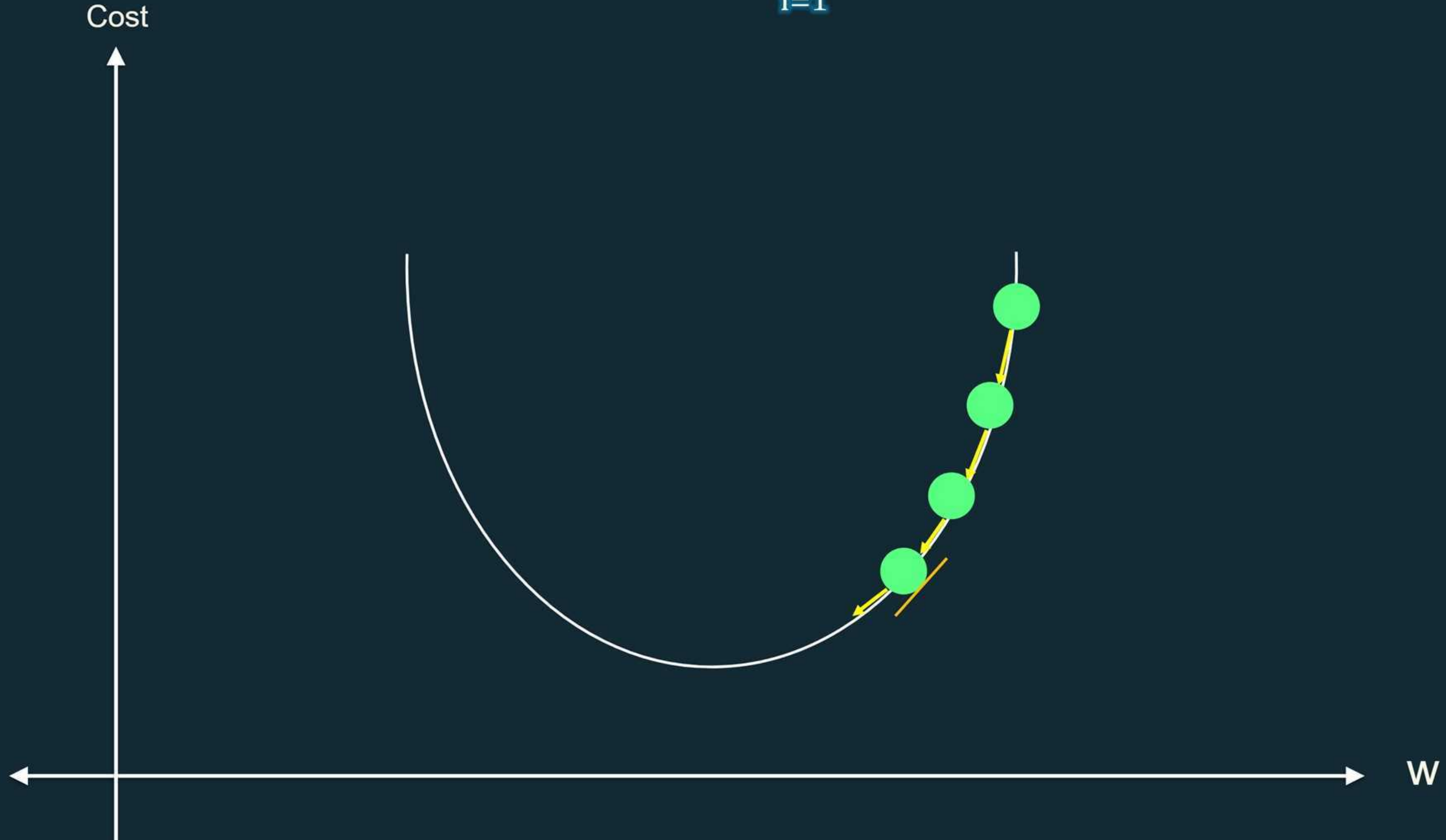
Linear Regression

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



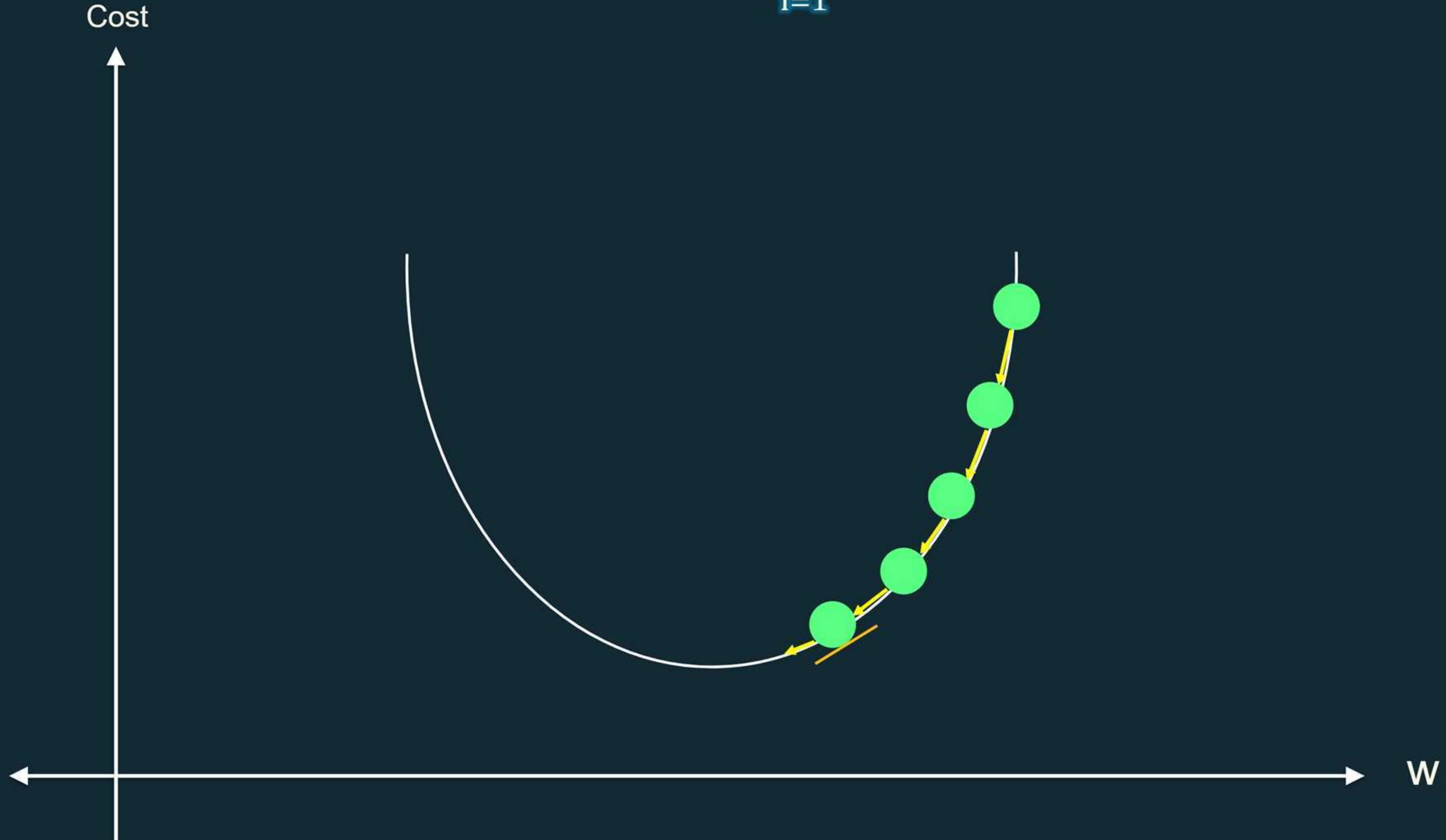
Linear Regression

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



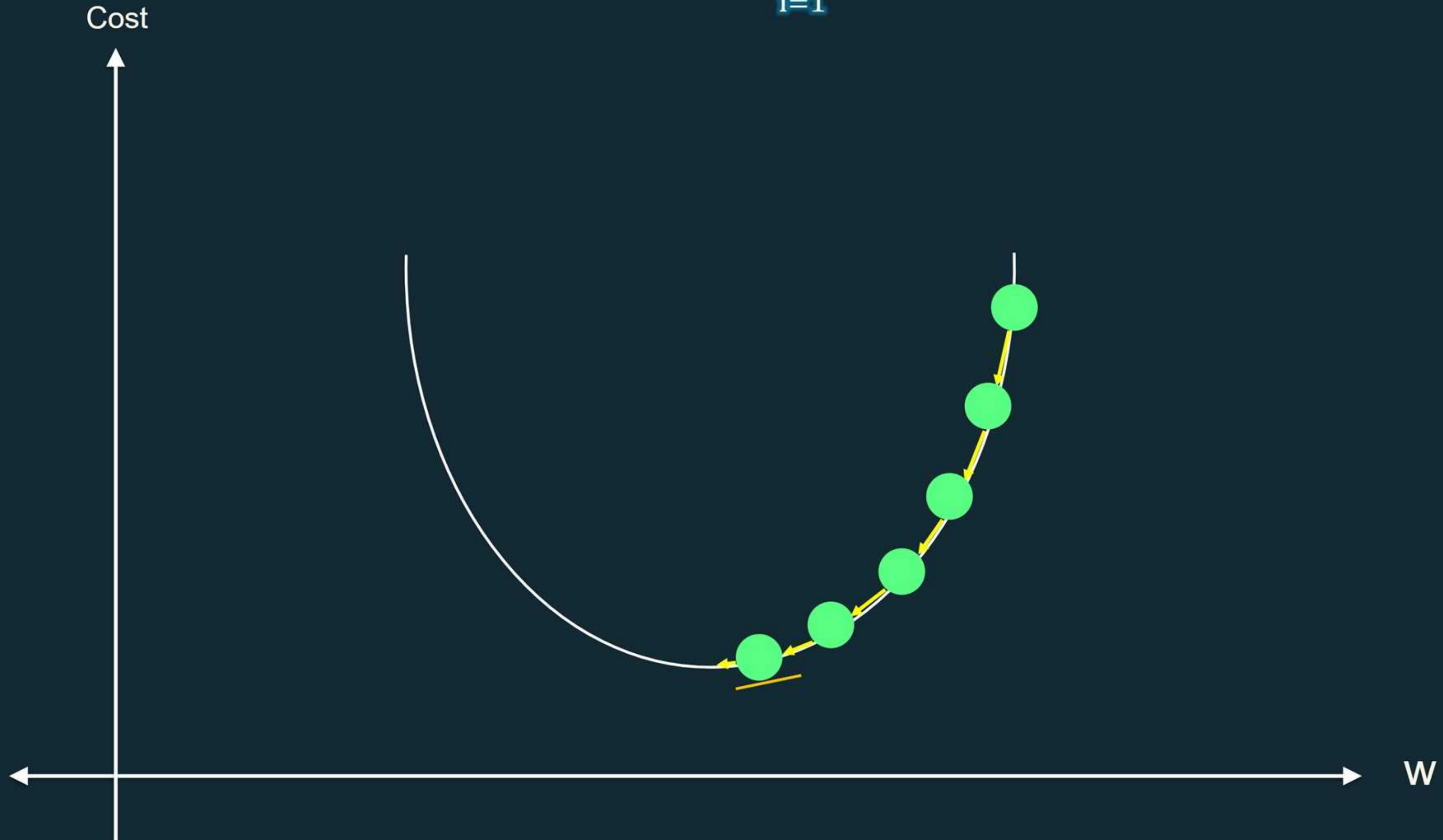
Linear Regression

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



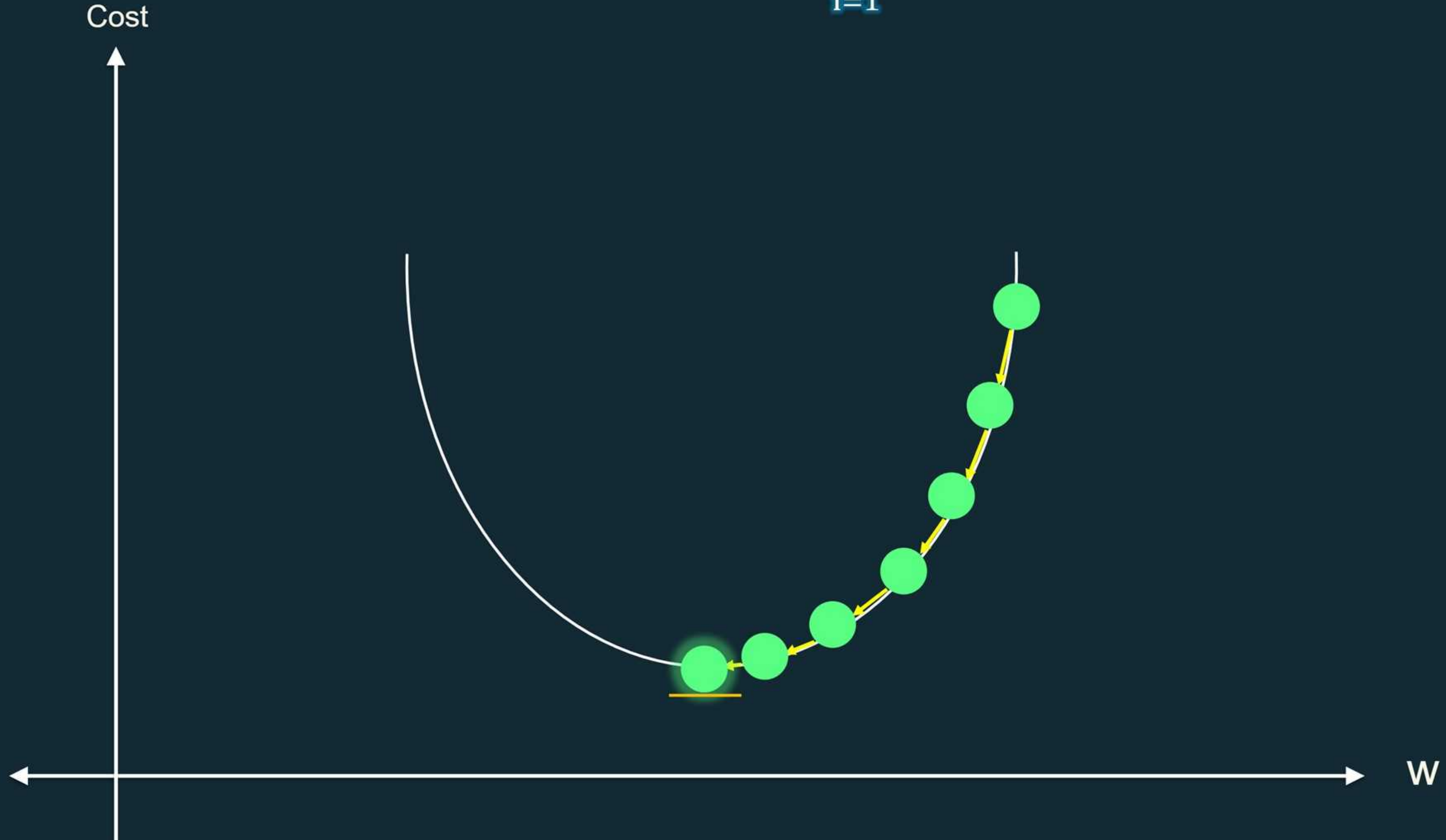
Linear Regression

$$\text{Cost Fun: } \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



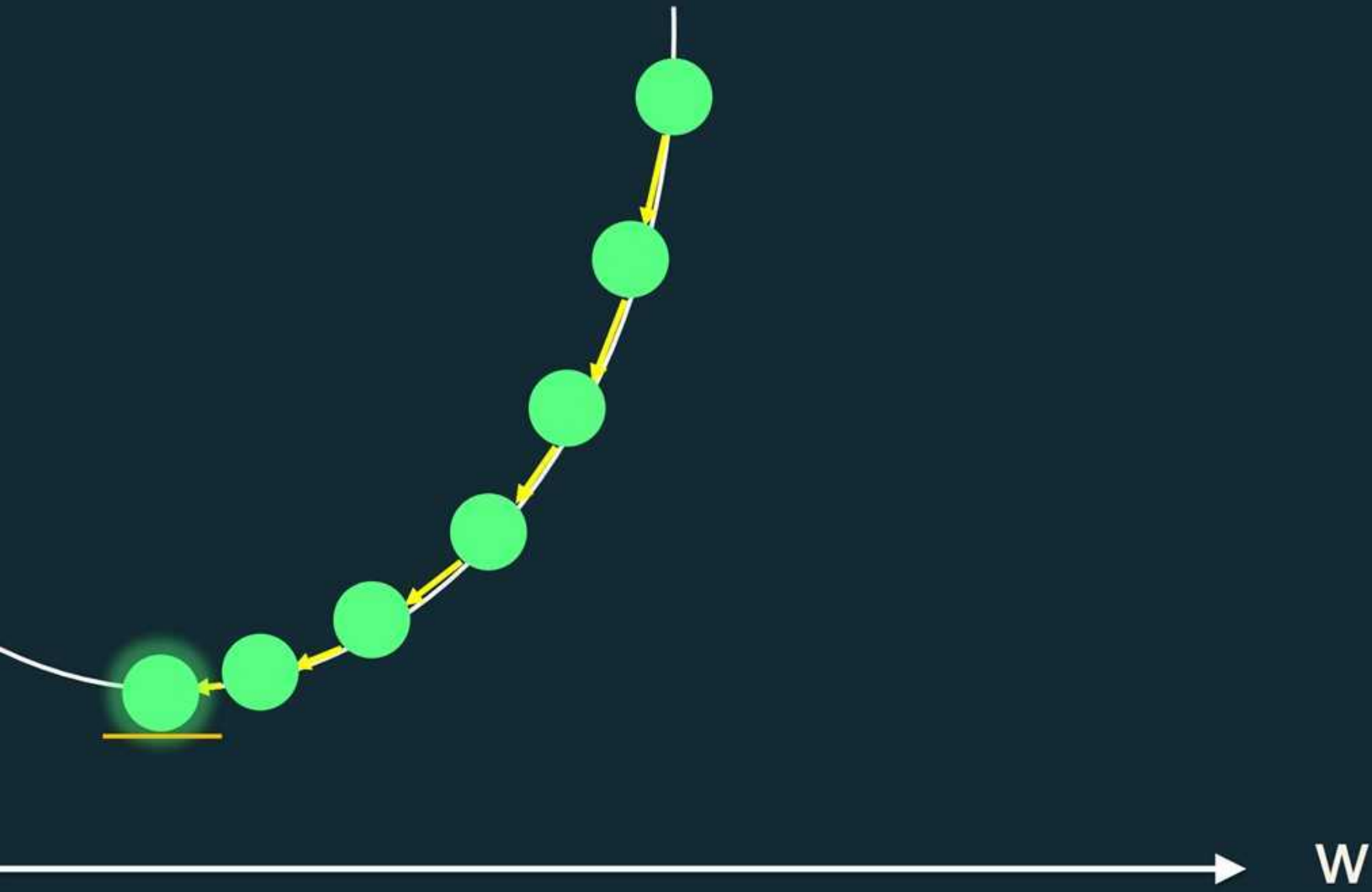
Linear Regression

$$Cost\ Fun = \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



Linear Regression

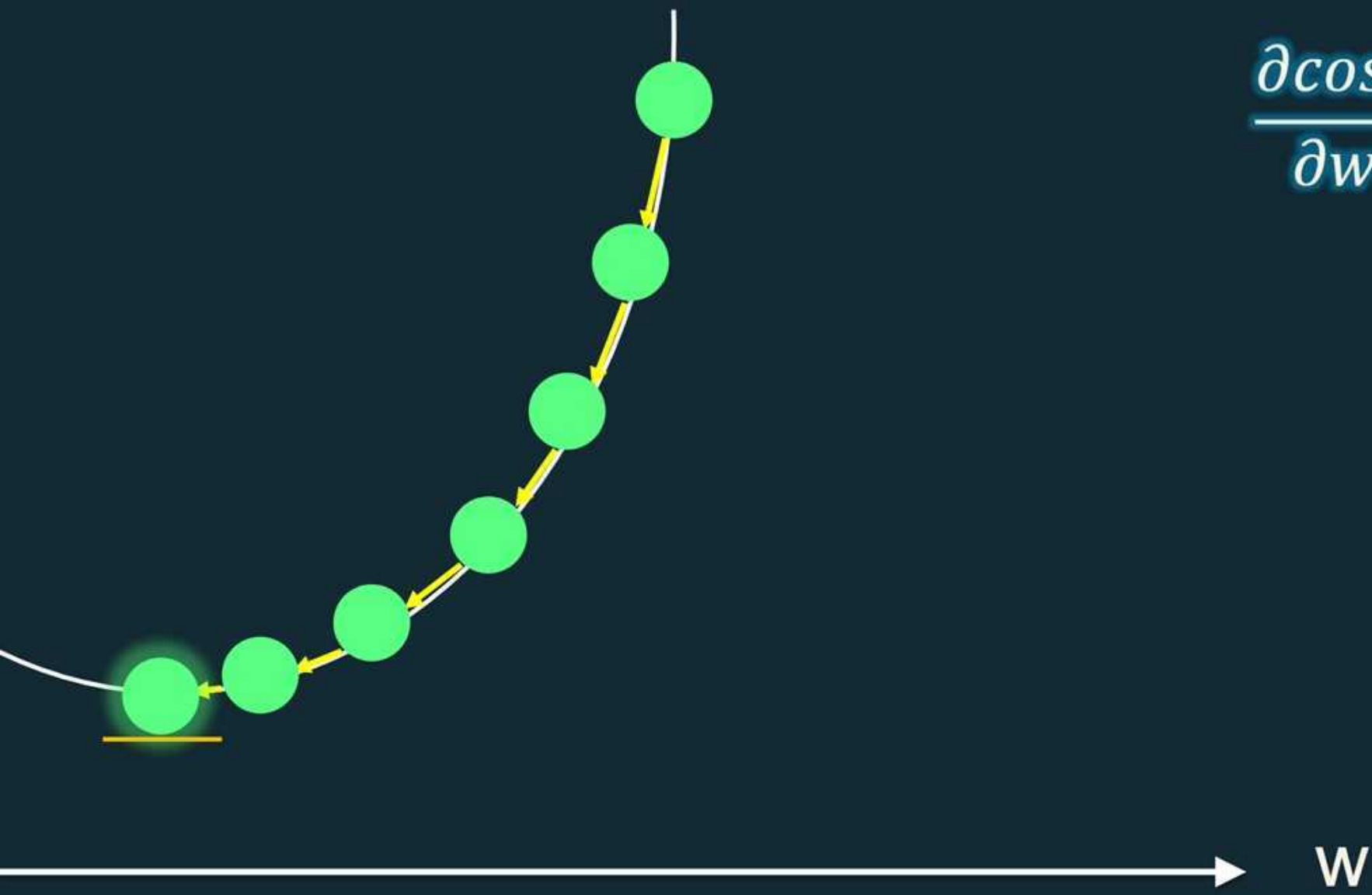
$$Cost\ Fun = \frac{1}{2m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$



Linear Regression

$$Cost\ Fun = \frac{1}{2m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$

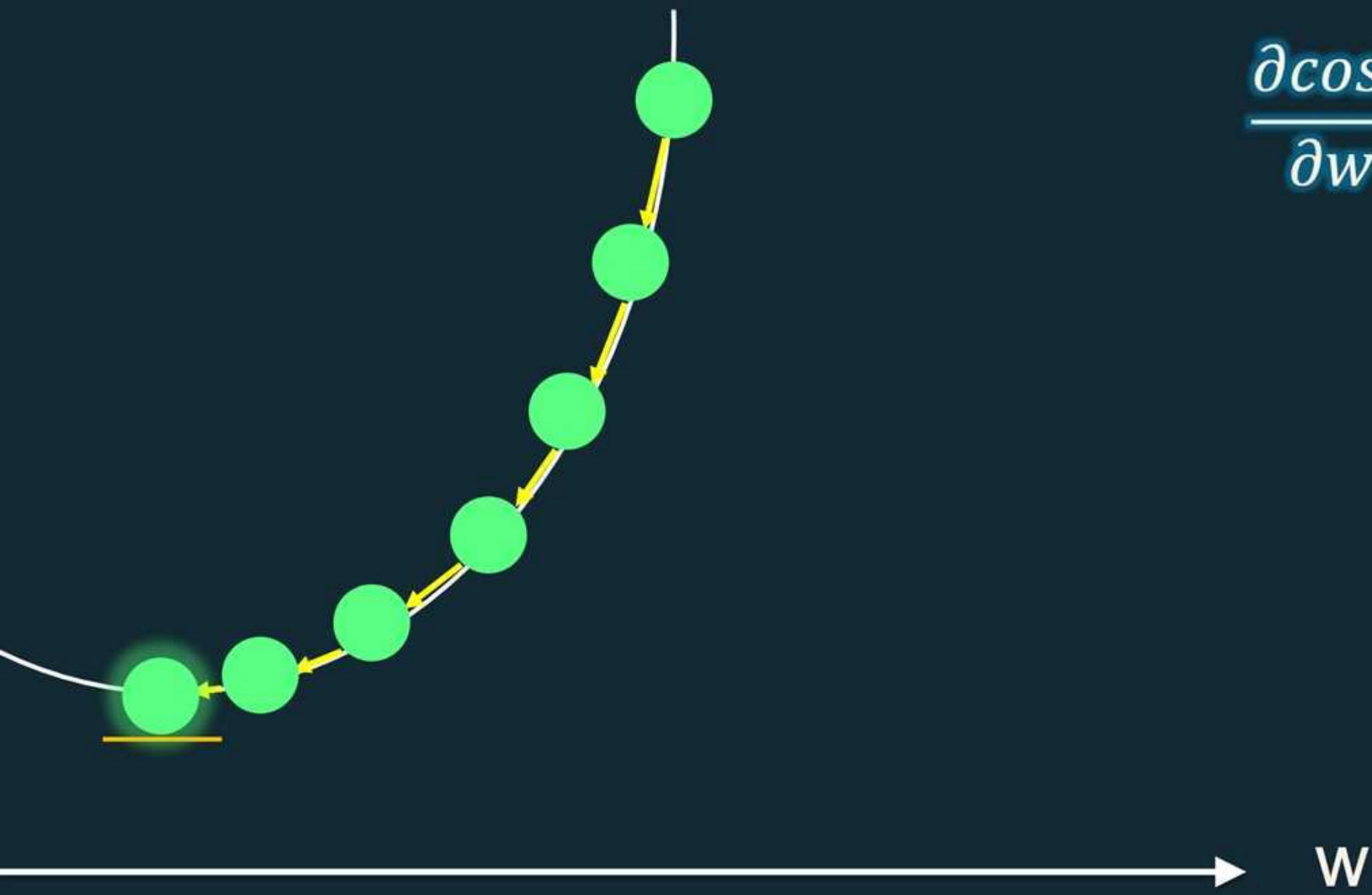
$$\frac{\partial cost}{\partial w} = \frac{2}{2m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})$$



Linear Regression

$$Cost\ Fun = \frac{1}{2m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$

$$\frac{\partial cost}{\partial w} = \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})X$$

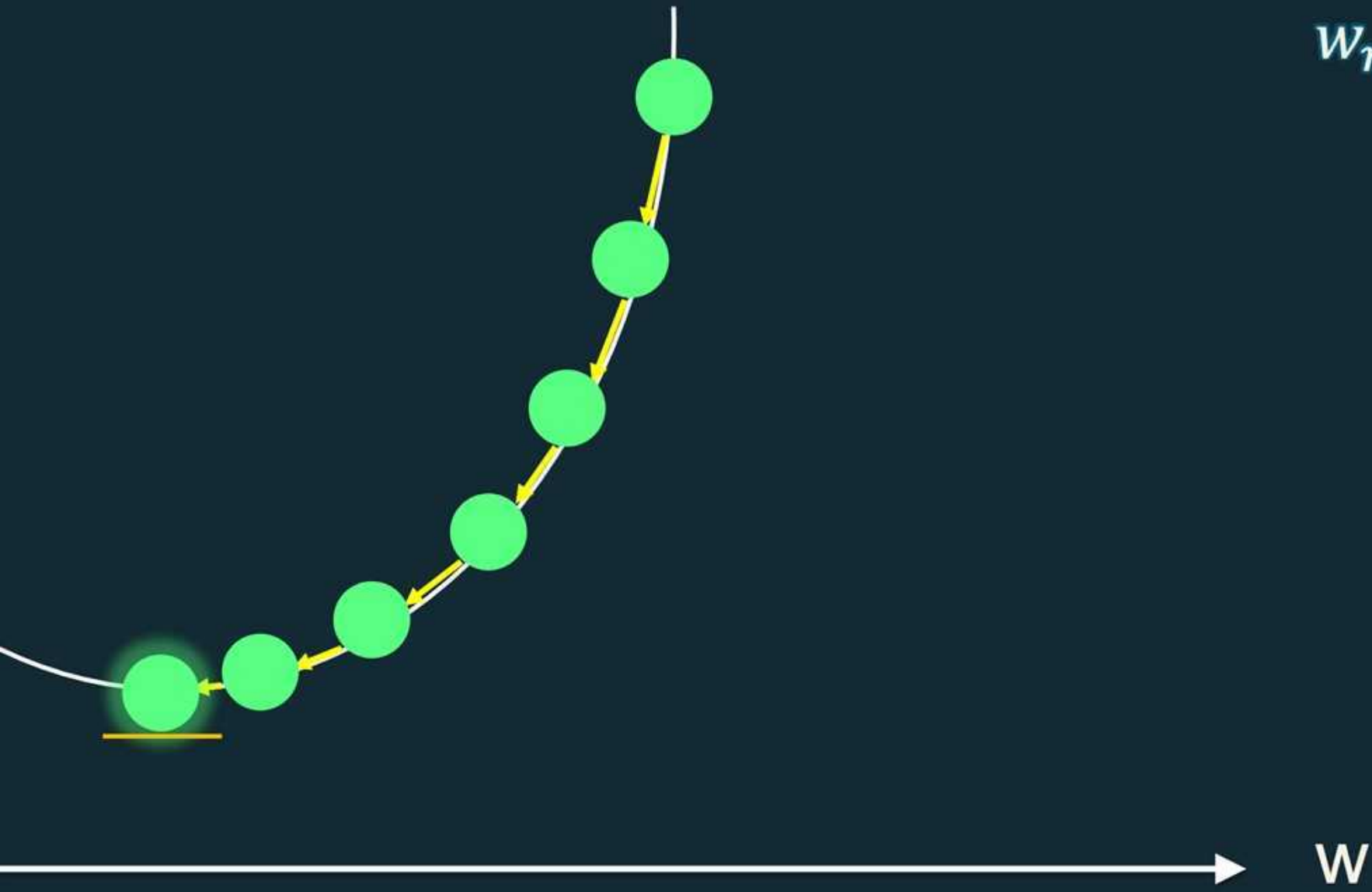


Linear Regression

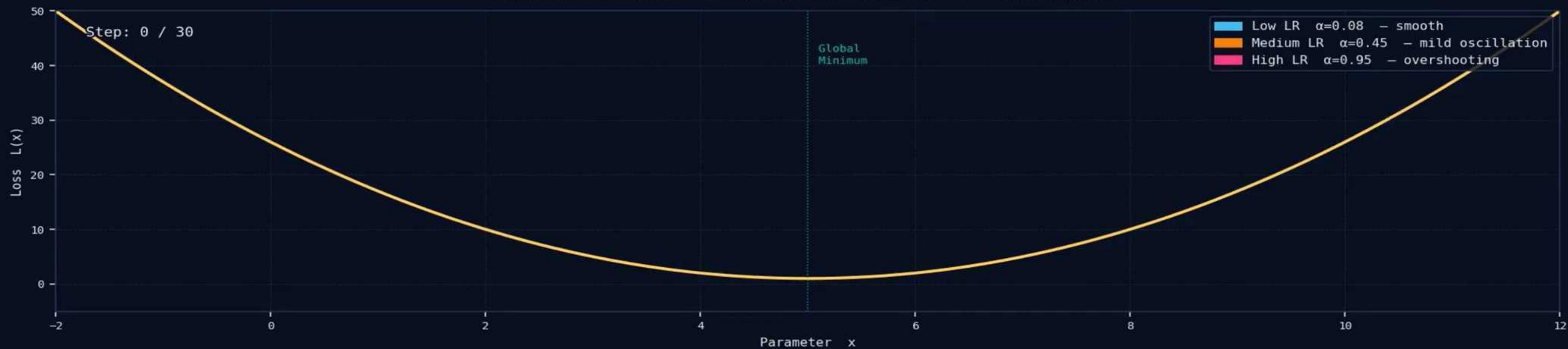
$$Cost\ Fun = \frac{1}{2m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$

$$\frac{\partial cost}{\partial w} = \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)}) X$$

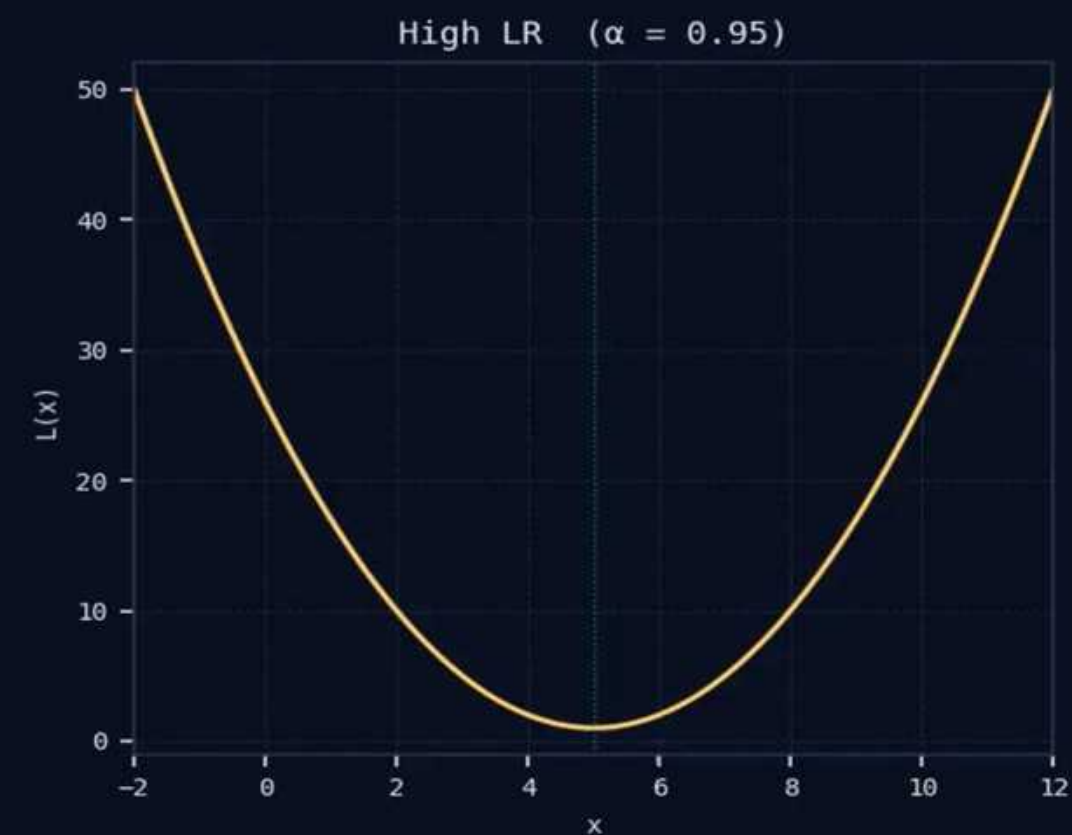
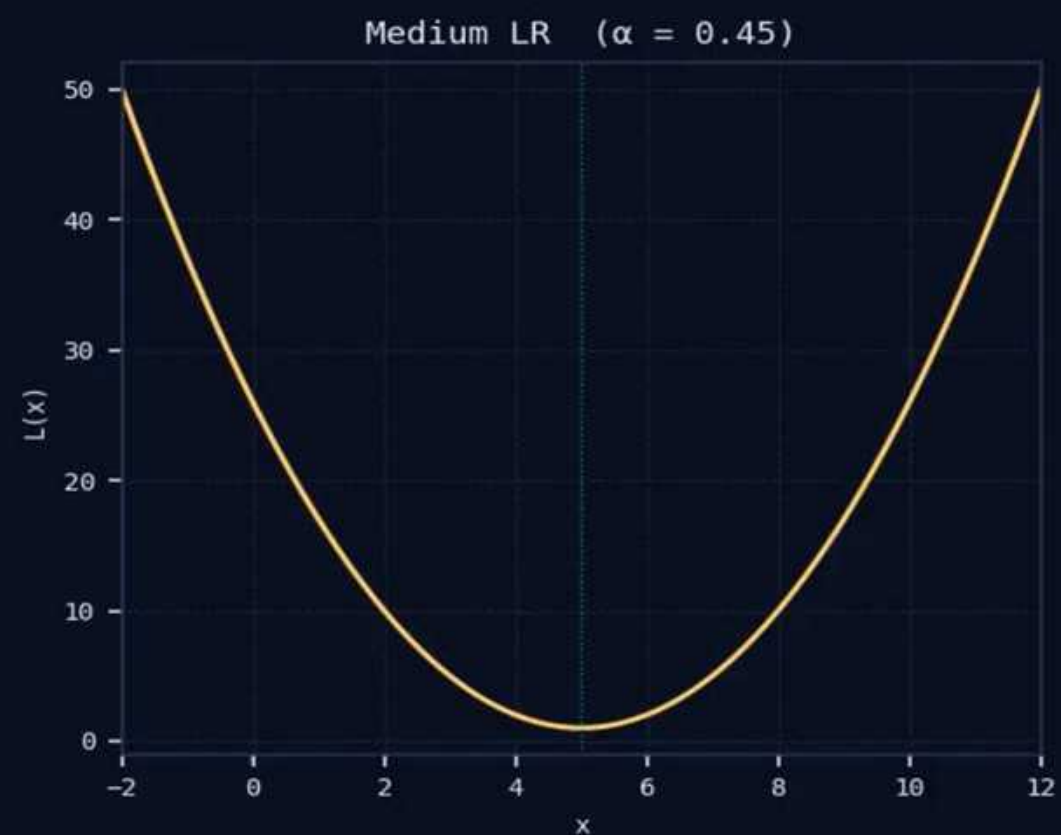
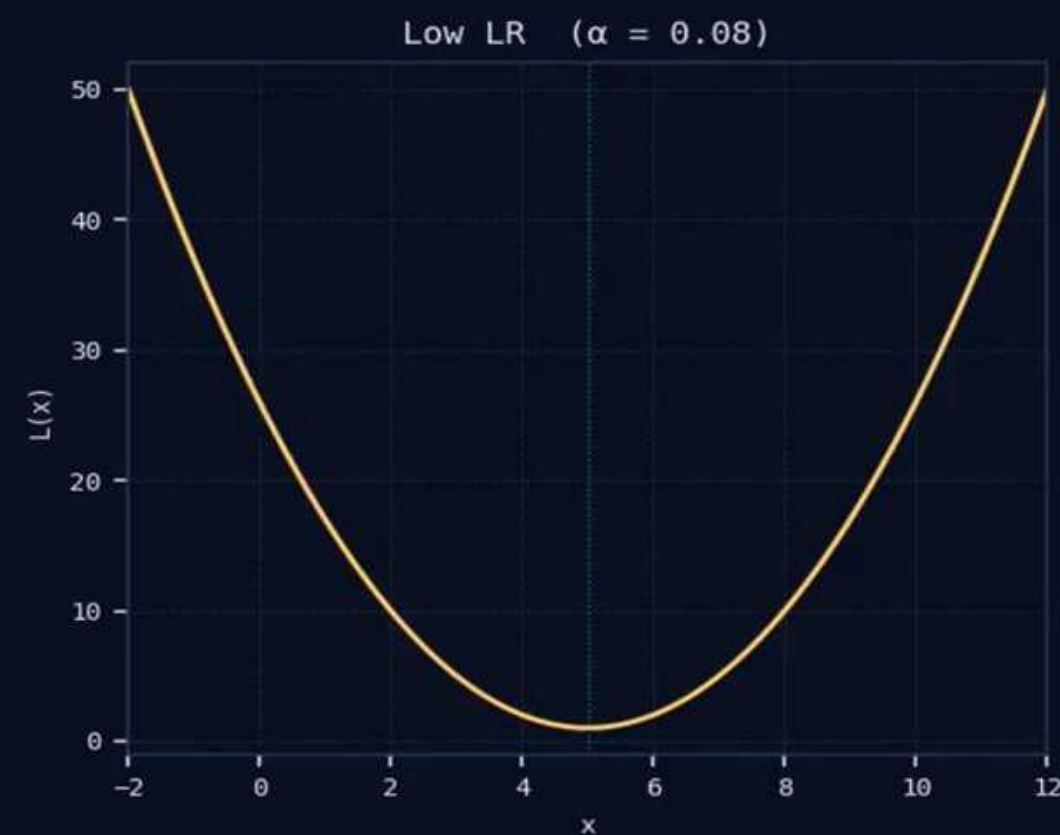
$$w_{new} = w_{old} - \eta (\hat{y}^{(i)} - y^{(i)}) X$$



Gradient Descent – Effect of Learning Rate on Convergence



Learning Rate



A high learning rate causes the optimizer to overshoot the minimum, oscillating or diverging | Low LR converges slowly but reliably

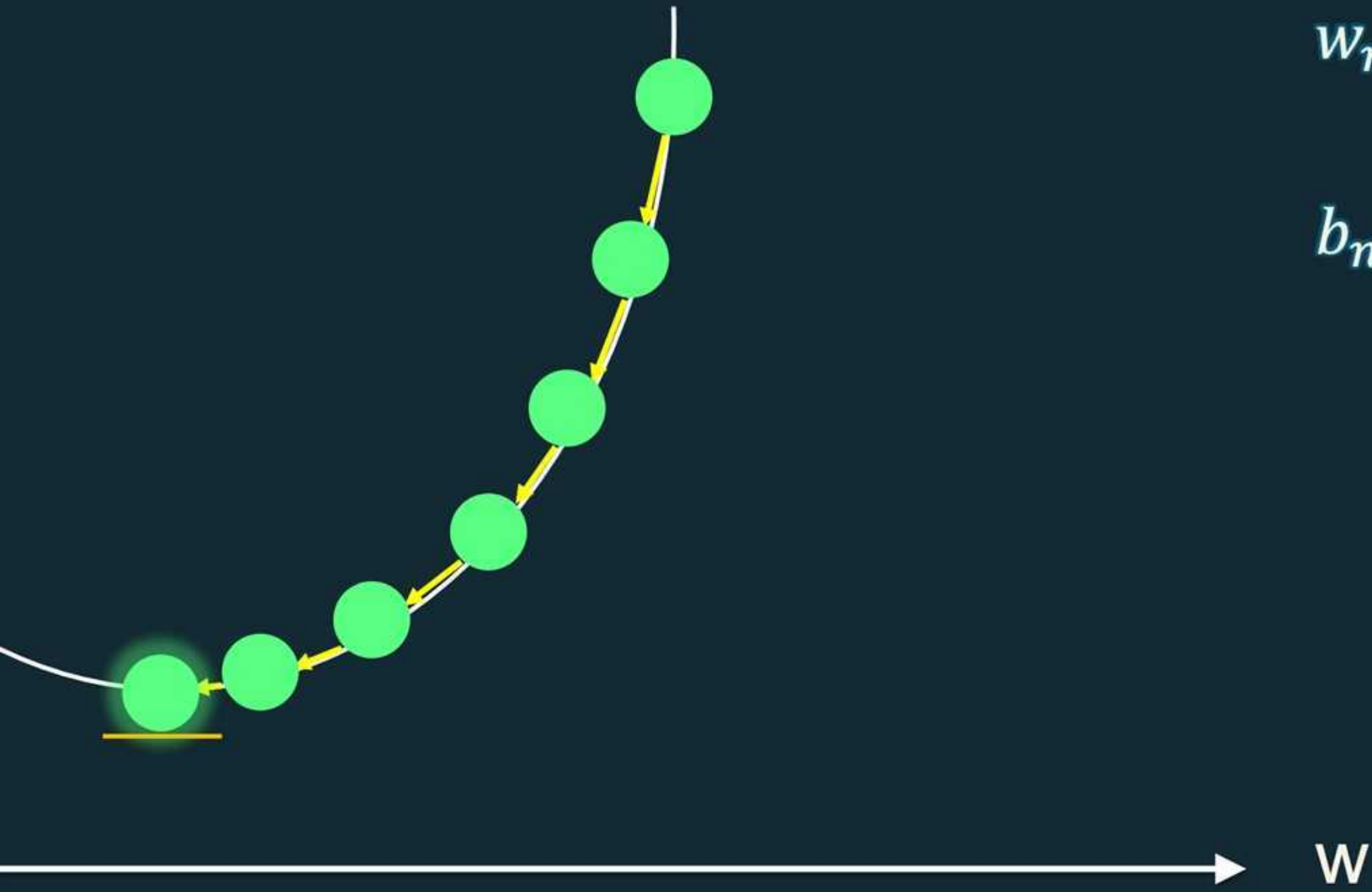
Linear Regression

$$Cost\ Fun = \frac{1}{2m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)})^2$$

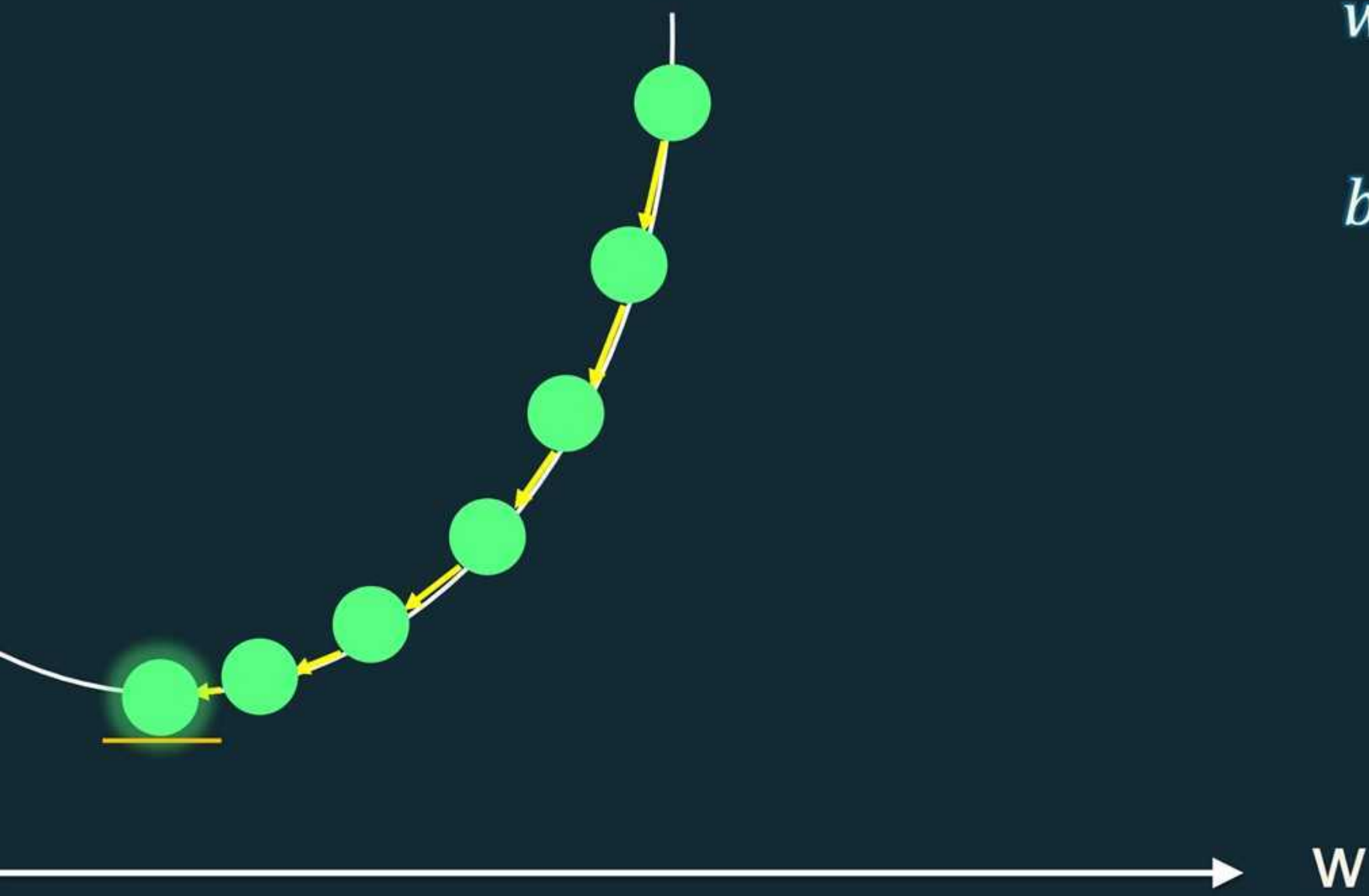
$$\frac{\partial cost}{\partial w} = \frac{1}{m} \sum_{i=1}^m ((wX + b)^{(i)} - y^{(i)}) X$$

$$w_{new} = w_{old} - \eta (\hat{y}^{(i)} - y^{(i)}) X$$

$$b_{new} = b_{old} - \eta (\hat{y}^{(i)} - y^{(i)})$$



Linear Regression



In general

$$w_{new} = w_{old} - \eta \frac{\partial}{\partial w} J(w, b)$$

$$b_{new} = b_{old} - \eta \frac{\partial}{\partial b} J(w, b)$$

Linear Regression

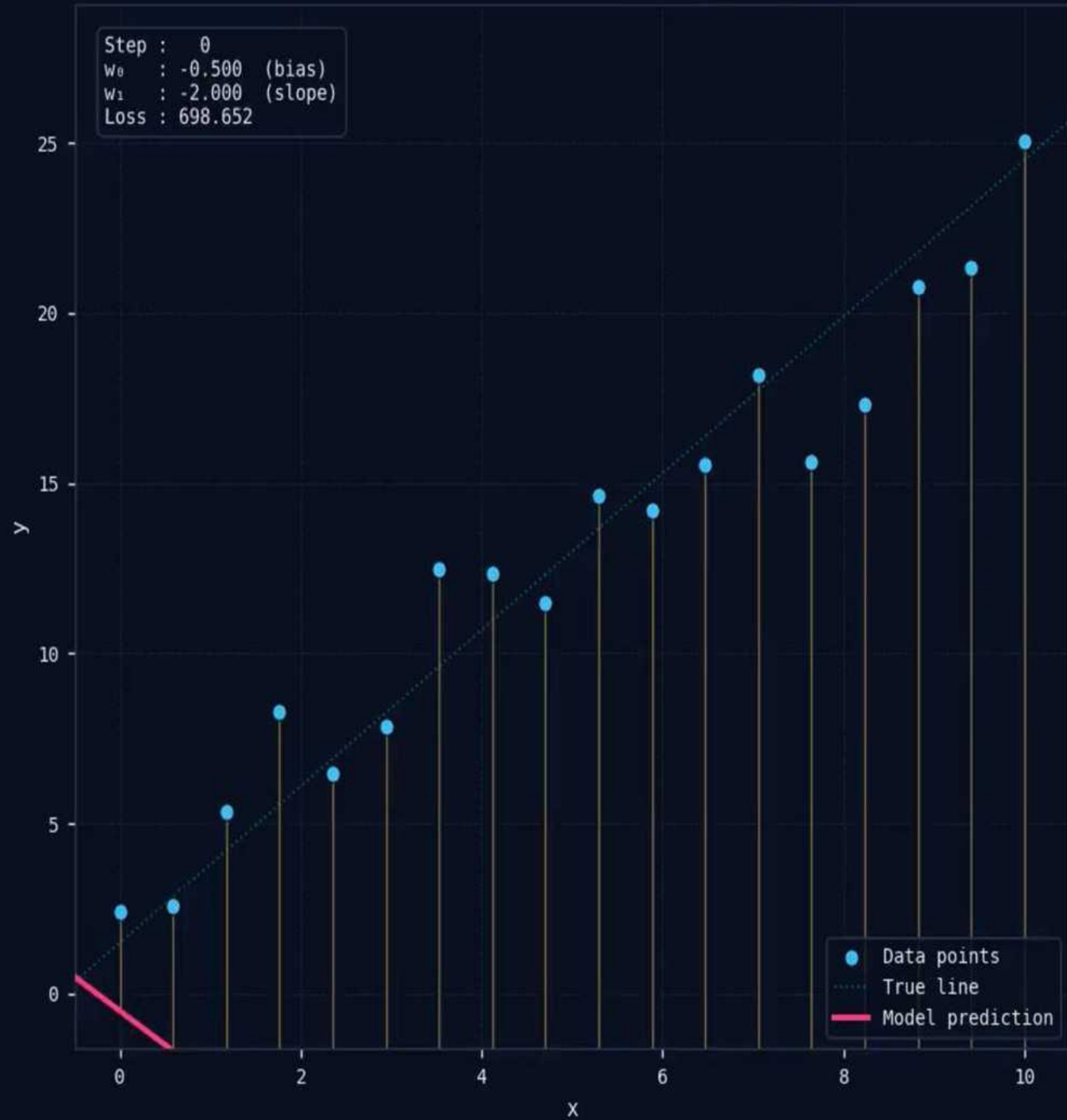
Gradient Descent

An optimization algorithm used to minimize the cost function by adjusting the model's parameters in the direction that reduces the cost.

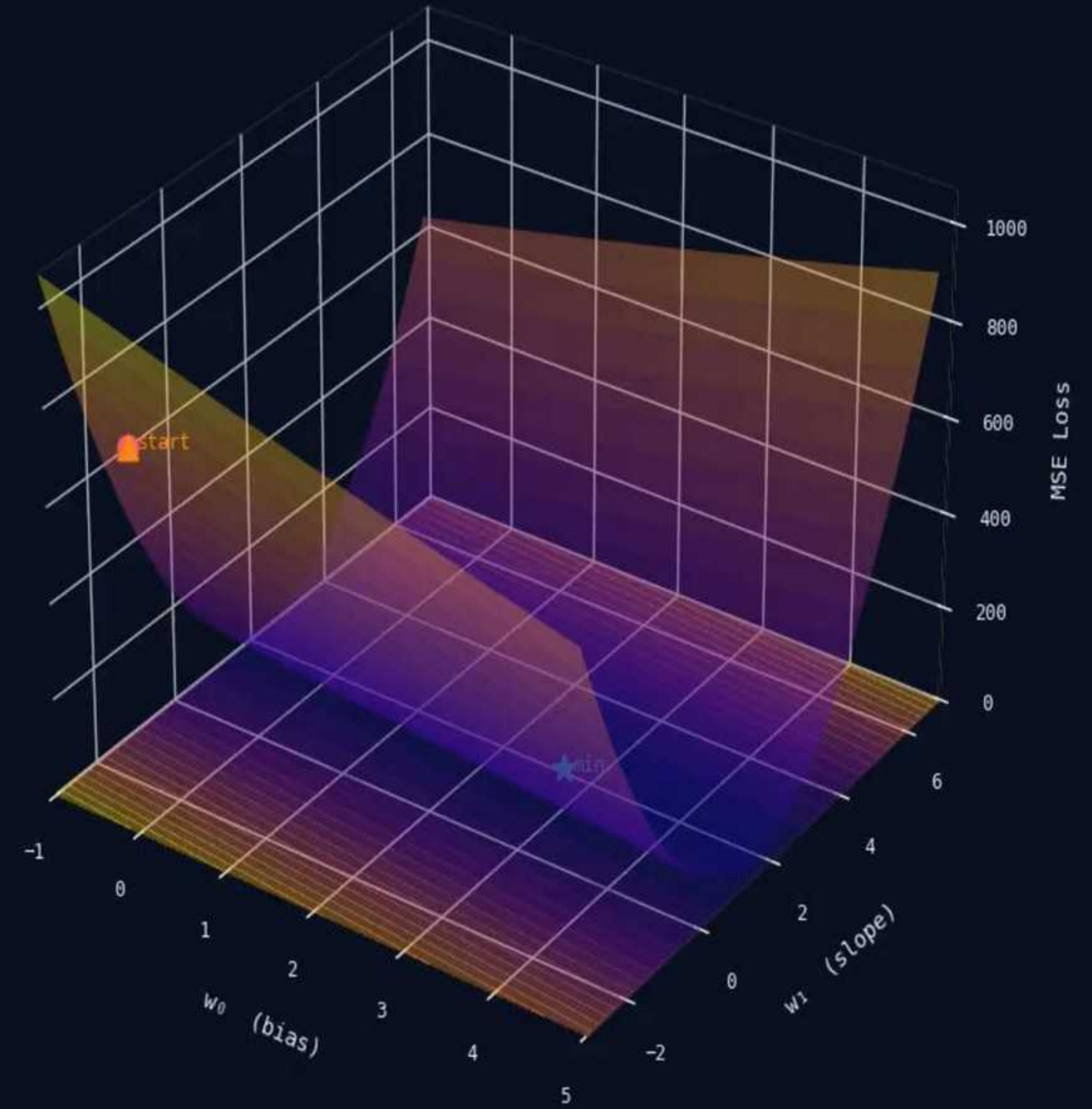
= Keep changing w , b to reduce cost function ($J(w, b)$) until we settle at or near a minimum.

Step 0 / 360 - Loss: 698.652

Model View - fitted line updating through data



Optimizer View - GD path on 3-D loss surface



Left: model (line) updates as weights change | Right: GD ball rolls down the 3-D loss surface - w_0 = bias, w_1 = slope

Multiple Linear Regression

Linear Regression

Feature (Independent variable)			Target (Dependent variable)
Engine Size	Brand	Miles	Price
1.2	A	120000	2.3
2	B	30800	4.2
1.6	A	42650	3.45
...	...	...	...
...	...	...	...

$$y = w_1X_1 + w_2X_2 + w_3X_3 + w_0 =$$

Linear Regression Assumptions

Linear Regression (Core Assumptions)

1. Linearity

2. Homoscedasticity

3. Independence of Errors

4. Normality of Residuals

5. No Multicollinearity

Linear Regression (Core Assumptions)

1. Linearity

2. Homoscedasticity

3. Independence of Errors

4. Normality of Residuals

5. No Multicollinearity

- **What it means:** The relationship between the independent variables (X) and the dependent variable (Y) must be linear.

- **Why it matters:** If you fit a straight line to curvilinear data, your model will underfit and produce highly inaccurate predictions.

- **How to test:** Examine a scatter plot of (X) vs. (Y), or a residual plot (residuals vs. predicted values). Look for random scatter; a U-shape or wave pattern indicates non-linearity

Linear Regression (Core Assumptions)

1. Linearity

2. Homoscedasticity

3. Independence of Errors

4. Normality of Residuals

5. No Multicollinearity

(Constant Residuals Variance)

- **What it means:** The variance of the residuals must remain constant across all levels of the independent variables.
- **Why it matters:** Heteroscedasticity (varying variance) inflates or deflates standard errors, making your hypothesis tests and p-values deceptive.
- **How to test:** Check a plot of residuals against predicted values. If the spread of data points fans out like a cone or funnel, homoscedasticity is violated. You can also run a formal statistical test like the Breusch-Pagan test.

Linear Regression (Core Assumptions)

1. Linearity

2. Homoscedasticity

3. Independence of Errors

4. Normality of Residuals

5. No Multicollinearity

(Constant Residuals Variance)



Linear Regression (Core Assumptions)

1. Linearity

2. Homoscedasticity

3. Independence of Errors

4. Normality of Residuals

5. No Multicollinearity

No Autocorrelation

- **What it means:** The residuals of one observation must not be correlated with the residuals of another observation. [1]

- **Why it matters:** This issue is highly common in time-series data, where yesterday's values heavily influence today's. Autocorrelation artificially inflates the model's statistical significance. [1, 2]

- **How to test:** Use the Durbin-Watson test. A test score near (2.0) indicates no autocorrelation, while scores far below or above (2.0) flag a violation. [1]

Linear Regression (Core Assumptions)

1. Linearity

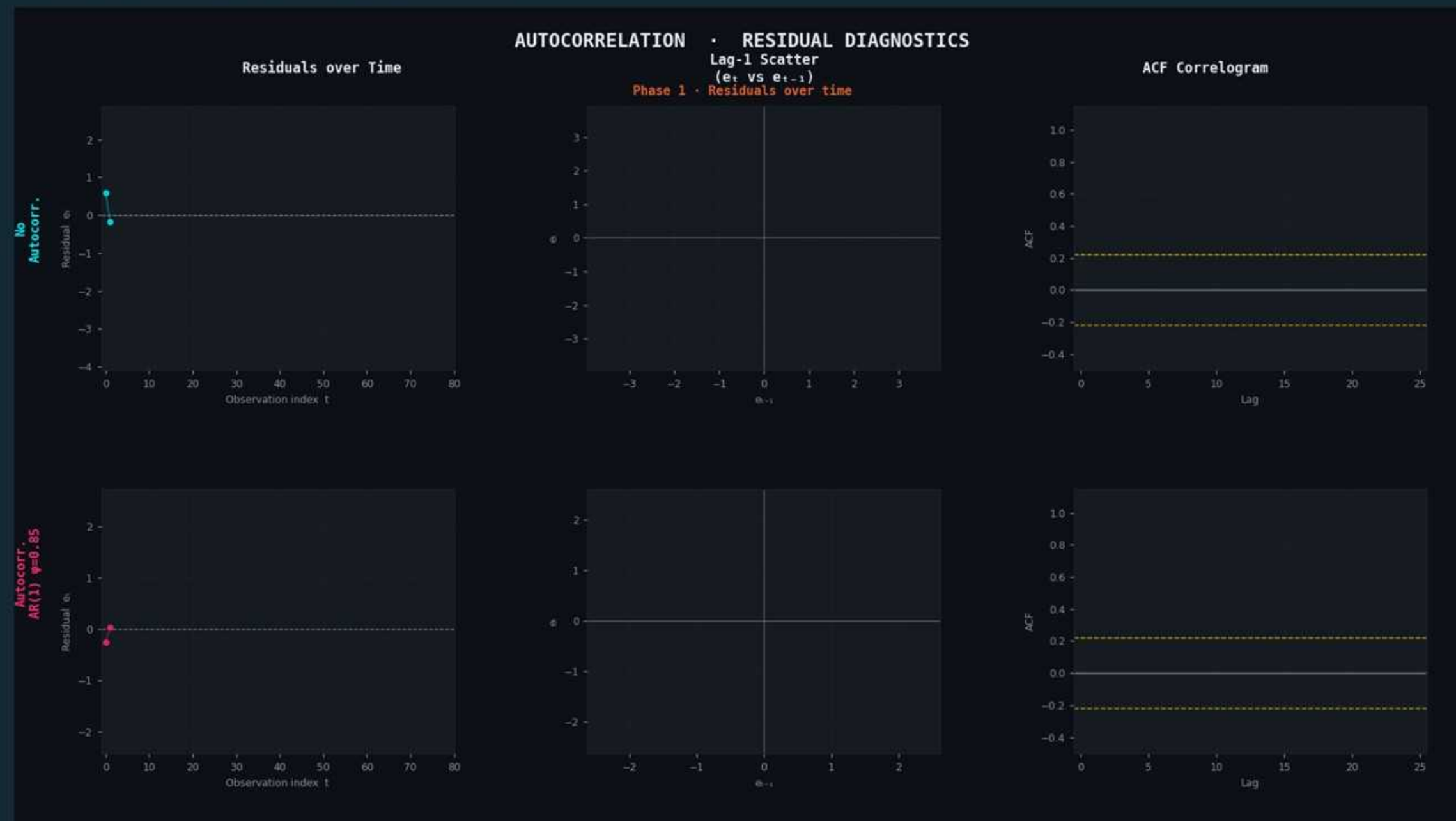
2. Homoscedasticity

3. Independence of Errors

4. Normality of Residuals

5. No Multicollinearity

No Autocorrelation



Linear Regression (Core Assumptions)

1. Linearity

2. Homoscedasticity

3. Independence of Errors

4. Normality of Residuals

5. No Multicollinearity

- **What it means:** The error terms (residuals) should be normally distributed around a mean of zero. Note that the input data (X) and (Y) does *not* need to be normally distributed only the errors do.

- **Why it matters:** While the central limit theorem minimizes this issue for massive sample sizes, normality is strictly required to calculate accurate confidence intervals and p-values in smaller datasets.

- **How to test:** Inspect a Quantile-Quantile (Q-Q) plot. If the residuals align tightly along the diagonal reference line, they are normally distributed. Alternatively, run a Shapiro-Wilk test. [1, 2, 3, 4, 5]

Linear Regression (Core Assumptions)

1. Linearity

2. Homoscedasticity

3. Independence of Errors

4. Normality of Residuals

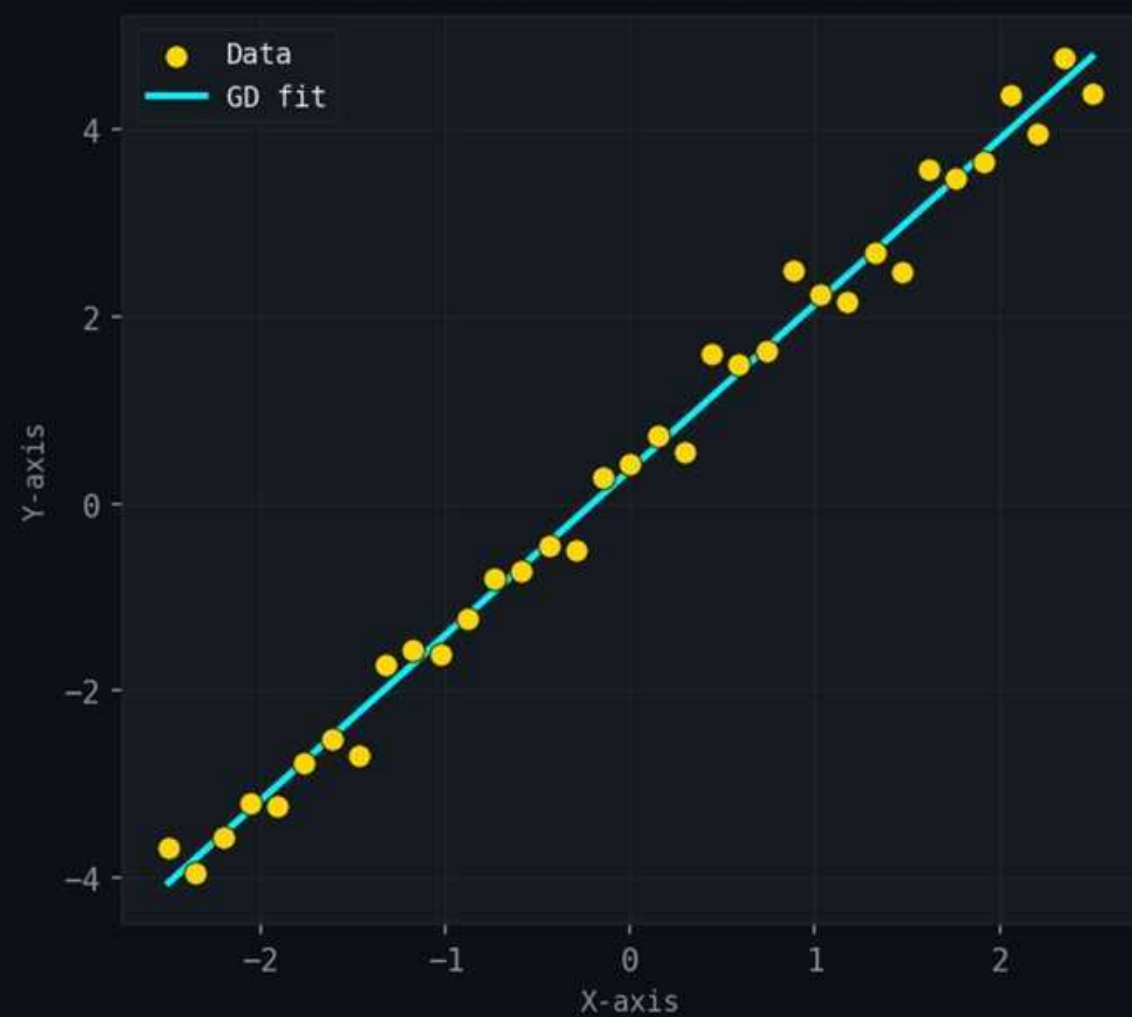
5. No Multicollinearity

- **What it means:** In multiple linear regression, the independent variables should not be highly correlated with one another.
- **Why it matters:** Severe multicollinearity makes it impossible for the model to isolate the true, individual impact of each independent variable on the target variable.
- **How to test:** Calculate the Variance Inflation Factor (VIF) for each predictor. A VIF value greater than (5) or (10) indicates problematic multicollinearity that requires fixing. [[1](#), [2](#)]

Polynomial Regression

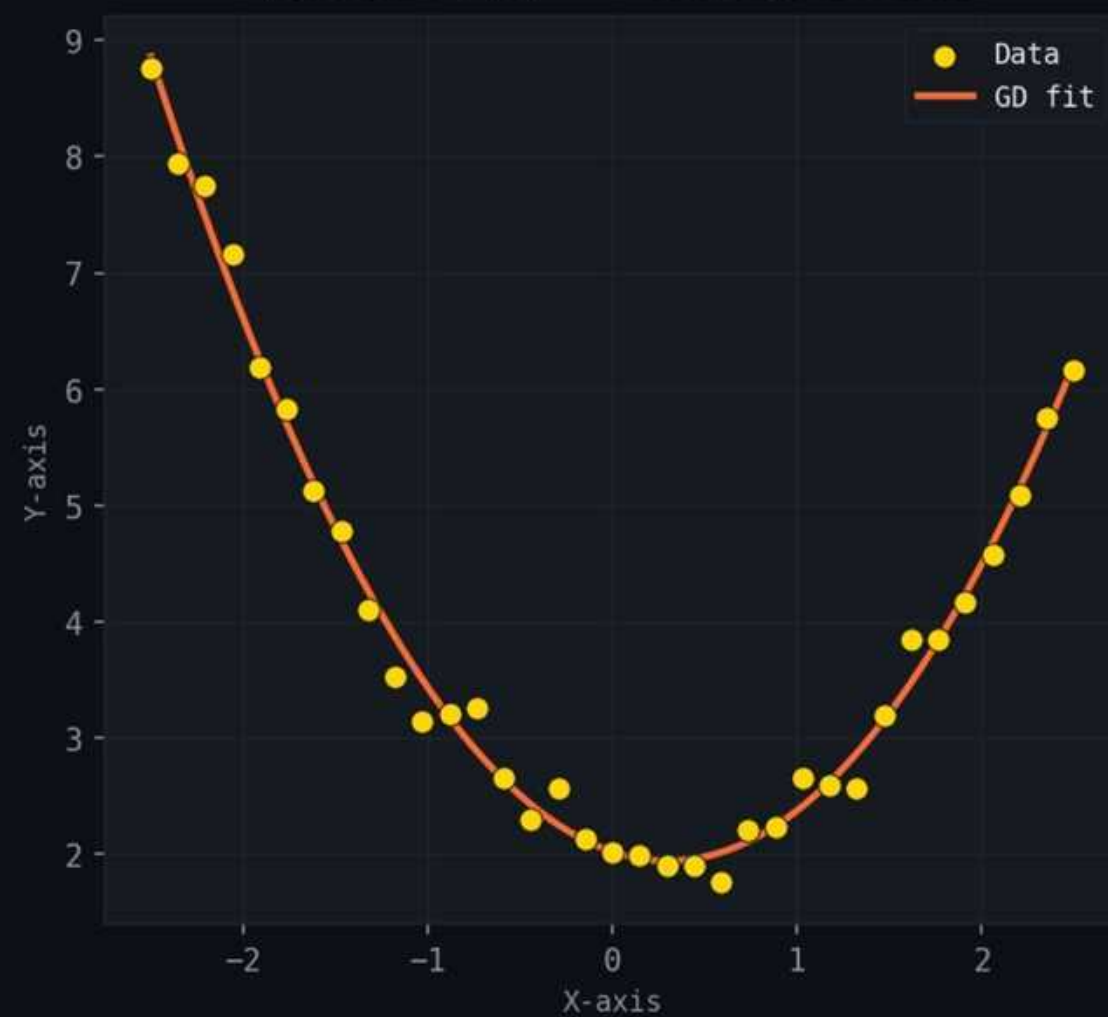
Polynomial Regression

Linear Data → Gradient Descent

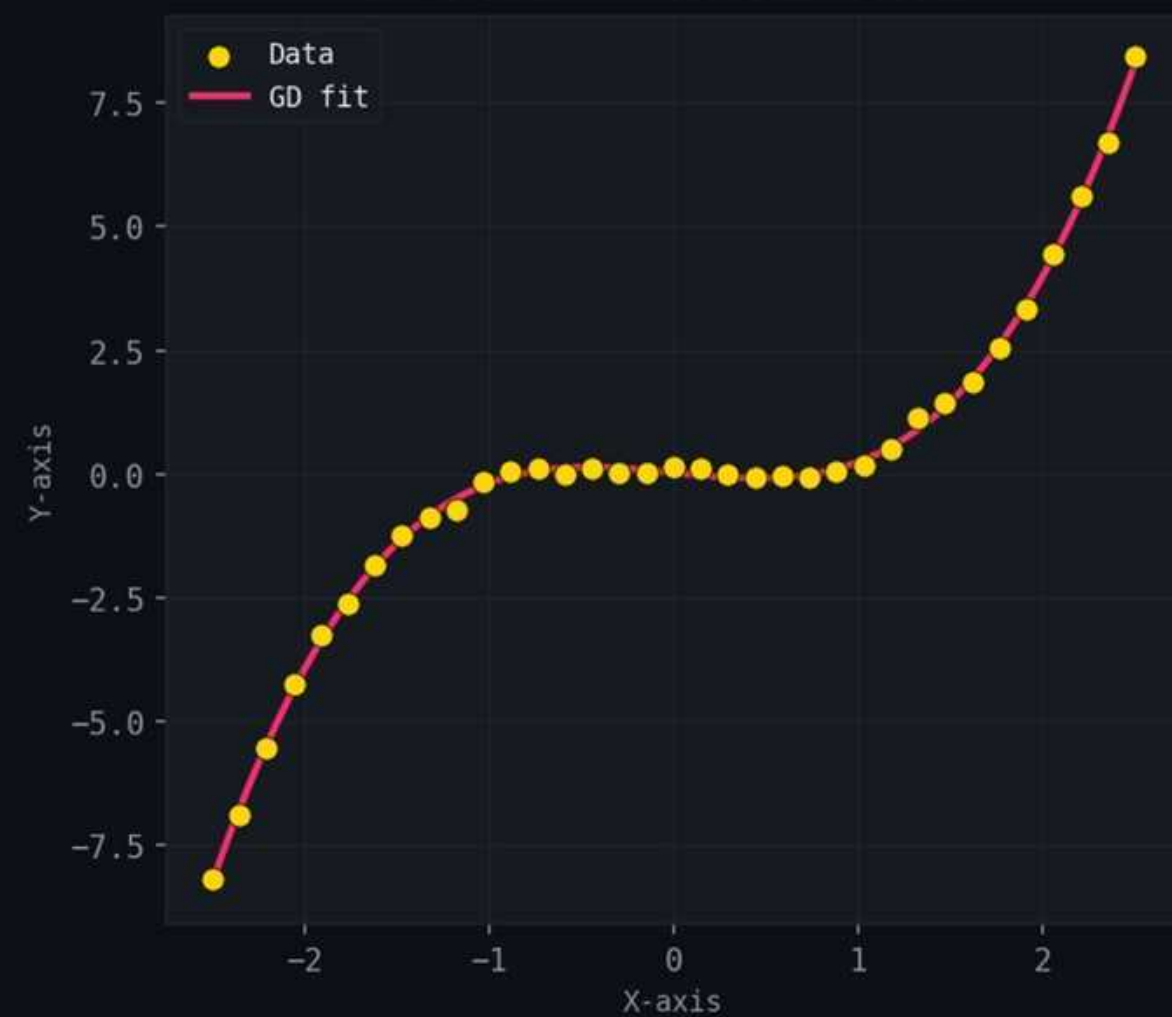


POLYNOMIAL FITS VIA GRADIENT DESCENT

Quadratic Data → Gradient Descent



Cubic Data → Gradient Descent



Polynomial Regression



Key Takeaways

-  Scaling numerical features
-  Always fit the scaler on training data only — never on test data
-  Encoding categorical features and use drop first in one-hot encoding to avoid multicollinearity
-  Use RobustScaler when your data has outliers
-  Use sklearn Pipelines in production to avoid leakage and simplify code
-  Check Linear model assumptions before fitting the model



THANK YOU

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